Magnetic-moment-constrained collisions

Equations, numerical design and verification notes

Project: `sato-morrison`

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Abstract

These notes derive and audit a small fixed-field JAX prototype of magnetic-moment-constrained collisions. Implemented paths include the local linear weak operator, a positive nodal nonlinear entropy step, five analytic vacuum fields, tangent toroidal Hamiltonian transport, and an explicit finite-range surrogate. Independent homogeneous controls include Lorentz and Dougherty evolution, physical three-velocity Landau weak moments, and direct screened binary encounters. A checked finite-range Fourier calculation shows how spatial localization creates a density nullspace and how Gaussian range lifts it with quartic long-wave decay. Geometric identities, finite-grid conservation and the selected toroidal observable refinements are verified. Full three-velocity Landau trajectories pass the declared refinements through normalized time 0.2. Selected screened-pair controls resolve sensitive paths and test individual magnetic-moment exchange. Long-time Landau accuracy, nonuniform velocity-tail convergence, a universal zero-set prescription, calibrated plasma rates and dipole confinement remain open.

1 Scope and equation map

The primary source is Ref. [1]. Its Eq. (35) gives a one-species noncanonical collision structure. Equations (111) and (119) are different levels of description: the former evolves a ten-coordinate pair distribution, while the latter closes two five-coordinate species distributions after a pair-factorization assumption. Equation (181) is a localized linear approximation with a prescribed scalar collision coefficient. This prototype must identify which of these models it implements.

The starting model has a prescribed, nonzero vacuum magnetic field, no magnetic perturbation, a fixed electrostatic reference potential, and one species. The nonlinear local-kernel extension is named separately from literal Eq. (181). No absolute collision rate or laboratory confinement result is established by these calculations.

Source equations	Implementation role
SM25 (35), (39)–(59)	Weak form, boundary conditions, invariants and stationary family.
SM25 (73)–(100)	Guiding-center bracket, vacuum assumption and invariant measure.
SM25 (118)–(119)	Pair closure and multispecies kernel; later stage.
SM25 (139)–(150)	Constrained stationary states and initial magnetic-moment marginal.
SM25 (164)–(183)	Linearization, kernel approximations and the first numerical target.
BS25, Sec. V	Structural comparison to a guiding-center Landau bracket.

2 Coordinates, measures and the bracket

We use SI variables before nondimensionalization. Let u be parallel velocity, $\mu = mv_\perp^2/(2B)$ the magnetic moment, and $\eta = \mu B$ the perpendicular kinetic energy. With $\boldsymbol{b} = \boldsymbol{B}/B$,

$$\boldsymbol{B}^* = \boldsymbol{B} + \frac{mu}{q}\nabla \times \boldsymbol{b}, \qquad B_\parallel^* = \boldsymbol{b} \cdot \boldsymbol{B}^*. \tag{1}$$

For the vacuum backgrounds used in Ref. [1], $\boldsymbol{b} \cdot \nabla \times \boldsymbol{b} = 0$, hence $B_\parallel^* = B$. This does not imply $\nabla \times \boldsymbol{b} = 0$: curvature terms in $\boldsymbol{B}^*$ generally remain.

In coordinates $z = (\mathbf{X}, u, \mu)$ the gyrophase-independent bracket acts on a covector $a = (\boldsymbol{a}_X, a_u, a_\mu)$ as

$$\mathcal{J}_\mu a = \begin{pmatrix} (\boldsymbol{b} \times \boldsymbol{a}_X)/(qB) + \boldsymbol{B}^* a_u/(mB) \\ -\boldsymbol{B}^* \cdot \boldsymbol{a}_X/(mB) \\ 0 \end{pmatrix}. \tag{2}$$

The Hamiltonian and characteristic equations are

$$E = \frac{1}{2}mu^2 + \mu B + q\Phi, \tag{3}$$

$$\dot{\mathbf{X}} = u\frac{\boldsymbol{B}^*}{B} + \frac{\boldsymbol{b} \times (\mu\nabla B + q\nabla\Phi)}{qB}, \tag{4}$$

$$\dot{u} = -\frac{\boldsymbol{B}^* \cdot (\mu\nabla B + q\nabla\Phi)}{mB}, \qquad \dot{\mu} = 0. \tag{5}$$

The signs in (2) must be checked against the cross-field drift in (4), not inferred from a memorized matrix convention.

The phase-space measures are

$$\mathrm{d}\Gamma = B\,\mathrm{d}^3X\,\mathrm{d}u\,\mathrm{d}\mu = \mathrm{d}^3X\,\mathrm{d}u\,\mathrm{d}\eta. \tag{6}$$

We follow the article's normalization: $n(\mathbf{X}) = \int f\,\mathrm{d}u\,\mathrm{d}\eta$. If F is an ordinary gyrotropic velocity density normalized with d^3v, then $\mathrm{d}^3v = (2\pi B/m)\mathrm{d}u\mathrm{d}\mu$ and $f = (2\pi/m)F$. Track this constant, particularly in multispecies comparisons.

The coordinate change $\zeta = (\mathbf{X}, u, \eta)$ has derivative

$$T = \frac{\partial\zeta}{\partial z} = \begin{pmatrix} I_3 & 0 & 0 \\ 0 & 1 & 0 \\ \mu(\nabla B)^T & 0 & B \end{pmatrix}, \qquad \mathcal{J}_\eta = T\mathcal{J}_\mu T^T. \tag{7}$$

This constructs the matrix in SM25 (93), equivalently (182), without duplicating its entries by hand. It also gives

$$\mathcal{J}_\eta \nabla_\zeta(\eta/B) = 0, \qquad \dot{\eta} = \mu\dot{\mathbf{X}} \cdot \nabla B. \tag{8}$$

Thus a discretization that freezes η during spatial transport is not the magnetic-moment-preserving model in a nonuniform field.

With general coordinates and measure $\rho(z)\mathrm{d}z$, conservative divergence means $\rho^{-1}\partial_a(\rho F^a)$, not $\partial_a F^a$. Verify the Liouville identity $\partial_a(\rho\mathcal{J}^{ab}) = 0$. The prototype uses $\rho = B$ in the μ chart and $\rho = 1$ in the article's η chart.

Continuous geometric audit

The implemented analytic fields pass automatic-differentiation checks of $\nabla \cdot \boldsymbol{B} = 0$, $\nabla \times \boldsymbol{B} = 0$, antisymmetry, the magnetic-moment Casimir, the Liouville density, and the Jacobi identity in both charts, at independent off-axis points and for both charge signs. Centered differences of $\boldsymbol{B}$ show second-order convergence to the automatic derivatives for the rational fields. These sampled identities audit the continuous formulas; they do not prove a discrete Jacobi identity for the collision mesh or positivity of B throughout an unsampled domain. Host validation rejects unsupported fields, nonfinite inputs, zero field, excluded dipole/toroidal sources, nonpositive mass, zero charge and negative moment.

For the vacuum toroidal field $B = C/R$, the characteristic equations give $\dot{R} = \dot{u} = \dot{\mu} = 0$, $\dot{\theta} = u/R$ and $\dot{z} = (mu^2 + \mu B)/(qC)$. Thus annular radial boundaries are tangent to Hamiltonian flow; a periodic z and toroidal angle are compatible with this geometry. The library checks this identity and the uniform electrostatic drift independently of collision evolution.

Normalization is part of the closure

Choose fixed scales L_*, v_*, E_*, t_* and a diagonal coordinate scale matrix S. For $\zeta = S\hat{\zeta}$, $E = E_*\hat{E}$ and $t = t_*\hat{t}$,

$$\hat{\mathcal{J}} = t_* E_* S^{-1} \mathcal{J} S^{-T}. \tag{9}$$

State these scales with every dimensional result. A Euclidean phase-space projector applied to a vector containing spatial velocities, accelerations and energy rates is not defined independently of those scales.

For a purely affine chart change $y = Az$ with time and energy conventions fixed, a pair vector transforms as $d_y = Ad_z$. Its quadratic form is unchanged only if

$$\Pi_y = A^{-T}\Pi_z A^{-1}, \qquad d_y^T \Pi_y e_y = d_z^T \Pi_z e_z. \tag{10}$$

Recreating an identity matrix and Euclidean projector in the new chart generally specifies a different kernel. For the position-dependent map (7), compute the two vectors in one common chart before subtracting:

$$\Delta_\eta \phi = T(z)\mathcal{J}_\mu(z)\nabla_z \phi - T(z')\mathcal{J}_\mu(z')\nabla_{z'}\phi'. \tag{11}$$

This is the safe route when storing the state on μ nodes but reproducing a kernel defined in the η chart.

3 Nonlinear collision structure and conservation

For a single species define

$$\Delta\phi = \mathcal{J}\nabla\phi - \mathcal{J}'\nabla'\phi', \qquad \xi = \Delta E. \tag{12}$$

A specified kernel must obey exchange symmetry, symmetry of its quadratic form, positive semidefiniteness and the energy degeneracy:

$$\Pi = \Pi^T \succeq 0, \qquad \Pi\xi = 0. \tag{13}$$

The source strong form is

$$C[f] = \nabla_z \cdot \left[f\mathcal{J} \int f'\Pi(\mathcal{J}'\nabla' \ln f' - \mathcal{J}\nabla \ln f)\, \mathrm{d}z' \right] \tag{14}$$

in a flat invariant-measure chart. Integration by parts, antisymmetry of $\mathcal{J}$ and exchange of the two labels give

$$\int \phi C[f] \mathrm{d}\Gamma = -\frac{1}{2} \iint f f' \, \Delta\phi^T \Pi \Delta \ln f \, \mathrm{d}\Gamma \mathrm{d}\Gamma'. \tag{15}$$

All following equalities require the boundary terms to vanish. The paper states conditions for both the total flux and the Hamiltonian part; a finite numerical box does not satisfy them automatically.

Set $\phi = 1$ in (15) to obtain particle conservation. Set $\phi = E$ and use (13) for collision energy conservation. Since $\mathcal{J}\nabla g(\mu) = 0$ for every smooth g,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int f g(\mu)\mathrm{d}\Gamma = 0. \tag{16}$$

For closed evolution this is equivalent to conserving the full marginal

$$G(\mu) = \int B f(\mathbf{X}, u, \mu)\mathrm{d}^3X\,\mathrm{d}u. \tag{17}$$

It is stronger than conservation of $\int \mu f \mathrm{d}\Gamma$ alone.

Define $S[f] = -\int f \ln(f/f_{\mathrm{ref}})\mathrm{d}\Gamma$ with a constant reference scale inside the logarithm. Particle conservation removes that scale from the entropy rate. Equation (15) yields

$$\dot{S} = \frac{1}{2}\iint f f' \Delta \ln f^T \Pi \Delta \ln f \,\mathrm{d}\Gamma\mathrm{d}\Gamma' \geq 0. \tag{18}$$

These are structural statements from the model, not a proof of a unique final state. Additional null directions of Π can prevent mixing.

A candidate observable p is conserved by the full kinetic model only if the Hamiltonian flow preserves it and the collision form annihilates Δp. An external dipole field does not imply conservation of all Cartesian momenta. For an axisymmetric model, check canonical angular momentum using the actual kernel and boundaries rather than inferring it from the field name.

4 The fixed-field linear model

Write $f = f_0(1 + h)$, where $f_0 > 0$ is stationary. For the source stationary family,

$$f_0 = A\exp[-\beta E_0 + g(\mu)], \qquad \mathcal{J}\nabla \ln f_0 = -\beta \mathcal{J}\nabla E_0. \tag{19}$$

With the fields and kernel fixed, the zeroth-order collision term vanishes. This supplied energy/moment family is not claimed to exhaust stationary states. For the exactly local poloidal surrogate, (81) adds an arbitrary conserved-flux function. The first-order weak form is

$$\int \phi C_L[f_0 h]\mathrm{d}\Gamma = -\frac{1}{2}\iint f_0 f_0' \Delta\phi^T \Pi_0 \Delta h \,\mathrm{d}\Gamma\mathrm{d}\Gamma'. \tag{20}$$

The local scalar approximation in SM25 (181) uses

$$\Pi_0 = DPI_xP\,\delta^3(\mathbf{X} - \mathbf{X}'), \quad P = I - \frac{\xi_0\xi_0^T}{\xi_0^T\xi_0}, \quad I_x = \mathrm{diag}(1,1,1,0,0). \tag{21}$$

Here D is a prescribed constant, and the projector is evaluated in a specified normalized common chart. The two velocity integrals remain after the spatial delta function is used. In a μ chart, the local pair measure contains two velocity Jacobians; do not accidentally retain only one factor of B.

The perturbation norm

$$Q[h] = \frac{1}{2}\int f_0 h^2 \mathrm{d}\Gamma \tag{22}$$

satisfies

$$\dot{Q} = -\frac{1}{2}\iint f_0 f_0' \Delta h^T \Pi_0 \Delta h \, \mathrm{d}\Gamma \mathrm{d}\Gamma' \leq 0 \tag{23}$$

for collision-only dynamics. A compatible stationary Hamiltonian flow also preserves Q under suitable boundary conditions.

For self-consistent fields, (20) is not a complete linearization unless the omitted terms have been ordered out. Differentiating the exact energy-degeneracy condition gives

$$\delta\Pi \, \xi_0 + \Pi_0 \delta\xi = 0. \tag{24}$$

The approximations in SM25 (164)–(170) must be carried explicitly into any extension. The fixed-field implementation checks its linear operator against a directional derivative of the nonlinear residual; this does not validate omitted self-consistent-field terms.

5 Uniform-field reduction and exact null modes

This section is an independent derivation from the source Eq. (181). Take a uniform field, a constant reference potential and $f_0 = f_0(u, \eta)$. For $\boldsymbol{b} = \boldsymbol{e}_z$,

$$\mathcal{J}a = \left(-\frac{a_y}{qB}, \frac{a_x}{qB}, \frac{a_u}{m}, -\frac{a_z}{m}, 0\right)^T, \qquad \mathcal{J}\nabla E_0 = (0, 0, u, 0, 0)^T. \tag{25}$$

Consequently $\xi_0 = (u - u')\boldsymbol{e}_z$ in the spatial block, and

$$PI_xP = \mathrm{diag}(1, 1, 0, 0, 0). \tag{26}$$

At $u = u'$ use the continuous limit (26); this prescription does not extend automatically to a general nonuniform field.

Only perpendicular spatial derivatives survive. Let

$$n_0 = \int f_0 \mathrm{d}u\mathrm{d}\eta, \qquad \delta n = \int \delta f \mathrm{d}u\mathrm{d}\eta. \tag{27}$$

Two applications of the cross-product matrix give $(\boldsymbol{b}\times)^2 = -I_\perp$. Substitution into the strong form gives

$$\boxed{C_L[\delta f] = \frac{D}{(qB)^2}\nabla_\perp^2(n_0\delta f - f_0\delta n).} \tag{28}$$

This is spatial mixing of velocity-dependent structure, not ordinary homogeneous velocity-space thermalization.

For a Fourier mode with $\boldsymbol{k}_\perp = \boldsymbol{k} - (\boldsymbol{k}\cdot\boldsymbol{b})\boldsymbol{b}$, define

$$\lambda = \frac{Dn_0|\boldsymbol{k}_\perp|^2}{(qB)^2}, \qquad \bar{h} = \frac{1}{n_0}\int f_0 h \mathrm{d}u\mathrm{d}\eta. \tag{29}$$

The collision-only equation is

$$\partial_t h = -\lambda(h - \bar{h}), \qquad h(t) = \bar{h} + [h(0) - \bar{h}]e^{-\lambda t}. \tag{30}$$

Thus a density-like perturbation is undamped. If $\boldsymbol{k}_\perp = 0$, every velocity perturbation is collisionally undamped. A homogeneous anisotropic distribution is therefore not an isotropization benchmark for this model.

With positive velocity quadrature weights w_i, set

$$m_i = w_i f_{0i}, \quad s_i = \sqrt{m_i}, \quad n_0 = \sum_i m_i. \tag{31}$$

The matrix for h and its entropy-scaled form are

$$L_h = -\lambda \left(I - \frac{\mathbf{1}\boldsymbol{m}^T}{n_0} \right), \tag{32}$$

$$M_v^{1/2} L_h M_v^{-1/2} = -\lambda \left(I - \frac{\boldsymbol{s}\boldsymbol{s}^T}{n_0} \right). \tag{33}$$

For $\lambda > 0$ the spectrum has one zero eigenvalue and $N_v - 1$ copies of $-\lambda$. This is an exact finite-quadrature statement when the same n_0 is used throughout. It provides a stringent factor, sign and normalization test.

When parallel streaming is added, $\partial_t h = -ik_\parallel u h + L_h h$. A density-like collision null mode need not remain in that nullspace under streaming. Distinguish the collision spectrum from the spectrum of the combined generator. With $\boldsymbol{k}_\perp = 0$ and frozen fields the exact streaming solution is $h(t, u, \eta) = h(0, u, \eta) e^{-ik_\parallel u t}$; this is not self-consistent Landau damping. An executed distribution regression integrates the Fourier generator obtained from the actual uniform Poisson characteristic with an independent DOP853 integrator and compares it with this analytic phase at four times through $t = 1.1$. It uses $m = 1.7$, $B = 2.3$, charges -0.8 and 1.3, and $k_\parallel = 2$. The positive reconstructed distribution agrees within the stated 2×10^{-12} relative/3×10^{-13} absolute test tolerances; each velocity-bin spatial population is preserved. The independent pair oracle gives zero parallel collision action at nonzero D. A separate active perpendicular-grid regression sets $D = 0$ and verifies that the implicit collision step is the identity. These finite-time controls do not assert long-time damping or resolve a kinetic recurrence limit.

6 A finite-range surrogate and its exact density branch

This section specifies a new calculation for an explicit surrogate, not a Coulomb scattering tensor or a claim of publication priority. Retain the uniform field and constant-potential assumptions of (25), and replace the spatial delta function in (21) by a nonnegative, even, translation invariant kernel $W(\mathbf{X} - \mathbf{X}')$ with $\int W \mathrm{d}^3X = 1$. Use the same uniform $PI_xP = I_\perp$ prescription, including its continuous equal-u limit. The weak form then gives

$$C_W[\delta f] = \frac{D}{(qB)^2} \nabla_\perp^2 \left[n_0 \delta f - f_0 (W * \delta n) \right]. \tag{34}$$

To check the sign and convolution placement, the projected pair difference for h is $(\boldsymbol{b}\times)/(qB)$ applied to $\nabla_\perp h(\mathbf{X}, v) - \nabla'_\perp h(\mathbf{X}', v')$. The first self term integrates to $n_0 \nabla_\perp h$; the cross term is $\nabla_\perp W * \delta n$. The second cross-product operation contributes $-I_\perp$, giving (34) with the displayed diffusion sign.

Finite-quadrature proposition. Let $m_i = w_i f_{0i} > 0$, $n_0 = \sum_i m_i$, $s_i = \sqrt{m_i}$, $r(\boldsymbol{k}) = \widehat{W}(\boldsymbol{k})$, and $\lambda = D n_0 |\boldsymbol{k}_\perp|^2 / (qB)^2$. In the entropy coordinates,

$$-M_v^{1/2} L_{W,\boldsymbol{k}} M_v^{-1/2} = \lambda \left[I - r(\boldsymbol{k}) \frac{\boldsymbol{s}\boldsymbol{s}^T}{n_0} \right]. \tag{35}$$

Its exact decay rates are λ on the $N_v - 1$ dimensional velocity-neutral subspace and $\lambda[1 - r(\boldsymbol{k})]$ on the density direction $\boldsymbol{s}$. Indeed, $\boldsymbol{s}^T \boldsymbol{s} = n_0$, so the normalized outer product is an orthogonal rank-one

projector. An even nonnegative kernel has real $r \leq 1$; thus all rates are nonnegative. This proof does not require $r \geq 0$. For a discrete periodic spatial quadrature the identical statement holds with r equal to the Fourier multiplier of the sampled, symmetrically normalized kernel, rather than the continuum multiplier.

For a normalized periodized isotropic Gaussian of covariance $\ell^2 I$, $r = e^{-\ell^2|\boldsymbol{k}|^2/2}$ and hence

$$\lambda_{\text{density}} = \frac{Dn_0|\boldsymbol{k}_\perp|^2}{(qB)^2}\left(1 - e^{-\ell^2|\boldsymbol{k}|^2/2}\right) = \frac{Dn_0\ell^2}{2(qB)^2}|\boldsymbol{k}_\perp|^2|\boldsymbol{k}|^2 + O(|\boldsymbol{k}_\perp|^2\ell^4|\boldsymbol{k}|^4). \tag{36}$$

A transverse long wave therefore has quartic rather than quadratic density decay. Every finite nonzero Gaussian range lifts the nonzero-transverse-wave local density null mode, but the lifted rate vanishes as ℓ^2 in the local limit. Homogeneous and purely parallel modes remain fully collisionally null. Consequently this calculation does not establish a whole-space spectral gap, or a gap for all modes on a periodic domain containing parallel-only modes. On a fixed finite set of transverse Fourier modes, it establishes the stated positive density rates. The limits $t \to \infty$ and $\ell \to 0$ need not commute for the density branch.

The independent NumPy check in `examples/06_range.py` assembles the ordered spatial/velocity pair weak sum, including the factor 1/2. The Fourier basis is differentiated exactly; the Gaussian is evaluated by positive periodic image sums and normalized with one scalar to retain pair symmetry. Three spatial resolutions, three velocity node counts and three kernel ranges are compared with (35). The velocity weights are normalized once at setup to fixed n_0; this verifies the finite-quadrature theorem rather than physical velocity-tail convergence. The result is independent of the production JAX pair implementation.

Literature comparison and novelty limits

SM25 supplies the noncanonical weak structure and its local scalar approximation. SM24 supplies the earlier noncanonical construction [2]. BS25, Eqs. (99)–(103), retains magnetic-moment friction and diffusion through its gyrophase-dependent projection Δ^{μ}_{gc}, and explains the structural reduction obtained by removing that component and replacing the relative velocity by the reduced Hamiltonian pair velocity. That passage does not identify every scalar/local SM25 approximation with the complete conventional Landau tensor. Our uniform surrogate remains unable to isotropize a homogeneous distribution.

KH17 supplies the discrete-gradient conservation and entropy method; those properties do not supply a coercivity constant. Bressan, Kraus, Maj and Morrison [3] analyze sufficient relaxation conditions, including additional finite-dimensional inequalities and tentative infinite-dimensional extensions. Their framework motivates testing the actual nullspace before assigning asymptotic decay to a constrained maximum-entropy state.

On 7 October 2026, primary-source searches of the SM25 title combined with finite range, nullspace and Gaussian terms, and searches for noncanonical finite-range collisions and magnetic-moment quartic relaxation, found no matching statement of (35)–(36). Local full-text checks covered SM25, SM24, BS25, KH17 and BBQ15. This is a scoped independently checked derivation for the specified surrogate; absence from these searches is not proof of publication priority. No general nonuniform finite-range theorem, Coulomb rate, or physical metastability is inferred.

7 Stationary states and initial constraints

The source stationary family (19) follows from energy degeneracy and the magnetic-moment Casimir. For $g(\mu) = -\beta\gamma\mu$,

$$f_* = A \exp\left[-\beta\left(\frac{1}{2}mu^2 + \mu(B+\gamma) + q\Phi\right)\right], \quad B + \gamma > 0. \tag{37}$$

Integrating over u and μ using (6) gives

$$n_*(\mathbf{X}) = A\sqrt{\frac{2\pi}{\beta m}}\, e^{-\beta q\Phi} \frac{B}{\beta(B+\gamma)}. \tag{38}$$

The corresponding temperature parameters in energy units are

$$T_\parallel = \beta^{-1}, \qquad T_\perp = \frac{B}{\beta(B+\gamma)}. \tag{39}$$

These formulas are useful checks of velocity integration and the definition of pressure. They do not establish that an arbitrary initial state evolves to this particular family member.

Two further velocity-integral regressions check explicit limits of the known stationary family. With $f = \exp[-\beta(mu^2/2 + \mu(B+\gamma))]$ and the specified measure $B\mathrm{d}u\mathrm{d}\mu$, its density is $n(B) = c_\beta B/(B+\gamma)$, where $c_\beta = \sqrt{2\pi/(\beta m)}/\beta$. At $\gamma = 0$, independent positive-quadrature integrals give the constant c_β at three points in each of the five fields, including nonunit mass. For the harmonic mirror of strength B_0 and amplitude a,

$$B(a) = B_0 + a(z^2 - R^2/2) + O(a^2), \qquad n(B(a)) = \frac{c_\beta B_0}{B_0+\gamma} + a\frac{c_\beta\gamma(z^2 - R^2/2)}{(B_0+\gamma)^2} + O(a^2).$$

The quadrature density agrees with the independently integrated analytic formula, and its divided-difference slope approaches this expansion under three amplitude halvings. These tests concern the known density family, not attraction to it, physical rate calibration or the near-uniform projector limit.

For a prescribed fixed-field energy and a known initial marginal $G_0(\mu)$, let

$$Z_\mu(\beta) = \int B(\mathbf{X}) e^{-\beta E(\mathbf{X},u,\mu)} \mathrm{d}^3X\,\mathrm{d}u. \tag{40}$$

A candidate constrained maximum-entropy state is

$$f_*(\mathbf{X}, u, \mu) = \frac{G_0(\mu)}{Z_\mu(\beta)} e^{-\beta E(\mathbf{X},u,\mu)}. \tag{41}$$

Solve for β from total energy. Add other invariant multipliers if they actually survive the chosen dynamics. The same initial inverse-temperature parameter need not equal the final multiplier. Extra kernel invariants can make (41) inaccessible.

At fixed G_0, differentiation of the mean energy of (41) gives

$$\frac{\mathrm{d}E_*}{\mathrm{d}\beta} = -\int G_0(\mu)\,\mathrm{Var}_{\beta,\mu}(E)\mathrm{d}\mu \le 0. \tag{42}$$

It is strictly negative if at least one populated moment bin has nonconstant energy. Thus an energy target within the attainable bracket selects a unique β in that bracket. Positive-temperature existence is not automatic for an arbitrary truncated initial distribution; the implementation rejects an unbracketed

target. The same calculation holds for positive finite quadrature. With matching energy and full moment marginal, relative entropy gives $S[f_*]-S[f]=\int f\ln(f/f_*)\mathrm{d}\Gamma\geq 0$. This identifies a constrained maximum-entropy candidate; it does not establish dynamic accessibility under an operator possessing additional invariants.

For self-consistent electrostatics,

$$-\epsilon_0\nabla^2\Phi=\sum_s q_s n_s+\rho_{\mathrm{fixed}}. \tag{43}$$

The self-field energy is $\epsilon_0\int|\nabla\Phi|^2\mathrm{d}^3X/2$, with boundary terms treated consistently. A prescribed external potential contributes $\sum_s\int q_s\Phi_{\mathrm{ext}}f_s\mathrm{d}\Gamma_s$ without the self-interaction half-factor. A periodic field solve requires net neutrality and a gauge choice.

8 Comparison models and what each tests

The conventional reduced Landau literature retains magnetic-moment friction and diffusion even when the distribution is gyrophase independent [4, 5, 6, 7]. Brizard and Sugama explicitly discuss a structural reduction with $\boldsymbol{\Delta}^{\mu}_{\mathrm{gc}}=0$ and a reduced relative velocity in Sec. V of Ref. [7]. This is a bridge for comparison, not a verification that every simplified Sato–Morrison kernel is identical to their Landau tensor. The papers use different conventions; convert all masses, charges and unit factors before comparing implementations.

Pitch-angle scattering

With $v=|\boldsymbol{v}|$ and $\xi=u/v$, use

$$C_{\mathrm{Lz}}[F]=\frac{\nu(v)}{2}\partial_\xi[(1-\xi^2)\partial_\xi F]. \tag{44}$$

The Legendre identity gives $C_{\mathrm{Lz}}[P_\ell]=-\nu\ell(\ell+1)P_\ell/2$ for constant ν on a speed shell. Hence degrees $0,1,2,3$ have rates $0,\nu,3\nu,6\nu$. Speed-shell population and energy are conserved, but momentum of the evolved species can be transferred to the fixed reservoir. Since $\mu=mv^2(1-\xi^2)/(2B)$, this is a moment-changing comparison. Derive its coordinate transform; it is not bare diffusion at fixed u in μ.

A nonlinear conserving Fokker–Planck model

The single-species Dougherty form used as a comparison is

$$C_D[F]=\nu\nabla_v\cdot[(\boldsymbol{v}-\boldsymbol{U})F+\theta\nabla_v F],\quad \boldsymbol{U}=\frac{1}{n}\int\boldsymbol{v}F\mathrm{d}^3v,\quad \theta=\frac{1}{3n}\int|\boldsymbol{v}-\boldsymbol{U}|^2F\mathrm{d}^3v. \tag{45}$$

Integration by parts shows number and momentum conservation. The trace of the centered covariance is constant. Its matrix satisfies

$$\dot{\Theta}=-2\nu(\Theta-\theta I),\qquad \Theta(t)=\theta I+[\Theta(0)-\theta I]e^{-2\nu t}. \tag{46}$$

For an initially Gaussian homogeneous state this specifies the full Gaussian solution. Use it as an exact code test. A fixed bath temperature or drift would define a different model. Neither version is the exact Coulomb operator.

Physical three-velocity Landau reference

A one-species normalized reference is

$$U(\boldsymbol{w}) = \frac{I - \hat{\boldsymbol{w}}\hat{\boldsymbol{w}}^T}{|\boldsymbol{w}|}, \tag{47}$$

$$C_{\mathrm{La}}[F] = \Gamma \nabla_v \cdot \int F F' U(\boldsymbol{v} - \boldsymbol{v}')(\nabla_v \ln F - \nabla_{v'} \ln F') \mathrm{d}^3 v'. \tag{48}$$

Derive the dimensional prefactor for the selected normalization separately. Its positive kernel and velocity-difference null direction support conventional number, momentum, energy and entropy tests. Use an independently assembled small 3V problem before implementing a general transformed operator.

A gyrotropic two-coordinate distribution still has three-dimensional collision geometry. Converge the gyrophase integral or derive its analytic average; a literal 2V Landau kernel is not an equivalent comparator. Maxwell-molecule and BKW-type tests in Ref. [8] can test a different kernel's numerical machinery, but do not establish Coulomb rates.

9 Analytic magnetic fields and boundaries

A vacuum field can be specified locally as the gradient of a harmonic scalar potential. Use the following fields with explicit domains.

Uniform field. $\boldsymbol{B} = B_c \boldsymbol{b}_0$, with $|\boldsymbol{b}_0| = 1$.

Harmonic mirror. Let $R^2 = x^2 + y^2$ and

$$\Psi_B = B_c z + a\left(z^3/3 - zR^2/2\right), \tag{49}$$

$$\boldsymbol{B} = \nabla \Psi_B = (-azx, -azy, B_c + a[z^2 - R^2/2]). \tag{50}$$

Its curl and divergence vanish identically. Choose a domain on which B stays strictly positive. The dimensionless variation is controlled by aL^2/B_c. An arbitrary field $B(z)\boldsymbol{e}_z$ is not an admissible substitute.

Toroidal vacuum field. On $R > R_{\min} > 0$,

$$\boldsymbol{B} = \frac{C}{R^2}(-y, x, 0). \tag{51}$$

This provides curvature and field-strength variation without a parallel gradient of B. Its noncontractible domain is part of the problem.

Point dipole. Away from the source,

$$\boldsymbol{B} = \frac{C_d}{r^5}(3xz, 3yz, 2z^2 - x^2 - y^2), \quad \Psi_B = -C_d z/r^3. \tag{52}$$

This is a field test, not a model of a finite winding or material boundary. Exclude $r = 0$ and any near-source region outside the guiding-center ordering.

Symmetry breaking. A controlled addition $\delta\Psi_B = \varepsilon B_c xy/L$ is harmonic. Its gradient can be added to a vacuum background; check the resulting nonzero-field domain and boundary setup.

Do not make a nonperiodic magnetic field periodic by wrapping its array. Begin with collision-only nonuniform tests under natural zero-collision-flux boundary conditions. For full dynamics, require tangent Hamiltonian flux, a derived invariant-compatible reflection, a specified confining potential, or an open system with measured flux budgets. Arbitrary specular reflection need not preserve μ. A density-profile test without these choices is not a closed relaxation experiment.

Complete smooth nullspace of the local toroidal surrogate

The following classification concerns the specified local PI_XP closure, not a conventional Coulomb operator. Assume $B = (C/R)\boldsymbol{e}_\theta$, $C > 0$, $m > 0$, $q \neq 0$, constant electrostatic potential and $D > 0$. Take a connected radial interval away from $R = 0$, periodic θ, z, and a product velocity domain $I_u \times I_\mu$ with both intervals of positive length. The equilibrium weight is strictly positive in this domain. Consider real C^2 test functions in the weighted form domain, so the pair quadratic integral is finite. Periodicity and natural zero collision flux on nonperiodic faces give the weak identity (20).

Proposition. The smooth collision-null functions are exactly

$$h = \phi(R, \theta, z) + g(\theta, \mu) + a(\theta)E + b(\theta)mRu, \qquad E = \tfrac{1}{2}mu^2 + \mu C/R. \tag{53}$$

Here the functions are arbitrary subject to the stated regularity, periodicity and integrability. Redundant angle-only terms may be moved between ϕ and g. The common collision and ideal stationary functions are exactly

$$h = \phi(R) + g(\mu) + aE + bmRu, \qquad a, b \text{ constant}. \tag{54}$$

For a stationary positive f_0, skew Hamiltonian transport and the collision norm identity imply that these also exhaust the smooth zero modes of the combined linear generator. This classifies stationary null modes; it does not prove a decay rate, coercivity constant, or spectral gap.

Proof of the collision statement. In the common Cartesian η chart, write the spatial action in the local orthonormal basis $(\boldsymbol{e}_R, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$:

$$A_X h = \left(\frac{Rh_z}{qC}, \frac{h_u}{m}, \frac{-Rh_R + uh_u}{qC} \right), \qquad V_X = A_X E = \left(0, u, \frac{mu^2 + \mu C/R}{qC} \right). \tag{55}$$

The factor m in $\boldsymbol{B}^*$ cancels the bracket denominator m in the curvature term, so the last numerator contains uh_u, not muh_u. The energy characteristic has $\dot{u} = \dot{\eta} = 0$; thus its pair vector ξ is spatial-only and PI_XP discards all u, η components. Zero dissipation is consequently equivalent to $A_X h(v) - A_X h(v')$ being parallel to $V_X(v) - V_X(v')$ for all local velocity pairs. Positivity of the pair weight first gives the statement almost everywhere; smoothness extends it throughout the interior.

The radial component gives h_z independent of (u, μ). The map $(u, \mu) \mapsto (u, (mu^2 + \mu C/R)/(qC))$ is invertible onto an open two-dimensional set. On such a set, a map whose every chord is parallel to the corresponding argument chord has the form $F(v) = \alpha v + c$ with scalar α constant on that velocity set. To prove this elementary lemma, fix one point and compare its chords with two noncollinear chords: parallelism of the third chord forces their scalar factors to agree. An open set supplies a noncollinear comparison for each additional point. Applied to the angular and vertical components of (55), this gives

$$\frac{h_u}{m} = \alpha(\mathbf{X})u + c_\theta(\mathbf{X}), \quad \frac{-Rh_R + uh_u}{qC} = \alpha(\mathbf{X})\frac{mu^2 + \mu C/R}{qC} + c_z(\mathbf{X}). \tag{56}$$

Integrate the first identity in u to obtain $h = m\alpha u^2/2 + mc_\theta u + d(\mathbf{X}, \mu)$. Comparing the u^2, u and constant terms in the second yields

$$\alpha_R = 0, \qquad (c_\theta)_R = c_\theta/R, \qquad d_R = -\alpha\mu C/R^2 - qCc_z/R. \tag{57}$$

Hence $c_\theta = Rb(\theta, z)$ and $h = \alpha(\theta, z)E + b(\theta, z)mRu + g(\theta, z, \mu) + \phi(R, \theta, z)$. The already obtained velocity independence of h_z forces $\alpha_z = b_z = 0$ and makes g_z independent of μ; its remaining velocity-independent part can be absorbed into ϕ. This proves necessity of (53). The product velocity domain ensures the integration constant is a globally single-valued function of μ; connected support alone would not exclude branchwise functions on disconnected u-fibers.

For sufficiency, a spatial-only ϕ has velocity-independent spatial action, a $g(\theta,\mu)$ has zero spatial action, an $a(\theta)E$ has pair spatial action $a\xi$, and a $b(\theta)mRu$ has spatial action $b(\theta)R\boldsymbol{e}_\theta$ independent of velocity. In the last identity, $\boldsymbol{b}\times(mu\boldsymbol{e}_R)/(qB)$ cancels the curvature part from $\partial_u(mRu)=mR$. Their projected pair differences therefore all vanish. This also proves that all spatial populations are conserved by local collisions, including the nonlinear surrogate. Since the local energy-flow map is injective, distinct toroidal velocity states never have $\xi=0$; the coincident diagonal has vanishing test-function difference.

Proof of the common stationary statement. The ideal derivative is $\mathcal{T}h=(u/R)h_\theta+[(mu^2+\mu C/R)/(qC)]h_z$. Substitute (53) and regard it as a polynomial in u. The cubic coefficient forces $a'=0$; the constant coefficient $\mu\phi_z/(qR)$ forces $\phi_z=0$, since the moment interval has nonzero length. The quadratic coefficient then forces $b'=0$. The remaining linear coefficient is $\phi_\theta+g_\theta=0$; angle-only redundant terms can be absorbed between ϕ and g, giving (54). Periodic z supplies the closed boundary for the norm argument, but is not needed for this pointwise algebraic intersection. A proposed nonperiodic compensator $\phi=-qCb'(\theta)z$ cancels a quadratic term but leaves $-\mu Cb'/R$, and is not stationary on open moment support.

Thus combined dynamics retain the full radial population, the full moment marginal, energy and toroidal canonical momentum mRu. A general joint (R,μ) marginal is not implied. Arbitrary radial profiles cannot approach a unique state determined only by energy and $G(\mu)$. The same argument classifies positive nonlinear collision equilibria by taking $h=\ln f$ when the entropy form is finite; their common ideal stationary family is $\ln f=\phi(R)+g(\mu)+aE+bmRu$. Open boundaries with imposed flux, velocity support of dimension one, vanishing pair weights, disconnected velocity fibers, a nonlocal spatial kernel, or an electrostatic field that changes the energy-flow components are outside this proposition.

On the $3^5=243$-node test, 27 spatial indicators, nine angular–moment indicators, three angular–energy functions and three angular–momentum functions have rank 39 after three duplicated angle-only functions are counted once. Their production actions vanish to roundoff and account for all 39 observed near-null eigenvalues of the unmodified matrix. The continuum proof above is independent of that finite-grid rank check; eigenvalues are not altered. A separate continuous-bracket check at $m=1.7$, $q=-0.8$ and three off-axis points verifies random classified functions and the cylindrical action formula.

Primary-source searches of the authors with toroidal collision nullspace, radial populations and moment constraints, and local review of SM25 and BS25, did not locate this specific local-kernel classification. Local density and symmetry-generated invariants are established ideas; publication priority for this closure-specific classification remains unresolved. No claim about physical Coulomb transport, general magnetic geometry, or a measured lifetime follows from it.

Nonzero-field bounds for the executed domains

For the collision-only Cartesian box $0.8\le x\le 1.2$, $|y|\le 0.2$, $0.1\le z\le 0.5$, the mirror with $B_c=1$, $a=0.15$ has $B\ge B_z\ge 1+0.15[0.1^2-(1.2^2+0.2^2)/2]=0.8905$. For the unit dipole, $B=r^{-3}\sqrt{1+3z^2/r^2}\ge(1.2^2+0.2^2+0.5^2)^{-3/2}>0.439$. The symmetry-breaking addition with $\varepsilon=0.03$, $B_c=L=1$ has norm at most $0.03\sqrt{1.2^2+0.2^2}<0.037$, so the perturbed dipole remains nonzero throughout that box, with $B>0.402$. These conservative analytic bounds certify the geometry over the domain, beyond sampled minima. They do not certify a guiding-center ordering or a confined Hamiltonian boundary. For the toroidal annulus $1\le R\le 1.5$, $C=1$, $B\ge 2/3$ exactly.

10 Vacuum energy-flow zero sets and spatial population constraints

The results in this section concern the prescribed local scalar DPI_XP closure, a time-independent nonzero vacuum field, constant electrostatic potential, $m > 0$, $q \neq 0$, and the common Cartesian η chart. The positive scalar D is a symmetric local pair weight and may depend on the local velocity pair. Statements about the nullspace require it and the equilibrium density to be strictly positive almost everywhere. Global conservation statements additionally require the closed weak collision boundaries used in the entropy identity. These are continuous statements; a modified discrete energy gradient is a separate approximation.

Injectivity away from critical magnetic strength

Set $\mathbf{g} = \nabla B$, $\mathbf{t} = \mathbf{b} \times \mathbf{g}$ and $s = \mathbf{b} \cdot \mathbf{g}$. The vacuum identity $\nabla \times \mathbf{b} = \mathbf{t}/B$ follows by differentiating $\mathbf{b} = \mathbf{B}/B$ and using $\nabla \times \mathbf{B} = 0$. For $E = mu^2/2 + \mu B$, the energy flow in the common chart is exactly

$$V = T\mathcal{J}_\mu \nabla E = \left(u\mathbf{b} + \frac{mu^2 + \mu B}{qB^2}\mathbf{t}, -\frac{\mu s}{m},\ \mu u s \right). \tag{58}$$

Here mu^2 denotes mass times u^2, whereas μ is the magnetic moment. At fixed spatial position, $\mathbf{b} \cdot V_X = u$. Equality of two flow vectors therefore fixes u. If $\mathbf{t} \neq 0$, the remaining spatial component fixes μ. If $\mathbf{t} = 0$ but $s \neq 0$, the parallel-velocity component $V_u = -\mu s/m$ fixes μ. Thus the map $(u, \mu) \mapsto V$ is injective whenever $\nabla B \neq 0$. Its derivative has rank two: the u column has $\mathbf{b} \cdot \partial_u V_X = 1$, while the μ column has zero such projection and is nonzero. Conversely, at $\nabla B = 0$ the map is $V = (u\mathbf{b}, 0, 0)$, so distinct equal-u moment states have equal flows. This classifies the fixed-position zero set without regularizing $|\xi|^2$. The fixed-position qualification is essential. In the unit dipole with $m = q = 1$, the distinct equatorial states $(\mathbf{X}, u, \mu) = ((1, 0, 0), 0.2, 0.5)$ and $((2, 0, 0), 0.2, 0.76)$ both have common-chart energy flow $(0, 1.62, -0.2, 0, 0)$, even though ∇B is nonzero at both positions. At the equator $B = C/R^3$, $\mathbf{b} = -\mathbf{e}_z$, and the azimuthal flow is $3[mu^2R^2/C + \mu/R]/q$. This distinct remote pair also has a nonremovable directional weak limit for the spatial scalar projector. For $h = u$, $(Au)_X = -\mathbf{e}_z + 3uR^2\mathbf{e}_\theta$, so the pair action difference is $-1.8\mathbf{e}_\theta$. Perturbing only μ at the first state makes ξ parallel to $\mathbf{e}_\theta$ and the weak factor vanish. Perturbing only u by δ instead gives

$$|I_X P \Delta A u|^2 = \frac{(1.8 - 3\delta)^2}{1 + (1.2 + 3\delta)^2} \longrightarrow \frac{3.24}{2.44} = 1.327868852\ldots. \tag{59}$$

The actual common-chart action and projection are tested along three nonsingular values of each perturbation; the singular pair is not assigned a projector. Thus local injectivity supplies no general equal-flow exclusion or continuous zero-vector prescription for a nonlocal scalar extension. This observation alone does not analyze the full interaction tensor in the nonlocal source Eq. (119).

The point dipole has no critical strength point away from its source: homogeneity gives $\mathbf{X} \cdot \nabla B = -3B$. For the mirror, writing $B_z = C + a(z^2 - R^2/2)$ gives

$$\partial_R(B^2) = 2aR(-C + aR^2/2), \qquad \partial_z(B^2) = 4az(C + az^2). \tag{60}$$

For $C > 0$ and $a \neq 0$, simultaneous zeros in the nonzero-field region occur only at the origin. The executed positive-z box excludes it. For a dipole plus the degree-one harmonic perturbation, write $\mathbf{B} = \mathbf{B}_d + \mathbf{B}_p$. Euler homogeneity yields

$$\frac{1}{2}\mathbf{X} \cdot \nabla(B^2) = \mathbf{B} \cdot (-3\mathbf{B}_d + \mathbf{B}_p) \leq -3|\mathbf{B}_d|^2 + 2|\mathbf{B}_d||\mathbf{B}_p| + |\mathbf{B}_p|^2 < 0 \tag{61}$$

whenever $|\mathbf{B}_p| < |\mathbf{B}_d|$. The bounds above establish this on the executed perturbed-dipole box. Hence all three nonuniform Cartesian test domains have injective continuous local energy-flow maps. This does not by itself certify injectivity of a finite-grid energy-flow approximation.

For a smooth test function, $\Delta Ah = O(|\Delta(u,\mu)|)$ near a coincident pair. Since PI_XP is bounded, its weak quadratic factor is $O(|\Delta(u,\mu)|^2)$ and extends uniquely by zero on the diagonal, even though the projector itself has a directional limit. In contrast, a distinct equal-flow pair at a critical spatial point need not have a continuous weak extension.

A nonremovable critical-point weak limit

Consider the mirror near its nonzero-field origin, equal $u \neq 0$, $\mu \neq \mu'$, and $h = \mu x$. Approach along the x axis with $z = 0$. Then $\mathbf{b} = \mathbf{e}_z$, $\mathbf{t}$ is parallel to $\mathbf{e}_y$, and both ξ and ΔAh are spatial and parallel. Therefore $|I_X P \Delta Ah|^2 = 0$. Approach instead along the z axis. Now $\mathbf{t} = 0$ and ξ has only u and η components, while $\Delta Ah = (\mu - \mu')\mathbf{e}_y/(qB)$. Consequently,

$$\lim_{x\to 0,\ z=0} |I_X P \Delta Ah|^2 = 0, \qquad \lim_{z\to 0,\ x=0} |I_X P \Delta Ah|^2 = \frac{(\mu-\mu')^2}{q^2C^2} > 0. \tag{62}$$

Varying $u - u'$ at the exact origin gives the latter uniform-field prescription. Thus a simultaneous small-gradient and equal-u limit is not equivalent to first imposing the uniform model. There is no universal continuous extension at these distinct-state critical pairs. An arbitrary bounded assignment on a measure-zero critical spatial set does not change an absolutely continuous weak integral, but it cannot be described as its unique continuous limit. Positive quadrature nodes on such a set need a separately declared prescription. No denominator softening is justified by this argument.

Near-uniform mirror fields and tensor diagonal weights. Example 18 tests a finite-quadrature limitation using exact continuous local actions, independently checked against the common-chart implementation. Its observable is the frozen positive pair Gram Q_h for $h = \mu x$, evaluated at $f = e^{-E-0.2\mu}$; it is not this equilibrium's entropy production, which is zero. At $X = (1, 0, 0.3)$, $m = q = C = D = 1$, the positive-amplitude mirror field has $\Delta Ah \to (\mu - \mu')e_y$. For $u \neq u'$, $\Delta V \to (u - u')e_z$ and the projector approaches the literal uniform prescription. For $u = u'$ and $\mu \neq \mu'$, however,

$$\Delta V = a(\mu - \mu')L(u) + O(a^2), \qquad L(u) = (0, -1, 0, -0.6, 0.6u).$$

Consequently the limiting spatial squared projected action is $(\mu - \mu')^2 r(u)$, with $r(u) = [0.36(1 + u^2)/(1 + 0.36(1 + u^2))]^2 < 1$, whereas the uniform prescription gives $(\mu - \mu')^2$. For positive tensor quadrature weights w_i in u and v_j in μ, put $M_k = \sum_j v_j e^{-1.2\mu_j}\mu_j^k$ and $H = M_0M_2 - M_1^2 > 0$. The unordered pair sum gives exactly

$$Q_n(0) - \lim_{a\to 0^+} Q_n(a) = H \sum_i w_i^2 e^{-u_i^2}[1 - r(u_i)] \le H \max_i w_i \sum_i w_i e^{-u_i^2}.$$

Thus a fixed quadrature retains a positive discrepancy. Positive Gauss refinement has $\sum_i w_i = 2U_{\max}$ and $\max_i w_i \to 0$, so the displayed bound vanishes when the moment rule is held fixed. On the fixed compact velocity domain the projector and analytic test actions are bounded. Dominated convergence therefore recovers the same uniform continuum form because $u = u'$ has zero product measure. This is a finite-quadrature discontinuity, not a physical discontinuity or a claim that the two infinite sequential limits differ. The example holds the moment quadrature and velocity tails fixed, isolates the parallel-order effect, and reports the direct positive-amplitude checks and proved bound. It does not certify the modified discrete-energy projector or a full spatial evolution scheme uniformly as $a \to 0$, and makes no priority claim for this numerical limitation.

Spatial invariants of local collisions

For a spatial observable $\phi(\mathbf{X})$ independent of u, μ, its common-chart action is

$$A\phi = \left(\frac{\mathbf{b} \times \nabla\phi}{qB}, -\frac{\mathbf{b} \cdot \nabla\phi}{m} - \frac{u\,\mathbf{t} \cdot \nabla\phi}{qB^2}, -\frac{\mu\,\mathbf{t} \cdot \nabla\phi}{qB} \right). \tag{63}$$

If $\mathbf{t} \cdot \nabla\phi = 0$, this entire vector is independent of the velocity pair at the same spatial position, so $\Delta A\phi = 0$. Every such observable is conserved by the linear and nonlinear local collision surrogates, regardless of the energy projector.

There is also a precise local necessity statement for spatial observables. At a point with $s \neq 0$ and $\mathbf{t} \neq 0$, let $\kappa = \mathbf{t} \cdot \nabla\phi$. Choose equal nonzero u and unequal μ, μ' in a product velocity domain of nonzero widths. Their action difference has only an η component, while $\xi_X = (\mu - \mu')\mathbf{t}/(qB)$ and $\xi_\eta = us(\mu - \mu')$. If $\kappa \neq 0$, then $\xi \cdot \Delta A\phi = -us\kappa(\mu - \mu')^2/(qB) \neq 0$ and the spatial block of $P\Delta A\phi$ is nonzero. Continuity gives a positive weak contribution on a neighborhood of this pair. Thus a spatial observable is locally null precisely when $\kappa = 0$ at such a point. If $s = 0$, the energy-flow difference is spatial, and the velocity-dependent part of $\Delta A\phi$ is nonspatial; every spatial observable is locally null. The case $\mathbf{t} = 0$ already has $\kappa = 0$. These statements classify spatial test functions, not the full nonuniform nullspace.

Locally a vacuum field has $\mathbf{B} = \nabla\chi$. Since $\mathbf{t}$ is orthogonal to both ∇B and $\nabla\chi$, every spatial $F(B, \chi)$ is a collision invariant. When χ is globally single valued, closed collisions therefore preserve the joint spatial pushforward population in (B, χ). The executed scalar potentials are

$$\chi_m = Cz + a(z^3/3 - zR^2/2), \tag{64}$$

$$\chi_d = -Cz/r^3, \qquad \chi_{na} = \chi_d + \varepsilon Cxy/L. \tag{65}$$

Ideal flow generally changes these populations. They are not interior Casimirs of the Poisson tensor and are not claimed to be invariants of the full combined dynamics.

Angular collocation nulls and polynomial reconstruction

An observed discrete null need not have a null tensor-polynomial interpolant. This matters when interpreting a small Cartesian-box spectrum. In an axisymmetric poloidal field, the spatial sufficient condition becomes $\mathbf{t} \cdot \nabla\phi = (t_\theta/R)L\phi = 0$, where $L = x\partial_y - y\partial_x$. Consider three Gauss nodes on $x \in [c - a, c + a]$, $y \in [-a, a]$, with $c > a > 0$ and $k = 3a^2/5$. Their monic node polynomials are

$$p_x = (x - c)^3 - k(x - c), \qquad p_y = y^3 - ky.$$

The tensor-quadratic polynomial

$$\phi_{\mathrm{a}} = (x^2 - 3cx)y^2 - 3c^2x^2 + c(2c^2 + k)x \tag{66}$$

satisfies the exact identity

$$L\phi_{\mathrm{a}} = 2yp_x - (2x - 3c)p_y. \tag{67}$$

The tensor derivative differentiates ϕ_{a} exactly, but its angular derivative vanishes at every collocation node. It is not identically zero between them: with $c = 1$, $a = 0.2$, at $(x, y) = (1.18, 0.18)$ the value is 0.001512. Thus the sampled local pair form vanishes for this interpolant, while its continuum form is positive on off-grid neighborhoods where $L\phi_{\mathrm{a}} \neq 0$, $s \neq 0$, $t_\theta \neq 0$, and the velocity-pair measure is positive.

There is a complete elementary classification of this planar collocation kernel within the tensor-quadratic space. Split a polynomial into the y-even part $A(x) + B(x)y^2$ and the odd part $C(x)y$, with all three coefficient polynomials of degree at most two. The angular derivative flips parity, so the two equations separate on the symmetric node set. At $y = 0$ the odd part gives $xC(x) = 0$ at three nonzero x nodes, hence $C = 0$. At the two nonzero y nodes, $y^2 = k$, so the even equation is

$$2xB - kB' - A' = \lambda p_x.$$

Equating polynomial coefficients leaves exactly three free coefficients. The kernel is therefore $\mathrm{span}\{1, R^2, \phi_\mathrm{a}\}$. With three independent z nodes, multiplication by $1, z, z^2$ gives nine spatial collocation nulls. Adding three moment-bin indicators and energy, and accounting for the shared constant, gives twelve independent known null vectors for the specified mirror and dipole grids. This is a statement about the sampled operator and the stated reconstruction, not a continuum nullspace classification. The same nodal values could also be interpolated by a different, nonpolynomial function of R^2.

The missed angular derivative has a positive exact squared integral:

$$\int_{c-a}^{c+a}\int_{-a}^{a}(L\phi_\mathrm{a})^2\,\mathrm{d}y\mathrm{d}x = \frac{128a^{10}}{525} + \frac{16c^2a^8}{175}. \tag{68}$$

To obtain it, use $\int p_y^2 \mathrm{d}y = 8a^7/175$ and the orthogonality of p_y to constants and linear polynomials; the cross term in (67) integrates to zero. The three-node tensor rule returns zero, whereas four or more Gauss nodes integrate the fixed squared polynomial exactly, since its degree in either coordinate is at most six. This is an overintegration diagnostic of the fixed interpolant; it does not modify or repair the production collision operator.

The executed diagnostic `examples/21_collocation_nullspace.py` uses $3^5 = 243$ nodes and 972 unordered pairs, $m = q = 1$, $D = 0.1$, $x \in [0.8, 1.2]$, $y \in [-0.2, 0.2]$, $z \in [0.1, 0.5]$, $u \in [-4, 4]$, $\mu \in [0, 20]$, and $f = \exp[-E - 0.2\mu + 0.1\sin(\pi y/0.4)u]$. It independently lifts tensor derivatives and assembles unordered pair projectors and measures, while using the production geometry's common-chart coefficients. Three deterministic probe actions agree with the unchanged production operator within 1.64×10^{-14} relatively. The unweighted rectangular factors have respectively $12, 12, 4$ numerical null vectors in the mirror, dipole and perturbed dipole boxes for relative thresholds 10^{-10} through 10^{-14}. Their candidate spans exhaust the measured nullspaces. These are finite-dimensional counts, not continuum coercivity. All singular values are saved before any threshold interpretation.

Positive physical weights preserve the same counts here; entropy scaling reports six perturbed-field nulls at the loose 10^{-10} threshold but four at 10^{-12} through 10^{-14}. This sensitivity is retained in the second figure. Squaring a factor to form a Gram matrix can further obscure small singular values. The maximum normalized axisymmetric alias-factor residual is 5.82×10^{-17}. For the fixed polynomial above, Gauss three gives 2.79×10^{-37} for the angular squared integral, whereas orders four, five, seven and nine reproduce $2.590232380952381 \times 10^{-7}$ within 3.3×10^{-15} relatively. Independent exact-rational integration gives $1328/5126953125$. No null shift or spectral projection is applied to the operator.

The producing commit is `4094e89`; the full identifier, script and array hashes, inputs, units, versions and hardware are in `results/collocation_nullspace/summary.json`. The checks took 6.236 seconds on the shared Apple M4, including setup, factor decompositions, compilation and probes, but excluding imports, metadata and figures; this is not an isolated runtime benchmark. Six regression cases also cover three translated/scaled planar boxes and all three physical field choices.

A mixed local moment and its angular obstruction

The spatial classification above does not exhaust velocity-dependent local collision invariants. A further sufficient construction follows directly from the common-chart action. Assume a smooth nonvanishing

vacuum field, $m > 0$, $q \neq 0$, and the same zero-potential convention. On a spatial patch where a smooth ϕ solves

$$\mathbf{t} \cdot \nabla\phi = qs, \qquad h_C(\mathbf{X}, u, \mu) = \frac{mu}{B} + \phi(\mathbf{X}), \tag{69}$$

the common-chart vector Ah_C is independent of velocity. Indeed, $\nabla h_C = -mu\nabla B/B^2 + \nabla\phi$ and $(h_C)_u = m/B$ give

$$(Ah_C)_X = \frac{\mathbf{b}}{B} + \frac{\mathbf{b} \times \nabla\phi}{qB}, \tag{70}$$

$$(Ah_C)_u = \frac{us}{B^2} - \frac{\mathbf{b} \cdot \nabla\phi}{m} - \frac{u\,\mathbf{t} \cdot \nabla\phi}{qB^2}, \tag{71}$$

$$(Ah_C)_\eta = \mu\left(\frac{s}{B} - \frac{\mathbf{t} \cdot \nabla\phi}{qB}\right). \tag{72}$$

The terms proportional to $mu\mathbf{t}/(qB^3)$ cancel in the spatial component. Substitution of (69) yields

$$Ah_C = \left(\frac{\mathbf{b}}{B} + \frac{\mathbf{b} \times \nabla\phi}{qB}, -\frac{\mathbf{b} \cdot \nabla\phi}{m}, 0\right) =: a(\mathbf{X}). \tag{73}$$

Thus every same-position pair satisfies $\Delta Ah_C = 0$, before applying any energy projector. Under natural zero collision flux, or other boundaries that admit this test function and remove the weak boundary term, $\int h_C f \, \mathrm{d}\Gamma$ is conserved by the local surrogate with symmetric mobility. This is a sufficient statement, not a full nullspace classification. Preservation of the full h_C marginal does not follow from this moment identity: $AF(h_C) = F'(h_C)a(\mathbf{X})$ generally has a nonzero local pair difference, whose projection must be checked separately.

Where $\mathbf{t} \neq 0$, the flow-box construction supplies a local smooth solution of the first-order equation in (69). At $\mathbf{t} = 0$, its necessary compatibility condition is $s = 0$; no extension across such points is presumed. Global single-valuedness imposes further conditions. For an axisymmetric poloidal field, $\mathbf{t} = t_\theta \mathbf{e}_\theta$, choose a smooth angular branch and $t_\theta \neq 0$. Then

$$\phi = q\theta\frac{Rs}{t_\theta}, \qquad \frac{t_\theta}{R}\partial_\theta\phi = qs. \tag{74}$$

Since Rs/t_θ is independent of azimuth, integrating the equation around a complete periodic circle requires $0 = 2\pi qRs/t_\theta$. Where $s \neq 0$ the proposed moment is therefore inadmissible on the full periodic azimuth, even though it is smooth on the executed $x > 0$ boxes. For the positive-strength dipole this gives the explicit expression

$$h_C = \frac{mu}{B} - q\theta\frac{z(3R^2 + 8z^2)}{R^2 + 2z^2}. \tag{75}$$

It has dimensions of charge times length before normalization. This moment is distinct from canonical toroidal momentum, and the construction does not assume that it is a physical collision invariant. With common-chart energy $E = mu^2/2 + \eta$, antisymmetry gives the ideal derivative

$$\mathcal{T}h_C = -(Ah_C) \cdot \nabla_{X,u,\eta} E = u\,\mathbf{b} \cdot \nabla\phi. \tag{76}$$

It is an ideal constant for all velocities only if the additional condition $\mathbf{b} \cdot \nabla\phi = 0$ holds. The dipole branch expression generally does not satisfy this condition. The toroidal limit $s = 0$, $\phi = 0$ does: $h_C = mRu/C$ reduces to the already classified momentum mode. In a uniform field the same choice gives a constant multiple of parallel momentum. For separated positions, $\Delta Ah_C = a(\mathbf{X}) - a(\mathbf{X}')$ need not vanish; projected pair values must be checked before drawing a conclusion about a particular nonlocal kernel. No finite-range decay rate follows from the local cancellation alone.

This is an elementary sufficient consequence of the common-chart tensor and local pairing, following the weak-moment criterion of SM25, Eqs. (45)–(47), the tensor in Eq. (93), and the same-position approximation in Eq. (172) [1]. It is not a claim for the full source Eq. (119). The angular admissibility distinction is essential when comparing a Cartesian-box prototype with a closed dipole geometry.

Mixed velocity–position momenta have a direct structural antecedent in quasisymmetry [9, 10]. BKM20, published Section VI Eq. (41), gives $K = -q\psi + mu\boldsymbol{\xi}\cdot\mathbf{b}$ for its symmetry generator $\boldsymbol{\xi}$ and convention $\nabla\psi = \mathbf{B}\times\boldsymbol{\xi}$. Its Eq. (32) implies that $\boldsymbol{\xi}\cdot\mathbf{B} = c$ is constant in vacuum. If $c \neq 0$, then

$$\frac{K}{c} = \frac{mu}{B} - \frac{q\psi}{c}, \qquad \mathbf{t}\cdot\nabla\psi = B\boldsymbol{\xi}\cdot\nabla B - cs = -cs. \tag{77}$$

Thus the known quasisymmetric momentum supplies $\phi = -q\psi/c$ satisfying (69). It also has $\mathbf{b}\cdot\nabla\phi = 0$, and $\mathbf{b}\times\nabla\psi = c\mathbf{b} - B\boldsymbol{\xi}$ reduces its action to $Ah_C = (\boldsymbol{\xi}/c, 0, 0)$. The $c = 0$ case cannot be normalized this way. This result additionally requires preservation of the magnetic magnitude, flux form and direction covector; the branch equation alone does not establish these symmetry conditions. BKM20 explicitly distinguishes local and globally defined flux potentials. BKM21, checked arXiv v1 Eq. (32), derives more general mixed momenta from approximate phase-space symmetries. These are Hamiltonian invariants, unlike the generic dipole branch here. The BKM20 comparison uses the published journal equations; the BKM21 numbering refers to the stated author-preprint version.

The pair-generator argument also has an established collisional antecedent: BBQ15, checked arXiv v1 Eqs. (28)–(30), proves toroidal-momentum conservation by equality of rotation generators on its physical-position delta support [5]. Its gyrocenter position and collision kernel differ from the present same-guiding-center closure. Together with the canonical momentum and equilibrium literature [11, 12], these results rule out a broad novelty claim for mixed momenta or pair-generator cancellation. Here the contribution is the explicit sufficient branch test, its angular obstruction and the checked limits of its use in this prototype; publication priority for that restricted calculation is unresolved.

Continuous and discrete checks. Example 20 checks the explicit dipole formula with $m = 1.7$, $C = 1.3$, $q = 0.7, -1.3$, three spatial points and three velocity states against independently differentiated closed field and observable formulas. The largest sampled gradient, PDE, action or local pair discrepancy is 8.15×10^{-15}. With $m = q = C = 1$, choose

$$z_1 = (1, 0, 0.3, 0.4, 0.2), \qquad z_2 = (1.1, 0.05, 0.35, -0.6, 0.7).$$

In the stated normalized common-coordinate metric, $|I_X P\Delta Ah_C|^2 = 0.3078910313 > 0$ for this separated pair. At the same spatial position as z_1, changing only the second velocity still gives zero for h_C, but gives 7.0302375381 for h_C^2. This explicitly distinguishes the conserved dipole moment from its marginal. At z_2, $\mathcal{T}h_C = -0.062160603 \neq 0$ checks the ideal limitation. These are pair/observable tests, not an evolved population or physical rate.

A separate analytic-field check puts $\mathbf{X}_2 = \mathbf{X}_1 + \epsilon d$, $d = (0.1, 0.05, 0.05)$, keeping the two velocities fixed. Since $|\xi(0)|^2 = 5.2330351645 > 0$, the projector is smooth there, so Taylor expansion of (73) gives

$$|I_X P\Delta Ah_C|^2 = \epsilon^2 |I_X P_{\xi(0)}(\nabla a)d|^2 + O(\epsilon^3). \tag{78}$$

The independently evaluated coefficient is 0.2349432711; the ratio at $\epsilon = 1/128$ differs by 0.214% and approaches that coefficient. Continuity supplies positive projected values on an open neighborhood. A positive scalar nonlocal extension whose support includes that neighborhood therefore does not retain this null moment. This statement gives neither a coercivity bound nor a finite-range decay rate and does not cover the full source interaction tensor.

On the finite box used above, example 20 applies the existing polynomial spatial derivatives and unmodified discrete-energy projector, with $n = 3, 5, 7, 9$ nodes per spatial axis, three parallel Gauss nodes on $[-1.5, 1.5]$ and two moment nodes on $[0.1, 1.1]$. The positive pair weight is the implemented measure times $e^{-E_i - E_j}$. Its normalized RMS projected local residual decreases from 0.036978 to 1.2188×10^{-6}. The finite grid thus does not preserve this continuum moment exactly; no additional null mode is removed or imposed. This is sampled spatial differentiation refinement, with fixed velocity quadrature and finite tails, not a joint continuum convergence certificate.

The public calculation was executed at `1129b19` on an Apple M4, with 16.08 seconds for its checks including compilation on the shared machine; imports, metadata collection and figure rendering are excluded. Exact inputs, versions, hashes, raw pair values, four refinement rows and plotting data are in `results/local_mixed_null/`. The independent receipt identifies its separate earlier checker and frozen geometry explicitly. Both literal source comparisons and the additional local construction remain distinct.

Poloidal flux memory and inaccessible dipole candidates

In an axisymmetric purely poloidal field, $\mathbf{t}$ is azimuthal. Therefore all $\phi(R,z)$ satisfy $\mathbf{t}\cdot\nabla\phi = 0$ in (63): local collisions preserve every meridional spatial population. With $\mathbf{B} = \nabla\psi \times \mathbf{e}_\theta/R$, the poloidal fluxes are

$$\psi_d = \frac{CR^2}{(R^2+z^2)^{3/2}}, \qquad \psi_m = \frac{CR^2}{2} + \frac{aR^2z^2}{2} - \frac{aR^4}{8}. \tag{79}$$

Both $\mathbf{b}$ and $\mathbf{t}$ are tangent to these flux surfaces. For constant potential, ideal flow conserves ψ; the same remains true for an axisymmetric electrostatic potential. Under closed, flux-compatible ideal boundaries and closed weak collision boundaries,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int F(\psi) f \,\mathrm{d}\Gamma = 0 \quad \text{for every admissible smooth } F. \tag{80}$$

This is conservation of the full flux pushforward marginal, which obstructs accessibility of an energy/$G(\mu)$-only equilibrium from arbitrary initial data. It does not generally preserve a joint (ψ,μ) population: for example $h = \mu\psi$ has a nonzero projected pair action off the equatorial plane. The test suite checks that counterexample explicitly. In particular, under the same closed combined conditions, the stationary family includes

$$f = \exp[-\beta E + g(\mu) + F(\psi)] \tag{81}$$

whenever it is positive, normalizable and in the weak form domain. The energy term is annihilated by P, the moment term by $\mathcal{J}$, and the flux term has zero local pair difference; all three terms are ideal constants. The earlier energy/moment candidate is therefore a proper subset of this stationary family, not a complete equilibrium classification for this field and closure. No completeness statement for (81) is asserted.

Canonical toroidal momentum conservation and the ideal flux constraint in the purely poloidal limit have established antecedents [11]. Here $A\psi = (R\mathbf{e}_\theta/q, 0, 0)$ with the flux convention in (79); thus $q\psi$ is the lowest-order canonical rotation generator. The additional conclusion is that exact spatial localization makes *every* $F(\psi)$ collisionally invariant. This proof uses equality of the paired positions essentially. It does not apply to the generally spatially nonlocal source Eq. (119), to its full interaction tensor, or to finite-Larmor collision displacements. The source introduces the spatial scalar/delta approximation separately in Eqs. (171)–(172). Conventional guiding-center collision operators retain mechanisms absent from this closure [4, 7]. Flux- or canonical-momentum-dependent equilibria are not new objects: Di Troia [12] treats equilibria depending on canonical toroidal momentum, energy and moment, and modifies the Landau projection using the selected equilibrium score difference in Eqs. (44) and (46) of the checked arXiv version. That target-dependent construction differs from a nullspace constraint forced by the present fixed local pair geometry. Sato, Yoshida and Kawazura [13] construct inward diffusion from fluctuating electric fields on a moment-preserving leaf; their Eq. (14d) contains a nonzero Wiener increment in ψ, and Eq. (15) contains flux diffusion. Such forcing changes the flux population and lies outside the closed axisymmetric ideal-plus-local-collision assumptions here. These antecedents preclude a broad novelty claim for constrained equilibria or dipole transport. The present negative result concerns the specified closure's inaccessible states, not a general prohibition of inward diffusion. Focused primary-literature searches found no matching statement for this precise surrogate; no publication-priority claim follows. Neither the invariant theorem nor the critical-point counterexample provides a physical rate, a quantitative decay bound or a dipole lifetime.

Symmetry breaking removes spatial ideal constants

For the unit dipole plus $a\nabla(xy)$, $0 < |a| \le 0.03$, on the connected Cartesian box stated above, there is no nonconstant C^2 spatial function $\phi(\mathbf{X})$ that is an ideal constant for every velocity state in an open

product domain. Consequently there is no nonconstant spatial function in the common ideal/collision nullspace. This concerns spatial observables only; it does not classify functions depending on u, μ, establish coercivity, or imply a relaxation rate.

Indeed, comparing the μ and u coefficients of $V_X \cdot \nabla\phi = 0$ in (58) gives $\mathbf{b} \cdot \nabla\phi = \mathbf{t} \cdot \nabla\phi = 0$. Set $S = |\mathbf{B}|^2$, $p = \mathbf{B}$, and $v = \mathbf{B} \times \nabla S = 2B^2\mathbf{t}$. Both vector fields annihilate ϕ, so their Lie bracket $[p, v] = (\nabla v)p - (\nabla p)v$ also annihilates it. At the interior unit-radius point $(15/17, 0, 8/17)$, exact rational differentiation gives

$$\det[p, v, [p, v]] = \frac{900a(225a^2 + 481)(6709a^4 - 313674a^2 + 2686977)}{6975757441}. \tag{82}$$

The quartic factor is at least $2686977 - 313674(0.03)^2 > 2686694$ on the stated amplitude interval. Thus the determinant is nonzero. The field is real analytic and bounded away from zero throughout this box, and the analytic determinant is not identically zero. Its nonzero set is therefore dense. On that set the three annihilating vector fields span physical space, so $\nabla\phi = 0$; continuity and connectedness make ϕ constant throughout the box. For $a = 0.03$, the exact value is $1396271742281072028423/279030297640000000000 = 5.004014811619193\ldots$; direct field automatic differentiation gives 5.004014811619226.

The certificate was derived independently by exact rational field derivatives and by symbolic differentiation; regression tests check the actual field implementation at both perturbation signs. This two-vector Lie-rank argument is distinct from the one-field Pfaff/Frobenius condition $\mathbf{B} \cdot \nabla \times \mathbf{B} = 0$ discussed in SM24 [2]. The latter holds for all these vacuum fields. The result removes the axisymmetric spatial flux constraint, not all phase-space constraints or all possible obstructions to mixing. Focused primary-source searches found no matching statement for this fixed analytic example, but no publication-priority claim is made.

11 Linear weak discretization and SOLVAX

Represent $h = \sum_i a_i\chi_i$. Define the mass and collision matrices by

$$M_{ij} = \int f_0\chi_i\chi_j \mathrm{d}\Gamma, \tag{83}$$

$$K_{ij} = \frac{1}{2}\iint f_0 f_0' \Delta\chi_i^T \Pi_0 \Delta\chi_j \mathrm{d}\Gamma\mathrm{d}\Gamma'. \tag{84}$$

Positive quadrature gives a Gram form $K = B_p^T W_p B_p \succeq 0$. Implement its action without assembling the full pair tensor. A dense assembly on a tiny grid must remain an independent reference.

The executable nodal construction uses tensor differentiation followed by $A_i\phi = T_i\mathcal{J}_{\mu,i}(\nabla_h\phi)_i$ in a common Cartesian η chart. For every unordered pair (i, j), set $B_{ij}\phi = A_i\phi - A_j\phi$ and construct $\xi_{ij} = B_{ij}e$ from the same discrete energy vector e and derivative. The nonuniform projector uses this discrete ξ, rather than inserting an incompatible continuum energy gradient. This is an explicitly stated discrete energy projector; its consistency must be checked under refinement. The scalar pair weight is $w_X(Bw_v)_i(Bw_v)_j$ locally and $w_iw_jW_{ij}$ for a spatially nonlocal model. In a toroidal chart the spatial weight contains $R\mathrm{d}R\mathrm{d}\theta\mathrm{d}z$ while each local velocity factor contains $B\mathrm{d}u\mathrm{d}\mu$, even though the one-particle product $BR = C$ simplifies.

The ordered weak form in (84) counts (i, j) and (j, i) equally: exchange reverses both pair differences, leaving their quadratic product unchanged. Therefore its $1/2$ exactly cancels the two ordered contributions, and each unordered pair is inserted once with no $1/2$. Self-node pair differences are identically

zero and need no stored contribution. Distinct nodes with equal energy-flow vectors are a separate issue: the uniform field uses its derived continuous limit, while an exactly zero nonuniform pair vector is rejected. No $|\xi|^2 + \epsilon$ is used. Directional limits may depend on the approach. The continuous local zero set is classified by (58); the critical-point counterexample (62) prevents a universal continuous prescription. Successful finite-grid examples do not by themselves establish a valid extension for a modified discrete energy map.

Periodic derivatives are Fourier derivatives; the real even-grid Nyquist component is unresolved and given zero derivative. Nonperiodic derivatives support local three-point quadratic interpolation and, for the refined nonuniform boxes, the derivative of the full nodal Lagrange polynomial. The derivative and its transpose enter the weak collision assembly. The selected spatial quadrature and derivative method are recorded explicitly. This imposes the natural collision flux rather than wrapping nonperiodic field values. The preassembled reference path retains pair indices and directions with quadratic velocity-pair storage (and spatial pairs for nonlocal models). The compact local path instead stores energy flows and forms pair directions only inside bounded blocks, as described below.

If c represents a collision invariant and the derivative/kernel construction respects it, then $Kc = 0$. Include the discrete energy and all resolved magnetic-moment-bin constraints. Finite differences do not automatically obey the product rules needed for an arbitrary sampled μB. Use the same discrete energy gradient in the degeneracy direction and in the weak operator, and prove consistency as the mesh is refined.

The collision-only system and norm balance are

$$M\dot{a} = -Ka, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}a^T M a = -a^T K a \le 0. \tag{85}$$

With a compatible stationary Hamiltonian discretization,

$$M\dot{a} = (A - K)a, \qquad A^T = -A. \tag{86}$$

Also test $Ac = 0$ for the required linear invariants; skew symmetry alone does not imply those individual identities.

In the nonuniform combined example, exact Fourier half-translations in the periodic toroidal angle and z surround one backward-Euler collision step. The translation uses $\dot{\theta} = u/R$ and $\dot{z} = (mu^2 + \mu C/R)/(qC)$ at each fixed (R, u, μ). The measure $BR = C$, energy and moment marginal are invariant under this ideal flow, and radial walls are tangent. This symmetric composition remains first order because the collision substep is backward Euler. It is a linear, fixed-equilibrium combined evolution; the nonlinear examples are collision only.

A backward-Euler collision step solves

$$(M + \Delta t K)a^{n+1} = Ma^n. \tag{87}$$

For positive-definite M, this is an appropriate PCG problem. SOLVAX's `pcg_linear_solve` supplies a converged-system differentiation wrapper [14]. Do not pass $M^{-1}K$ to Euclidean PCG without mass scaling or a proof of symmetry. A coupled implicit step with A is generally nonsymmetric and calls for GMRES or another suitable method.

For a linear invariant c, multiply (87) by c^T to check its conservation. Backward Euler also satisfies

$$Q^{n+1} - Q^n = -\Delta t (a^{n+1})^T K a^{n+1} - \frac{1}{2}\left\|a^{n+1} - a^n\right\|_M^2 \le 0. \tag{88}$$

This does not prove positivity of a reconstructed finite-amplitude distribution. An implicit midpoint step can be second order, but is not automatically a good stiff-decay scheme or a nonlinear entropy-preserving integrator.

Local pair application costs $O(N_X N_v^2)$ in a direct quadrature. A spatially nonlocal direct application can cost $O(N_X^2 N_v^2)$. Chunk pair contractions, reuse geometry and quadrature, and keep derivatives sparse. Block-tridiagonal solves may be useful in a line preconditioner; the full velocity coupling is not generally tridiagonal. The current tensor derivative matrices are stored densely, including zeros outside the nonperiodic stencils. Their contraction adds $O(N \sum_\alpha n_\alpha)$ work, with $N = \prod_\alpha n_\alpha$; stencil sparsity is a mathematical property, not an implemented sparse matmul claim.

12 A nonlinear discrete entropy construction

This finite-dimensional fixed-field construction is implemented. It is not a guarantee for arbitrary interpolation, self-consistent fields, or nonlinear Hamiltonian transport. It follows the weak-structure approach developed for Landau discretizations [15, 16, 17, 18].

Let $f_i > 0$, $w_i > 0$, and $n_i = w_i f_i$. Construct a symmetric positive mobility $K(f)$ from the discrete pair weak form, with

$$K\mathbf{1} = 0, \quad Ke = 0, \quad Kc_\mu = 0, \tag{89}$$

where e is the fixed-field energy vector and c_μ spans the represented moment-marginal constraints. Define

$$S(n) = -\sum_i n_i \ln(n_i/w_i). \tag{90}$$

Then the collision ODE can be written

$$\dot{n} = K(f)\nabla_n S = -K(f)\ln f. \tag{91}$$

The constant term in $\nabla_n S = -\ln f - \mathbf{1}$ vanishes through (89). Consequently $\dot{S} \geq 0$ and $\mathrm{d}(e^T n)/\mathrm{d}t = 0$. This is an algebraic guarantee only for the actual constructed K.

For each component define a divided difference of $s_i(n) = -n\ln(n/w_i)$:

$$(\overline{\nabla} S)_i = \begin{cases} [s_i(n_i^+) - s_i(n_i^-)]/(n_i^+ - n_i^-), & n_i^+ \neq n_i^-, \\ -\ln(n_i^-/w_i) - 1, & n_i^+ = n_i^-. \end{cases} \tag{92}$$

Evaluate it stably near equal arguments using logarithmic divided differences, not cancellation-prone subtraction. A step

$$n^+ - n^- = \Delta t\, \overline{K}\, \overline{\nabla} S, \qquad \overline{K} = \overline{K}^T \succeq 0 \tag{93}$$

with the null directions in (89) gives

$$S(n^+) - S(n^-) = \Delta t\, (\overline{\nabla} S)^T \overline{K} \overline{\nabla} S \geq 0. \tag{94}$$

The conservation laws follow by left multiplication with each invariant vector. Use a positive mean state to construct $\overline{K}$ and a converged nonlinear solve. Parameterizing $n_i^+ = w_i e^{g_i^+}$ enforces positive nodal populations, but does not prove existence of the solve or positivity between interpolation nodes. Check all required quadrature/reconstruction locations.

Angular leaf components of the Cartesian box

The mirror and dipole collision-only boxes have an additional domain qualification. In the open rectangle $0.8 < x < 1.2$, $|y| < 0.2$, a fixed circle with $1.2 < R < \sqrt{1.48}$ meets the box in two disconnected angular arcs, with $\arccos(1.2/R) < |\theta| < \arcsin(0.2/R)$. For $g \in C_c^\infty((1.2, \sqrt{1.48}))$, the observable

$\phi = g(R)\,\mathrm{sign}(y)$ is smooth inside the box: its active support is separated from $y = 0$. Since $\mathbf{t}$ is azimuthal, $\mathbf{t} \cdot \nabla\phi = 0$. Closed local collisions therefore conserve these signed component populations as well as the single-valued meridional populations. This is a consequence of the natural closed boundary, not a failure of that boundary condition.

The corner region has unweighted geometric area fraction

$$\frac{2}{0.16}\int_{\sqrt{1.4}}^{1.2}[0.2 - \sqrt{1.44 - x^2}]\mathrm{d}x = 0.01394734125\ldots. \tag{95}$$

Its volume fraction is the same because the z interval is independent; its state-measure or particle fraction requires the actual B and density weights. Five-point Gauss nodes per axis have maximum $R = 1.19506$ and miss this region; seven-point nodes reach $R = 1.20487$ with two corner nodes in the xy grid. Thus independent bulk-observable refinements do not by themselves certify preservation of every localized leaf-component constraint or global continuum convergence. This analytical limitation is reported independently of the finite-time outcomes; it introduces no later acceptance criterion.

Matrix-free Newton–GMRES and residual control

The compact nonuniform evolution solves the same discrete-gradient equation as the small dense reference. Let $l = \ln f^+$, $l^- = \ln f^-$, $\bar{f} = (e^l + e^{l^-})/2$, and write the unscaled residual as

$$F(l) = w\,(e^l - f^-) - \Delta t K(\bar{f})\overline{\nabla} S(l^-, l). \tag{96}$$

Products with w are componentwise. Set $a_i = \sqrt{w_i f_i^-} > 0$, $r = F/a$, and parameterize the Newton correction as $\delta l = v/a$. The exact matrix-free operator is

$$v \longmapsto a^{-1} J_F(l)(v/a), \qquad a^{-1} J_F(l)(v/a) = -r. \tag{97}$$

For the compact path, the pair tangent is prepared exactly. If $\mathcal{A}h$ is the nodal common-chart action, a pair contribution to $K(\bar{f})h$ is proportional to $\bar{f}_i \bar{f}_j P I_X P \Delta\mathcal{A}h$. Its variation is

$$\delta[K(\bar{f})h] = \mathcal{A}^T \sum_{ij} D w_X v_i v_j \bar{f}_i \bar{f}_j P I_X P\,[\Delta\mathcal{A}\delta h + (r_i + r_j)\Delta\mathcal{A}h]\,, \qquad r_i = \frac{f_i^+}{f_i^+ + f_i^-}\delta l_i. \tag{98}$$

The sum denotes the same paired flux accumulation as the mobility, including the local velocity Jacobians v_i. Here $h = \overline{\nabla} S$ and its componentwise derivative gives $\delta h = h_i' \delta l_i$. The actions $\mathcal{A}h$, nodal mean populations and scalar h_i' are cached once per Newton iterate; each Krylov matvec applies one projected pair tangent. No frozen-mobility approximation is made. Independent forward Jacobian-vector products on the explicit-pair path differentiate all terms and check this prepared tangent. SOLVAX restarted GMRES solves this system; a finite-step Newton Jacobian need not be symmetric positive definite, so PCG is not substituted. Neither route retains a linearization tape containing every primal pair block.

The nonlinear relative stopping norm is $\|F/a\|_2/\sqrt{\sum_i w_i f_i^-}$, fixed by the previous state. The dense reference retains its original population residual norm; it solves the same unscaled equation with its own explicitly recorded tolerance. A backtracking line search decreases the chosen residual norm and rejects nonfinite or exponential-underflow candidates. It clips no population and applies no invariant correction. Since the divided difference gives $\Delta S = \overline{\nabla} S \cdot w(f^+ - f^-)$, the exact entropy defect is

$$\Delta S - \Delta t \overline{\nabla} S^T K(\bar{f}) \overline{\nabla} S = \overline{\nabla} S \cdot F, \qquad |\mathrm{defect}| \le \|a\,\overline{\nabla} S\|_2 \|F/a\|_2. \tag{99}$$

The implementation checks this bound with a floating-point allowance. Likewise, if $Kc = 0$, the invariant error is $c \cdot F$ and is bounded by $\|ac\|_2 \|F/a\|_2$. These are residual-dependent bounds, not exact conservation assertions for an unconverged root.

For the compact local Cartesian grid the stored coefficients are $N \times 5 \times 4$, and energy flows, populations, moment indices and local quadrature have $O(N)$ entries. Dense axis derivative matrices add $O(\sum_\alpha n_\alpha^2)$ storage. No local pair list or dense Newton Jacobian is stored. A target-row contraction processes all N_v partners; with pair budget C_p, temporary pair workspace is $O(\max(C_p, N_v))$ fixed-dimensional vectors. Restarted GMRES additionally stores $O(Nr)$ basis entries for restart length r. Thus the precise storage bound is $O(N + \sum_\alpha n_\alpha^2 + Nr + \max(C_p, N_v))$, apart from compiler/runtime storage. Measured grid bytes and process timings are reported independently. This does not supply an accelerated pair asymptotic: the direct local contraction still evaluates $O(N_X N_v^2)$ interactions per matvec.

For the executed nonlinear uniform and linear toroidal examples, continuous spatial Fourier positivity is checked with a sufficient certificate. If $f(x) = \sum_k \widehat{f}_k e^{ikx}$, then

$$f(x) \geq \widehat{f}_0 - \sum_{k \neq 0} |\widehat{f}_k|. \tag{100}$$

The same triangle bound holds for a double Fourier sum in (θ, z). The example rejects a nonpositive bound, and additionally oversamples the reconstruction. Positive piecewise-linear interpolation in the remaining coordinates then preserves positivity throughout that declared reconstruction. This certificate is conditional on the computed coefficients and is not a positivity theorem for arbitrary Fourier states or higher-order velocity interpolation. The collision derivative still uses its explicitly stated local quadratic differentiation, independently of the positivity-preserving display reconstruction. The Cartesian collision-only boxes use positive nodal values with piecewise-linear reconstruction.

For self-consistent fields the energy is nonlinear in n. Its discrete gradient must enter the collision degeneracy consistently; the fixed vector e above is not enough. Likewise, nonlinear Hamiltonian transport requires its own conservative and positivity-compatible treatment. A linear skew matrix alone is not a proof of a nonlinear H-theorem.

Lagged mobility and an implicit entropy-variable step

For a finite positive-quadrature grid, let $n_i = w_i f_i$ and freeze the constructed mobility at the old density: $K_- = K(f_-)$. Assume finite $w_i, n_{-,i} > 0$, $K_- = K_-^T \succeq 0$, $K_- \mathbf{1} = 0$, and $\Delta t \geq 0$. The implemented lagged-mobility step solves

$$n_+ - n_- = -\Delta t K_- g_+, \qquad g_{+,i} = \ln(f_{+,i}/f_{\mathrm{ref}}), \qquad n_{+,i} = w_i f_{\mathrm{ref}} e^{g_{+,i}}, \tag{101}$$

where f_{ref} is a fixed positive density scale. Computation uses the previously specified normalized units with $f_{\mathrm{ref}} = 1$. A constant change of this scale drops out through $K_- \mathbf{1} = 0$. This changes the time discretization of the same nodal collision ODE, not the kernel, its boundaries or its physical calibration.

The residual $R = w f_{\mathrm{ref}} e^g - n_- + \Delta t K_- g$ is the gradient of

$$\Psi(g) = \sum_i w_i f_{\mathrm{ref}} e^{g_i} - n_-^T g + \frac{\Delta t}{2} g^T K_- g. \tag{102}$$

Each exponential minus positive linear term tends to infinity at either end of the real line and has a finite lower bound. The finite sum is coercive; the nonnegative quadratic preserves this property. Moreover $\nabla^2 \Psi = \mathrm{diag}(n_+) + \Delta t K_- \succ 0$ at every finite g. There is therefore a unique finite minimizer and a unique positive root for every nonnegative timestep. This is a finite-dimensional existence statement, not a uniform conditioning or finite-precision convergence bound.

For $\Delta t > 0$, the equivalent primal problem is

$$\min_{n \geq 0,\ n - n_- \in \mathrm{Range}\, K_-} \left\{ \sum_i n_i [\ln(n_i/(w_i f_{\mathrm{ref}})) - 1] + \frac{(n - n_-)^T K_-^\dagger (n - n_-)}{2\Delta t} \right\}. \tag{103}$$

The range constraint retains every actual Gram null constraint. Since $\mathbf{1}$ is null, the admissible populations have fixed total mass and form a compact set. Strict convexity gives a unique minimizer. A boundary minimizer is impossible: the feasible direction toward the strictly positive n_- has entropy derivative tending to $-\infty$ at each zero component, while the quadratic derivative remains finite. At the interior minimizer, project the Euler equation onto $\operatorname{Range} K_-$ and multiply by K_- to recover (101). This is a fixed finite-dimensional Onsager metric; it is not an exact Wasserstein-distance or JKO step.

At an exact root, $c^T(n_+ - n_-) = 0$ for every $K_- c = 0$. With a residual, $c^T(n_+ - n_-) = c^T R$. Positive frozen pair weights retain the same actual discrete Gram nullspace, including number, discrete energy and resolved moment-bin populations. They do not restore continuum constraints lifted by the spatial approximation. A represented stationary density remains stationary.

For $S(n) = -\sum_i n_i \ln(n_i/(w_i f_{\mathrm{ref}}))$, define

$$\mathcal{D}(n_-\|n_+) = \sum_i [n_{-,i}\ln(n_{-,i}/n_{+,i}) - n_{-,i} + n_{+,i}] \geq 0. \tag{104}$$

Direct subtraction gives the exact budget, including a nonzero root residual,

$$S_+ - S_- = \Delta t\, g_+^T K_- g_+ + \mathcal{D}(n_-\|n_+) - (g_+ + \mathbf{1})^T R. \tag{105}$$

With $a_i = \sqrt{n_{-,i}}$, the residual entropy-defect bound is $|(g_+ + \mathbf{1})^T R| \leq \|a(g_+ + \mathbf{1})\|\,\|R/a\|$; invariant defects have the analogous bound $|c^T R| \leq \|ac\|\,\|R/a\|$. The first positive term is discrete mobility dissipation approximating the continuous nodal model, without a calibrated physical rate. The KL term is a separate contribution from time discretization. For a smooth positive solution it is $O(\Delta t^2)$ per step. Positivity and monotone entropy do not establish temporal accuracy or continuum convergence.

The implicit-function expansion at $\Delta t = 0$ uses the invertible derivative $\operatorname{diag}(n_-)$ and gives $n_+ - n_- = -\Delta t K(f_-)\ln(f_-/f_{\mathrm{ref}}) + O(\Delta t^2)$. Thus the step is first-order consistent with the same fixed spatial ODE. This local statement does not supply a mesh-uniform error bound or a convergence theorem near vanishing populations.

Fixed-reference congruence and dual residual bounds. A numerical population reference is distinct from the constant density scale f_{ref} in the logarithm. Choose finite $n_i^{\mathrm{r}} > 0$, fixed throughout the Newton solve and, when reused, the trajectory, with the same declared total population $N_{\mathrm{r}} = \sum_i n_i^{\mathrm{r}} = N_- := \sum_i n_{-,i}$. Set $A_{\mathrm{r}} = \operatorname{diag}(\sqrt{n^{\mathrm{r}}})$ and $M(g) = \operatorname{diag}(w f_{\mathrm{ref}} e^g)$. The Newton equation is transformed exactly by

$$H_{\mathrm{r}} v = -A_{\mathrm{r}}^{-1} R, \qquad v = A_{\mathrm{r}}\delta g, \qquad H_{\mathrm{r}} = A_{\mathrm{r}}^{-1}[M(g) + \Delta t K_-]A_{\mathrm{r}}^{-1} = \operatorname{diag}\left(\frac{n(g)}{n^{\mathrm{r}}}\right) + \Delta t A_{\mathrm{r}}^{-1} K_- A_{\mathrm{r}}^{-1}. \tag{106}$$

This is an SPD congruence in exact arithmetic. It changes neither R, its unique root, $K(f_-)$, the old-state channel covariances, the objective nor the entropy/KL identity. The log direction is recovered as $\delta g = A_{\mathrm{r}}^{-1} v$, and its objective descent is still $R^T \delta g = (A_{\mathrm{r}}^{-1} R)^T v$. A fixed-reference preconditioner uses the same congruence on its auxiliary matrix, rather than replacing the old density inside the physical mobility. No spectral or floating-point conditioning guarantee follows from positive definiteness alone.

The reported nonlinear norm for this option is

$$\rho_{\mathrm{r}}(R) = \frac{\|A_{\mathrm{r}}^{-1} R\|_2}{\sqrt{N_{\mathrm{r}}}}. \tag{107}$$

At a nonzero Newton residual, the independently checked linear residual is $\|A_{\mathrm{r}}^{-1}[R + (M + \Delta t K_-)\delta g]\|_2 / \|A_{\mathrm{r}}^{-1} R\|_2$. It uses the same original population equation in the chosen metric; native

recursive PCG convergence does not replace this check. The dual Cauchy bounds are

$$|(g_+ + \mathbf{1})^T R| \le \sqrt{N_\mathrm{r}}\, \|A_\mathrm{r}(g_+ + \mathbf{1})\|_2\, \rho_\mathrm{r}(R), \tag{108}$$

$$|c^T(n_+ - n_-)| = |c^T R| \le \sqrt{N_\mathrm{r}}\, \|A_\mathrm{r} c\|_2\, \rho_\mathrm{r}(R) \quad (K_- c = 0). \tag{109}$$

The numerical entropy guard additionally accounts for roundoff in the scalar budget. These bounds control solver defects, not discretization error or pointwise relative density accuracy.

Let ρ_- denote the old-population norm with $A_- = \mathrm{diag}(\sqrt{n_-})$. For equal totals,

$$\rho_\mathrm{r}(R) \le \sqrt{\max_i \frac{n_{-,i}}{n_i^\mathrm{r}}}\, \rho_-(R), \qquad \rho_-(R) \le \sqrt{\max_i \frac{n_i^\mathrm{r}}{n_{-,i}}}\, \rho_\mathrm{r}(R).$$

For a fixed finite positive grid these are norm-equivalence bounds, but their constants can become arbitrarily large as tail populations vanish. Equal nominal stopping tolerances therefore do not imply equivalent accuracy. Both the chosen norm and the unchanged population equation must be identified when comparing results. Rescaling n^r by a constant $c > 0$ would divide ρ_r by c; the total-population condition prevents this arbitrary loosening of the normalized criterion. The compiler copies and binds an optional finite positive reference with the exact nodal shape. Each host step rejects $|N_\mathrm{r}/N_- - 1| > 10^{-9}$ without renormalizing either population. Omitting the option retains the legacy $n^\mathrm{m} = n_-$ congruence and old-population stopping norm. No improved conditioning, successful refinement or new-integrator claim is implied by making the option available.

The positive Hessian permits PCG after entropy scaling, unlike the generally nonsymmetric finite-step discrete-gradient Jacobian. Candidate Newton directions must pass an objective-descent check, a true Hessian-residual check and objective backtracking. For a log increment s, use the stable exact objective change

$$\Psi(g + s) - \Psi(g) = \sum_i n_i(e^{s_i} - 1 - s_i) + R(g)^T s + \frac{\Delta t}{2} s^T K_- s. \tag{110}$$

The small-argument exponential remainder is evaluated by its series. Comparing independently evaluated objective values can erase the tiny Newton decrease near the root. No population clipping, invariant correction, nullspace removal or denominator softening is part of this step.

The method is described as lagged-mobility backward Euler in entropy variables, without a new-integrator claim. Shen and Xu [19] provide a close antecedent: their first-order scheme treats old density in the mobility and new logarithmic density implicitly, and their fully discrete positive-quadrature proof uses a PSD stiffness pseudoinverse and a constrained strictly convex entropy functional (Theorem 3.2, Eqs. 3.12–3.15). The present collision Gram matrix can have additional null directions, so the finite-dimensional argument above is stated independently; their PDE convergence results are not imported. General variational entropy-flow and metriplectic antecedents remain relevant context, but do not identify this frozen Gram metric with a Wasserstein distance.

Implementation and independent checks. The compiler accepts the old density dynamically and prepares the positive channel-lumped Jacobi approximation once per timestep. The approximation drops cross-derivative and pair cross terms only from the auxiliary inverse; the full Gram operator remains in the residual and Hessian. Original Hessian residuals are recomputed independently of recursive PCG stopping. The checked host step rejects nonfinite populations, failed linear corrections, non-descent directions, failed backtracking and inconsistent entropy budgets. A compiled operator is associated with its grid and solver configuration. For a derivative channel d, the positive auxiliary covariance is

$$q_{i,d} = Dw_X v_i f_i \sum_{j \ne i} v_j f_j |I_X P_{ij} C_{i,d}|^2 \ge 0,$$

where v_i includes the velocity Jacobian and $C_{i,d}$ is the corresponding common-chart action column. The diagonal approximation is $\sum_d (D_d \circ D_d)^T q_d$. The optional Cartesian x-line inverse instead retains $L_x = D_x^T \operatorname{diag}(q_x) D_x$ on each fixed (y, z, u, μ) line. With $n^{\mathrm{m}} = n_-$ for the legacy scaling or $n^{\mathrm{m}} = n^{\mathrm{r}}$ for the fixed-reference option, its scaled block is

$$B_x = \operatorname{diag}\left(\frac{n_+}{n^{\mathrm{m}}} + \Delta t \frac{\sum_{d \neq x} (D_d \circ D_d)^T q_d}{n^{\mathrm{m}}}\right) + \Delta t \operatorname{diag}((n^{\mathrm{m}})^{-1/2}) L_x \operatorname{diag}((n^{\mathrm{m}})^{-1/2}).$$

It is SPD for strictly positive finite populations: the first mass term is positive diagonal and the remaining terms are PSD. Batched Cholesky factors apply its inverse. This changes only PCG preconditioning; the full Hessian and all null constraints remain unchanged. Independent explicit-pair projectors check the covariance, and both inverse choices converge to the same separately assembled dense roots. No uniform conditioning bound follows. The archived line-preconditioned campaign uses this inverse and a 3000-iteration linear budget; failed residual checks remain fatal. API defaults retain the diagonal inverse and a 1000-iteration budget. For N state nodes and n_x nodes on a line, stored Cholesky factors use $O(N n_x)$ entries, their factorization costs $O(N n_x^2)$, and each inverse application costs $O(N n_x)$. This is additional solver storage beyond the earlier compact-grid arrays; the direct pair action still costs $O(N_X N_v^2)$.

Optional full spatial-block auxiliary. For the five-axis Cartesian tensor grids (x, y, z, u, μ), the option `preconditioner_axis=(0,1,2)` retains each spatial channel's Gram on the entire spatial tensor at fixed (u, μ). It requires coherent three-axis spatial and two-axis velocity shapes and one matching derivative channel on each selected spatial axis. Let D_d denote the corresponding derivative lifted to all state nodes, and define

$$Q_{\mathrm{sp}} = \sum_{d=x,y,z} D_d^T \operatorname{diag}(q_d) D_d, \qquad L_u = \operatorname{diag}((D_u \circ D_u)^T q_u), \qquad Q_{\mathrm{xyz}} = Q_{\mathrm{sp}} + L_u.$$

The implementation stores the nonnegative q_d and remaining-channel diagonal as prepared data, then constructs the spatial Gram blocks. Thus the existing finite, nonnegative prepared-data check applies without incorrectly requiring Gram off-diagonal entries to be nonnegative. Each spatial Gram is PSD and L_u is nonnegative diagonal. With $A_{\mathrm{m}} = \operatorname{diag}(\sqrt{n^{\mathrm{m}}})$, the auxiliary is

$$P_{\mathrm{xyz}} = \operatorname{diag}(n_+/n^{\mathrm{m}}) + \Delta t A_{\mathrm{m}}^{-1} Q_{\mathrm{xyz}} A_{\mathrm{m}}^{-1}.$$

It is SPD for positive finite n_+ and n^{m} and $\Delta t \geq 0$. Batched Cholesky factors apply its inverse. This retains spatial correlations within each derivative channel, rather than the cross-channel covariance $C_{i,d}^T P_{ij} I_X P_{ij} C_{i,e}$ for $d \neq e$ or the pair endpoint cross terms. All such terms remain in the exact residual and Hessian

$$H = \operatorname{diag}(n_+/n^{\mathrm{m}}) + \Delta t A_{\mathrm{m}}^{-1} K_- A_{\mathrm{m}}^{-1}.$$

Consequently this option changes neither the physical root equation nor its invariant nullspace, entropy budget or chosen residual norm. Mathematical SPD does not guarantee well-conditioned, representable floating-point factors; failed corrections and the independently recomputed original Hessian residual remain checked.

A one-sided algebraic comparison is available for the implemented four channels and three-point local quadratic u derivative. Write $Z_{ij} = I_X P_{ij}$ and $a_{i,d} = C_{i,d}(D_d h)_i$. Endpoint and channel Cauchy–Schwarz give

$$\left| Z_{ij} \sum_{d=1}^{4} (a_{i,d} - a_{j,d}) \right|^2 \leq 8 \sum_{d=1}^{4} (|Z_{ij} a_{i,d}|^2 + |Z_{ij} a_{j,d}|^2).$$

Summing with the positive symmetric unordered pair weights defining K_- produces each $q_{i,d}$ once, hence

$$K_- \preceq 8Q_{\mathrm{all}}, \qquad Q_{\mathrm{all}} = \sum_{d=x,y,z,u} D_d^T \operatorname{diag}(q_d) D_d.$$

Because each D_u row has at most three nonzero entries, $D_u^T \operatorname{diag}(q_u) D_u \preceq 3L_u$. Therefore

$$K_- \preceq 24Q_{\mathrm{xyz}}, \qquad H \preceq 24P_{\mathrm{xyz}}.$$

The constant uses the stated channel count and derivative row support; it is not asserted for arbitrary derivative schemes. There is no converse uniform bound: the auxiliary can penalize exact K_- null functions, and positive population ratios can vary without a uniform bound. No field-, state- or refinement-independent condition estimate or speedup follows.

With $N_s = n_x n_y n_z$ and $N_v = n_u n_\mu$, one dense spatial factor contains $N_v N_s^2$ entries, its factorization costs $O(N_v N_s^3)$, and applying its inverse costs $O(N_v N_s^2)$. Simultaneously retained Gram blocks require another $N_v N_s^2$ entries; prepared covariances, derivative lifts and compiled workspace are additional. This storage is not $O(N_s N_v)$ under balanced spatial refinement. A noncubic explicit-pair test independently assembles the covariance, spatial derivatives and dense auxiliary inverse under both population metrics, and checks the exact Hessian and the one-sided Gram bound. These finite-grid checks do not imply continuum convergence or a general conditioning guarantee.

Independent explicit-pair dense roots and a separately assembled pair ODE check tiny mirror, dipole and nonaxisymmetric grids. These method checks do not resolve the thermal velocity domain or establish a continuum limit.

13 Scattering physics and validity tests

The constrained model assumes negligible accumulated change of magnetic moment. That assumption is stronger than a small collision frequency compared with the gyrofrequency. At constant B, an energy change satisfies

$$\Delta E = \frac{m}{2}\Delta(u^2) + B\Delta\mu + q\Delta\Phi. \tag{111}$$

Small ΔE does not separately bound $\Delta\mu$. In particular, the approximate identification in SM25 (85), printed p. 10, requires an extra bound on parallel-energy exchange: at constant B,

$$B|\Delta\mu| \le |\Delta E| + \tfrac{1}{2}m|\Delta(u^2)| + |q\Delta\Phi|. \tag{112}$$

Spatial locality and nearly constant B alone supply no bound on the second term. This audit concerns the inference from small energy change; it does not rule out a stronger regime in which every relevant exchange is ordered small. For independent unbiased increments, zero mean accumulation can coexist with

$$\operatorname{Var}\left(\sum_{j=1}^{N} \delta\mu_j\right) = N \operatorname{Var}(\delta\mu). \tag{113}$$

For independent increments, both $N|\langle\delta\mu\rangle|/\mu_{\mathrm{ref}} \ll 1$ and $N\operatorname{Var}(\delta\mu)/\mu_{\mathrm{ref}}^2 \ll 1$ are needed for small accumulated change on a chosen nonzero scale. Compensating mean changes can still alter the full moment marginal through diffusion. These are audit points for the inference in SM25 (85) and the accumulation hypothesis (86), rather than replacements for its ordering. The source explicitly calls (84) and (86) working assumptions.

A controlled microscopic reference can integrate

$$m_s\dot{\boldsymbol{v}}_s = q_s\boldsymbol{v}_s \times \boldsymbol{B} - \nabla_{\boldsymbol{r}_s} V, \qquad \dot{\boldsymbol{r}}_s = \boldsymbol{v}_s, \tag{114}$$

with a specified screened potential, for example $V(r) = q_1 q_2 e^{-r/\lambda}/(4\pi\epsilon_0 r)$. Screening is a declared model choice, not a result of this two-body calculation. Resolve close encounters or exclude them with a stated grazing criterion. The generalized magnetized scattering theory in Ref. [20] is an important reference.

For uniform $\boldsymbol{B}$, compare with the unperturbed helical orbit at identical incoming conditions. Use

$$\boldsymbol{R}_s = \boldsymbol{r}_s + \frac{\boldsymbol{v}_s \times \boldsymbol{b}}{\Omega_s}, \qquad \mu_s = \frac{m_s|\boldsymbol{v}_{\perp s}|^2}{2B}, \quad \Omega_s = q_s B/m_s. \tag{115}$$

Extract asymptotic incoming/outgoing guiding-center data at a common reference time. A raw endpoint displacement can contain an arbitrary flight-time contribution after the encounter changes parallel velocity.

Sample incident flux, impact parameters, gyrophases and relative velocities with the appropriate measure. Check timestep, start/end distance, angular sampling and screening/cutoff convergence independently. Recover a Rutherford limit before using strong-field results. Attractive close encounters outside the grazing regime are not automatically described by the reduced kinetic closure.

For independent encounters at a stated rate ν_{enc}, diagnostic drift and diffusion coefficients are

$$A_\mu = \nu_{\mathrm{enc}}\langle\delta\mu\rangle, \qquad D_{\mu\mu} = \frac{\nu_{\mathrm{enc}}}{2}\langle(\delta\mu)^2\rangle. \tag{116}$$

Use both, together with cross-covariances and correlation tests. A timescale estimate must specify a nonzero population scale μ_{ref}; the quantities $\mu_{\mathrm{ref}}/|A_\mu|$ and $\mu_{\mathrm{ref}}^2/(2D_{\mu\mu})$ diagnose different failures of the constraint. Near $\mu = 0$, use a suitable distribution-level metric.

Compare these with an independently determined mixing time. A physically supported constrained interval requires

$$\tau_{\mathrm{mix}} \ll t \ll \tau_\mu. \tag{117}$$

Choosing a small input pitch-angle rate establishes a synthetic asymptotic test, not this physical inequality. Adding two collision terms also requires a nonoverlapping process decomposition; it can otherwise double count encounters.

14 Independent homogeneous controls and encounters

Lorentz pitch-angle evolution

For a fixed physical speed shell, the implemented operator is $C_{\mathrm{Lz}} = \nu\partial_\xi[(1-\xi^2)\partial_\xi]/2$, with natural endpoint flux at $\xi = \pm 1$. The Legendre equation gives eigenrates $\nu\ell(\ell+1)/2$. Gauss–Legendre projection and exact modal exponentials provide an independent solution; the tests additionally apply the differential operator to Legendre polynomials. Thus the $\ell = 0, 1, 2, 3$ rates are $0, \nu, 3\nu, 6\nu$. Shell population and energy are preserved, while the $\ell = 1$ momentum is transferred to the fixed reservoir. The coordinate map and full Jacobian are

$$u = v\xi, \quad \mu = \frac{mv^2(1-\xi^2)}{2B}, \quad \left|\frac{\partial(u,\mu)}{\partial(v,\xi)}\right| = \frac{mv^2}{B}, \quad \mathrm{d}^3v = \frac{2\pi B}{m}\mathrm{d}u\mathrm{d}\mu. \tag{118}$$

The shell comparison changes μ while preserving speed, as ordinary pitch scattering must; it is not an unbalanced bare moment diffusion term. All four evolved modes are also checked on actual mapped (u,μ) nodes with the full transformed measure, rather than only checking the Jacobian at one point. Take $F(v) = 1 + 0.3v^2$ on $0.4 \le v \le 2$, $m = 1.7$, $B = 2.3$, $\nu = 0.6$, $t = 0.8$, and amplitude $\epsilon = 0.15$. Twelve positive Gauss nodes in each of v and ξ are mapped to (u,μ); each speed shell is evolved by the production Legendre solver. An independent polynomial primitive $I_p = \int_{0.4}^{2} v^p F(v)\,\mathrm{d}v$ gives

$$N = 4\pi I_2(1+\epsilon\delta_{\ell 0}), \qquad E = 2\pi m I_4(1+\epsilon\delta_{\ell 0}), \qquad P_{\parallel}(t) = \delta_{\ell 1}\frac{4\pi m\epsilon}{3} I_3 e^{-\nu t}.$$

The mapped quadrature reproduces count and energy to 2×10^{-14} relative test tolerance and the parallel momentum to 2×10^{-13} absolute tolerance. The $\ell = 1$ momentum decreases, while all represented densities remain positive. This is a mapped shell quadrature check, not a rectangular (u,μ) Lorentz discretization.

Nonlinear conserving Dougherty evolution

The physical three-velocity operator is

$$C_D[F] = \nu\nabla_v\cdot[(\boldsymbol{v}-\boldsymbol{U})F + \theta\nabla_v F], \quad \boldsymbol{U} = \langle\boldsymbol{v}\rangle, \quad \theta = \tfrac{1}{3}\langle|\boldsymbol{v}-\boldsymbol{U}|^2\rangle. \tag{119}$$

Integration against 1, $\boldsymbol{v}$ and $|\boldsymbol{v}-\boldsymbol{U}|^2$ proves that $\boldsymbol{U}$ and θ are constant for closed homogeneous evolution. For a positive normalized Gaussian mixture with weights p_a, component means $\boldsymbol{U}_a$ and covariances A_a, the exact solution remains a positive mixture with

$$\boldsymbol{U}_a(t) = \boldsymbol{U} + e^{-\nu t}(\boldsymbol{U}_a(0)-\boldsymbol{U}), \quad A_a(t) = e^{-2\nu t}A_a(0) + \theta(1-e^{-2\nu t})I. \tag{120}$$

Substituting each Gaussian into the now constant-moment Fokker–Planck equation verifies this solution; conservation makes it a solution of the original nonlinear model. It includes a non-Gaussian control, without substituting prescribed reservoir moments. Exact Gaussian gradients and Laplacians evaluate the strong operator independently of a centered temporal difference.

In `03_controls.py`, Lorentz modal errors are below 6.5×10^{-15}. The two-component Dougherty mixture has nonzero drift and anisotropic covariance. A 40^3 physical Cartesian velocity quadrature on extent 9 gives particle error 2.0×10^{-11}, covariance error 7.82×10^{-10} and centered-energy error 4.76×10^{-10}. The strong equation versus centered time derivative error is 2.21×10^{-10}. The sampled entropy increases; positive mixture reconstruction is analytic, and the smallest sampled density is 1.96×10^{-64}.

Physical three-velocity Landau weak reference

The conventional one-species normalization used here is

$$C[F] = \Gamma\nabla_v\cdot\int FF'U(\boldsymbol{w})[\nabla_v\ln F - \nabla_{v'}\ln F']\mathrm{d}^3v', \quad U(\boldsymbol{w}) = \frac{I-\hat{\boldsymbol{w}}\hat{\boldsymbol{w}}^T}{|\boldsymbol{w}|}, \quad \boldsymbol{w} = \boldsymbol{v}-\boldsymbol{v}'. \tag{121}$$

With $\int F\mathrm{d}^3v = n$, the SI coefficient is $\Gamma = q^4\ln\Lambda/(8\pi\epsilon_0^2 m^2)$, matching the mass convention of Fitzpatrick's Eqs. (3.93)–(3.94) [21]. Writing $F = (n/v_*^3)\widehat{F}$ gives the frequency scale $\Gamma n/v_*^3$. The Coulomb logarithm and its cutoffs are supplied parameters; this conversion does not calibrate the constrained scalar D.

For a normalized Gaussian of covariance A, $\boldsymbol{w}$ has covariance $2A$ and the pair score is $\boldsymbol{g} = -A^{-1}\boldsymbol{w}$. The exact weak covariance and entropy rates are

$$\dot{A} = -\frac{\Gamma}{2}\mathbb{E}[\boldsymbol{w} \otimes U\boldsymbol{g} + U\boldsymbol{g} \otimes \boldsymbol{w}], \tag{122}$$

$$\dot{S} = \frac{\Gamma}{2}\mathbb{E}[\boldsymbol{g} \cdot U\boldsymbol{g}] \geq 0. \tag{123}$$

Trace conservation follows pointwise from $U\boldsymbol{w} = 0$. The center velocity integrates analytically; radial/polar/gyrophase quadrature retains the full physical d^3w geometry. A separate Cartesian Gauss–Hermite expectation avoids $\boldsymbol{w} = 0$ by even quadrature order. A third independent check diagonalizes A and uses the Laplace integrals for $1/r$ and $1/r^3$. Define $Z(t) = \prod_j (1 + 4A_j t)^{-1/2}$ and $V_j(t) = 2A_j/(1 + 4A_j t)$; then

$$\dot{A}_i = \frac{\Gamma}{\sqrt{\pi}} \int_0^\infty Z(t) \left[t^{-1/2} \frac{V_i}{A_i} - 2t^{1/2} V_i \sum_j \frac{(1 + 2\delta_{ij})V_j}{A_j} \right] \mathrm{d}t. \tag{124}$$

Gaussian second and fourth moments after multiplication by e^{-tr^2} give V_i and $(1 + 2\delta_{ij})V_iV_j$, proving the integrand. The substitution $t = s^2$ removes its endpoint singularity before adaptive one-dimensional integration.

For $A = \mathrm{diag}(1.15, 1.15, 0.7)$ and $\Gamma = 1$, the covariance derivatives are

$$(-0.0740393, -0.0740393, 0.148079), \qquad \dot{S} = 0.0413884.$$

The reported energy rate residual is 2.1×10^{-18}. Radial quadrature refinement reaches a 4.14×10^{-14} discrepancy from (124) at order 28, while the independent Cartesian Hermite route converges more slowly from 8.83% at order 8 to 0.316% at order 48. Tail and denominator-softening scans are saved separately. Softening changes only $1/r$ and retains $U\boldsymbol{w} = 0$; it is a convergence diagnostic, not an unmodified Coulomb tensor. A small-anisotropy check takes $A = I + \epsilon\,\mathrm{diag}(1, 1, -2)$. At first order, $\mathbb{E}|\boldsymbol{w}| = 4/\sqrt{\pi}$ for covariance $2I$ and the angular moments are $\mathbb{E}[n_i^2] = 1/3$ and $\mathbb{E}[n_i^2 n_j^2] = (1 + 2\delta_{ij})/15$. Substitution in the weak covariance rate gives $\dot{A}_i = -4\Gamma\epsilon D_i/(5\sqrt{\pi}) + O(\epsilon^2)$, where $D = (1, 1, -2)$. The instantaneous normalized stress rate approaches $4\Gamma/(5\sqrt{\pi}) = 0.451351667$; for $\epsilon = 0.03, 0.01, 0.003$ the two independent references agree and give $0.457605, 0.453335, 0.451936$ respectively. This is a limiting conventional control, without a discovery claim.

Maxwellian score differences are parallel to $\boldsymbol{w}$ and annihilate the kernel for multiple drifts and temperatures. A separate positive non-Gaussian nodal pair calculation checks weak number/momentum/energy conservation and positive entropy production. Its omitted coincident-node contribution is a stated quadrature prescription, not a general directional limit. These controls verify instantaneous weak moments; no full nonlinear Landau kinetic time integration or Gaussian closure trajectory is claimed.

Direct screened encounter audit

The identical-particle relative and center equations and separate energies are established in Nersisyan and Zwicknagel [22], Eqs. (4)–(8). Their Sec. V, Eqs. (66)–(76), distinguishes transmission over the repulsive barrier from backscattering, including two velocity-transfer branches in Eq. (73). The finite-transverse-motion extension [23], Sec. IV, further limits the simple strong-field barrier picture. We inspected the stated author-preprint versions locally. Their potential is regularized at the origin; neither their strong-field threshold nor their scattering cross sections are substituted for the finite-field, singular screened potential and finite event planes used here. Mixed exit channels are thus an established antecedent, not a newly discovered mechanism.

`04_encounters.py` directly integrates equal-mass, equal-charge repulsive pairs in a uniform field with $V(r) = \kappa e^{-r/\lambda}/r$. It measures changes relative to the free helix at identical incoming conditions. Endpoint refinement holds fixed the incoming guiding center and phase at the $z = 0$ crossing; changing the initial plane at fixed raw position and phase would instead change the orbit and is not a valid endpoint convergence test. For relative guiding-center offset b, $v_\parallel$ and gyrophase φ at that crossing, the incoming plane $z = -L$ has phase $\varphi + \Omega L/v_\parallel$ and transverse position $(b - v_y/\Omega, v_x/\Omega)$. This follows directly from the invariant free $\boldsymbol{R} = \boldsymbol{r} + \boldsymbol{v} \times \boldsymbol{b}/\Omega$.

A documented case uses $b = 1.2$, $v_\parallel = 1$, $v_\perp = 0.35$, $\Omega = 2$, $\kappa = 0.05$, $\lambda = 4$ and 32 phases. Increasing L from 12 to 32 changes the mean moment increment from 5.0126×10^{-5} to 5.0074×10^{-5} and its second moment from 9.4135×10^{-7} to 9.4059×10^{-7}. The individual incoming μ is 0.00765625, so the rms increment is approximately 12.7% of that scale while the ensemble pair-energy error is below 3.4×10^{-11}. This is a controlled counterexample to inferring moment preservation solely from energy conservation; it does not characterize every weak encounter regime. For the same $B = 2$ case, the maximum relative velocity change from the free helix is 0.0340, the minimum sampled separation is 1.045, and $\rho/\lambda = 0.0219$, $\rho/b_{90} \simeq 0.982$, $\Omega\lambda/v_{\mathrm{rel}} \simeq 7.55$. Uniform B gives $\rho/L_B = 0$; ν/Ω is unassigned because no encounter rate is measured. The free-orbit control has moment drift below 1.2×10^{-14} and guiding center drift below 6×10^{-16}. In a weak-field unscreened Rutherford control, increasing endpoint distance from 12 to 96 reduces the angular error, but the final relative discrepancy remains 1.11×10^{-3} and is recorded as finite-endpoint error. A separate close-encounter timestep scan decreases pair-energy error from 5.40×10^{-4} to 3.57×10^{-9}. The two individual moment increments are correlated by the imposed symmetric center-of-mass initialization; saved joint covariances must not be interpreted as independent species statistics.

No incident-flux/impact-parameter ensemble, plasma encounter frequency, or independently measured lifetime is assigned to these phase statistics.

15 Research questions and evidence

Geometry and nullspace. Begin with the proven finite-quadrature spectrum (33). Track its changes under a valid mirror, toroidal or dipole field. Separate collision null modes from decay generated through coupling to Hamiltonian transport. A small positive eigenvalue on one finite grid is not a continuum gap. Vary all resolutions, domains and velocity tails.

Localization. Compare the local kernel with a specified symmetric finite-range kernel, using a derived normalization as the range shrinks. Boundary renormalization must retain pair symmetry. A Gaussian-range surrogate remains a surrogate unless independent scattering physics supports it.

Constrained dipole states. Ref. [24] studies maximum-entropy dipole states and additional adiabatic constraints. Here the result sought is a dynamical relaxation law, an extra invariant, a loss of mixing, or an independently measured lifetime. Reproducing (38) is verification rather than discovery.

For any geometry-amplitude scaling, compare competing fits and inspect projector degeneracies before using regular perturbation theory. For steady-state parameter derivatives specify which particle, energy and marginal constraints are held fixed; otherwise a nullspace can make the derivative underdetermined.

Executed prototype evidence and remaining limits

The public library has five substantive modules: `geometry`, `collisions`, `solver`, `controls` and the independent `reference`. Examples save inputs, boundary/model descriptions, package versions, source checksums and commit context with their results. All quoted values below are normalized, finite-grid

evidence from the saved runs; rates are not dimensionally calibrated to a plasma.

The earlier 216-node NumPy oracle verifies the local uniform rate 0.11742603550295856 and its exact velocity-nullspace structure. The JAX implementation is compared with that independent pair calculation, a dense weak assembly, analytic Fourier decay, nonlinear directional linearization, conservation and discrete entropy identities. Geometry tests additionally cover both charges, both coordinate charts, sampled Jacobi and Liouville identities, independent finite-difference convergence, measure transformation, electric cross-field drift and mirror orbit timestep convergence.

`01_relaxation.py` evolves a positive 105-node uniform nonlinear state for ten steps of $\Delta t = 0.08$. Its saved particle, energy and maximum moment-bin errors are below 7.1×10^{-17}; the minimum step entropy increase is 4.75×10^{-4} and the maximum discrete entropy identity error is 6.08×10^{-15}. Nodal values remain positive, and the continuous spatial Fourier certificate and positive piecewise-linear velocity reconstruction are checked by the example. This is the specified nonlinear local surrogate, without self-consistent fields or a claim that its additional null directions disappear.

`02_geometry.py` evolves collision-only, ideal-only and combined linear states on the nonuniform vacuum toroidal annulus. On the base $5 \times 3 \times 3 \times 11 \times 13$ grid at $t = 0.24$, the respective $Q/Q(0)$ values are 0.820926, 1.000000 and 0.810980. Number, energy and moment-bin errors are below 1.1×10^{-17} for these three controls. Three-level scans separate radial, angular, parallel-velocity, moment and timestep errors; velocity tails are varied separately. The last changes in both final Q and dissipated fraction pass the 1% target for every scan. Changes in the dissipated fraction are 0.332% radially, rounding-level in the two angular grids for this resolved initial mode, 0.709% in parallel velocity, 0.638% in moment resolution, 0.495% in time, 0.0578% in parallel-velocity tail extent, and 0.519% in moment-tail extent. These are separate refinements of a specified observable, not a mathematical continuum convergence theorem. The unmodified 243-node collision matrix has 39 near-null eigenvalues, compared with the 39-dimensional independent invariant span from spatial populations and angular-weighted invariants in (53), with scaled eigenpair residual 1.63×10^{-15}. Every observed tiny-grid null direction is accounted for by that span; the eigenvalues themselves are unchanged. No continuum gap or uniqueness follows from that finite matrix.

`06_range.py` checks 27 combinations of spatial grid, velocity count and Gaussian range. The finest spatial density-rate discrepancy is at most 3.03×10^{-13} relative to (35). Underresolved narrow ranges are marked unresolved: for $\ell = 0.15$, 16 spatial nodes give a 58.5% density-rate error, while 32 give 4.75×10^{-4} and 64 reach rounding accuracy. This explicitly demonstrates why the range must be resolved before interpreting density-nullspace lifting.

`07_equilibria.py` independently integrates the stationary density (38) in all five fields with three velocity quadrature orders and two tail extents. The largest finest error on $|u| \le 7$, $\mu \le 20$ is 1.61×10^{-7} (dipole). Its separate full-marginal finite-quadrature energy solve gives $\beta = 1.7310695$, zero reported energy residual and moment-bin residual 2.22×10^{-16}. This verifies candidate construction, not attraction from arbitrary initial data.

`08_fields.py` performs actual nonlinear collision-only steps in mirror, dipole and perturbed-dipole boxes with the bounds derived above. At three spatial resolutions, equilibrium residuals are below 2.8×10^{-14} and initial entropy production is positive. The last spatial changes in initial production are 0.336%, 0.255% and 0.257% respectively. Three positive nonlinear steps on each coarse box conserve number, energy and the full moment marginal to below 2.0×10^{-17} in the saved run, with positive entropy increments. Velocity resolution/tails and longer nonlinear time evolution remain unresolved for these boxes. These runs impose natural collision flux only; they do not impose a confined dipole Hamiltonian boundary or demonstrate dipole self-organization.

Independent controls and direct encounters are described in the preceding sections. Performance

measurements separate first-call compilation, warmed synchronization, pair chunk size and solver residual. Implicit parameter derivatives are checked against dense solves and centered finite differences. All performance claims remain specific to the saved device and accuracy; no unmeasured GPU speedup is claimed. The selected P1–P5 prototype examples have executable structural and convergence evidence. This does not certify every geometry or parameter regime; joint box time-evolution convergence remains specific to the executed scans. The continuous local zero set is classified above, including a critical-point failure of continuous extension; modified discrete energy maps require separate validation. Physical coefficient calibration, unequal-species closure, self-consistent electrostatics, and measured plasma mixing-versus-moment timescale separation are not implemented claims.

Focused trajectory, unit and adjoint verification

On the symmetry axis of the harmonic mirror, $B(z) = B_0 + az^2$, $\dot{z} = u$ and $m\dot{u} = -2\mu az$. For initial $z = 0$ and $u = u_0 > 0$, a separate scalar energy calculation locates the positive turning point from $\mu B(z_t) = mu_0^2/2 + \mu B_0$ by bracketing. The bounce integral is

$$T_b = 4\int_0^{z_t} \frac{\mathrm{d}z}{\sqrt{2[E - \mu B(z)]/m}}.$$

The substitution $z = z_t(1 - s^2)$ removes its endpoint singularity and gives $T_b = 4\int_0^1 2z_t\,[2\mu a z_t^2(2 - s^2)/m]^{-1/2}\mathrm{d}s$. Numerical quadrature is compared with turning-point events of a separately integrated five-coordinate Poisson flow. Both event position and period converge under integrator-tolerance refinement; the check uses nonunit mass and negative charge, and does not infer a collision rate from the orbit.

A uniform Fourier-pair regression also changes the base units rather than changing the physical input. Let the new units be L_*, T_*, M_*, Q_*, N_*, where N_* is a number-density unit. Then $U_* = L_*/T_*$, $E_* = M_*U_*^2$, $B_* = M_*/(Q_*T_*)$, and the prescribed uniform coefficient has unit $D_* = M_*^2L_*^2/(T_*^3N_*)$. In the (u, η) chart,

$$\widehat{u} = u/U_*, \quad \widehat{\eta} = \eta/E_*, \quad \widehat{w}_v = w_v/(U_*E_*), \quad \widehat{f} = fU_*E_*/N_*, \quad \widehat{k} = kL_*.$$

Mass, charge, field and coefficient are divided by their respective units, and $\partial_{\widehat{u}}h = U_*\partial_u h$. The independent pair calculation verifies $\widehat{C} = T_*U_*E_*C/N_*$ and $\widehat{\lambda} = T_*\lambda$ for the density-neutral observable $\lambda = -\langle h, C\rangle_v/\langle fh^2\rangle_v$, for two unit maps and both charge signs. This check is restricted to the uniform reduced kernel; it does not replace the metric pullback required for a general coordinate or units change in phase space.

For the linear implicit system $Ax = Mh$, $A = M + \Delta tK$, and observable $J = x^Tx$, the adjoint obeys $A^T\lambda = 2x$ and $\partial_D J = -\Delta t\lambda^T Kx$ at unit coefficient. The actual implicitly differentiated solver's VJP with respect to h returns $M\lambda$. Dividing by the positive diagonal mass recovers the adjoint, whose true residual is checked against the independently assembled dense matrix alongside the primal residual, dense derivative and centered finite-difference plateau. These are finite-grid derivative checks.

Directional zero-set check

At the off-axis dipole state $(1, 0.2, 0.3, 0.6, 0.4)$, the independent `test_orbits.py` calculation approaches coincidence along u and μ separately at separations $10^{-2}, 10^{-3}, 10^{-4}$. The two limiting PI_XP tensors remain different: there is no direction-independent continuous projector extension. For the smooth observable $x^2+u^3+\mu x$, however, $\Delta\phi^T PI_XP\Delta\phi$ decreases quadratically along both paths. Thus identical-node diagonal contributions vanish in the weak form despite the kernel's directional discontinuity. This

fact does not authorize deletion of distinct states with the same energy-flow vector; the general constructor rejects those ambiguous pairs. The uniform directional prescription is derived separately and tested against the exact oracle.

Measured costs at matched error

For four independent uniform Fourier blocks with 128 velocity nodes each, $\Delta t = 0.025$ and $\lambda = 0.3$, the common temporal error is 2.77×10^{-5}. On the Apple M4 CPU with JAX 0.11.2 and SOLVAX 0.28.2, five synchronized repetitions give the following measured values:

Route	Compile/first (s)	Median (μs)	Range (μs)	Temporary bytes
Dense	0.092	550.3	214.2–1297.8	262852
PCG	0.184	571.7	273.1–804.7	16696
Diagonal PCG	0.225	43.8	34.5–126.6	16832

The implicit solve residuals are below 6×10^{-13}; these separable uniform timings do not describe general pair-kernel complexity. For 945 actual weak pairs, a 16-pair chunk reduces XLA temporary buffers from 118913 to 16480 bytes, while the warm median increases from 96.3 to 158.2 μs. The complete benchmark process peaks near 332 MB RSS, including the runtime and compiled executables. The base combined toroidal evolution takes 2.38 s for six steps after 1.12 s compilation/first execution and 1.22 s setup. Its state is 51480 bytes and its stored pair list contains 456885 unordered pairs; pair-application chunk workspace and all version/input/commit details are in the saved JSON. No extrapolated speedup or isolated-RSS interpretation is asserted.

Nonuniform velocity quadrature and resolved tails

The additional `11_field_velocity.py` campaign evaluates the initial nonlinear entropy production on the same mirror, dipole and nonaxisymmetric boxes, with fixed 3^3 spatial quadrature. Its positive initial distribution is

$$f = \exp[-E - 0.2\mu + 0.1\sin(x)u + 0.04y\mu]. \tag{125}$$

Positive Gauss–Legendre velocity quadrature replaces both the population weight $w_X B w_u w_\mu$ and the local pair weight $w_X (B w_u w_\mu)(B w'_u w'_\mu)$ consistently. Derivatives and energy directions are unchanged. The three-point derivative is exact for the quadratic velocity energy and the linear-u logarithmic perturbation on either node family. Number and energy are checked independently using shifted Gaussian integrals in u and exponential integrals in μ.

The base velocity domain is $u \in [-4, 4]$, $\mu \in [0, 20]$, with orders 25 and 21. Separate order scans are $17, 21, 25$ in u and $13, 17, 21$ in μ. Tail extensions to $|u| = 5$ and $\mu = 24$ are separately checked at wider-domain orders 29 and 25. A Gauss tail extension changes node spacing; these extra order checks are essential and are not described as fixed-spacing tests. All 36 cases retain their inputs and status. Across all three fields, the largest final Gauss production change is 0.261%; the largest expanded-domain quadrature change is 0.180%. Base finite-domain number/energy quadrature errors are below 9.4×10^{-15}. Independent analytic omitted energy is about 0.050%, which is not itself a bound on production error.

Equidistant trapezoid controls at $9^2, 17^2, 25^2$ velocity nodes remain unresolved: the last production changes are 16.63%, 10.14%, and 10.15% for mirror, dipole and nonaxisymmetric fields. Their statuses are preserved. The successful Gauss checks establish the stated initial-production comparison on a fixed spatial grid, not joint spatial–velocity or finite-time nonlinear convergence. The constructor now rejects

zero pair directions without allocating and discarding a 25-component tensor per pair; the retained unit directions and collision action are unchanged and tested against explicit projector matrices.

Finite-time campaigns and provenance

The historical discrete-gradient campaign and the current lagged-mobility campaign are separate time discretizations. The recovered four mirror rows retain their original execution at commit `fdc625d`; the subsequent field-split execution uses `4ce8e5c`. Dipole and nonaxisymmetric first steps in that execution failed their original GMRES check, and those failed logs and partial summaries remain archived. A failed Newton iteration is not a proof that a positive root does not exist.

The historical resume validator checked exact input hashes, package-source bytes and scientific-function syntax against its fixed anchor, together with recorded commits, clean-source hashes, environment metadata, complete time histories, positivity and retained solver/invariant tolerances. Tests retain that validator from Git history to audit the archived rows. Its saved summaries lack stage states and gradient-dependent entropy bounds, so rechecking them cannot replay every original entropy acceptance decision. Original timings and run identifiers remain attached to their producing executions.

The completed historical mirror run contains thirteen cases. Its eight independent refinements all pass the original 1% target; the largest selected-observable change is 0.1126103%, from extending $|u| \le 4$ to $|u| \le 5$. The baseline relative entropy decreases by 43.7427% and the active moment by 25.0538% through $T = 0.02$. Across all accepted histories, number, energy and full moment-bin errors are at most 7.779×10^{-14}, 1.363×10^{-13} and 2.258×10^{-14}; the maximum nonlinear residual is 7.545×10^{-13} at target 10^{-12}. The retained minimum nodal density is 1.232×10^{-16} and the largest absolute entropy identity discrepancy is 5.877×10^{-15}. These are fixed-other-parameter comparisons, not joint continuum convergence or a classification of every spatially localized constraint.

The archived validator independently accepts all thirteen rows and exactly reproduces each saved comparison. Raw summaries, environment, CPU report and logs are in `results/nonuniform_mirror_dg/raw`; its separate audit retains exact inputs and per-case costs. The $7^3 \times 25 \times 21$ case has 180075 nodes and 41786320 bytes of stored grid arrays (not peak memory). Its four accepted steps took 2001.55 seconds including the first compilation, versus 366.68 seconds for the 5^3 case, on a four-logical-CPU AMD EPYC 7763 runner. These are observed run costs, not a matched-error speedup comparison.

The current `12_nonuniform_evolution.py` instead uses (101). Its fifteen predefined cases per field independently vary spatial order, parallel-velocity and magnetic-moment order, timestep, velocity tails and quadrature on each expanded domain. Baseline values are $5^3 \times 25 \times 21$ nodes, $D = 0.1$, $T = 0.02$, and $\Delta t = 0.005$; the first-order timestep sequence is $0.005, 0.0025, 0.00125, 0.000625, 0.0003125$. Every adjacent difference and the coarsest-to-finest difference are retained; a finest-pair pass does not certify the original baseline timestep or jointly refine the spatial and temporal discretizations. The common 1% observable-change target is retained. All units are normalized; these times use the freely prescribed coefficient. The current example binds the fixed initial population as its numerical reference and uses the optional three-axis spatial channel-Gram auxiliary; earlier line-preconditioned results retain their separate source revisions and result directories.

`SM_EVOLUTION_FIELDS` and `SM_EVOLUTION_CASES` select predefined independent execution jobs, leaving their scientific inputs fixed. Without filters the script runs all forty-five cases. The CI driver separates each field/case pair and preserves failed or partial evidence. Historical discrete-gradient checkpoints are explicitly rejected by this new campaign. A completed selected job is distinct from a passing refinement comparison. Every row records its actual parameters, full step diagnostics, Newton history, separate KL contribution, residual entropy bound and original run identifier; metadata record source, versions and hardware. Independent refinements do not establish joint continuum convergence. No final outcome for

the current campaign is asserted by this execution description.

Completed mirror lagged-mobility campaign. The retained collection contains 16 of 45 planned cases and remains unresolved overall. All fifteen independently executed mirror cases reach $T = 0.02$ with the lagged-mobility entropy step, retaining a separate provenance record from the historical discrete-gradient campaign. The eight final, fixed-other-parameter comparisons all meet the original 1% target. Their largest change is 0.197278%, in the active-moment change for the final timestep pair. At $\Delta t = 0.005$, the $5 \to 7$ spatial-order changes in entropy gain, final production and active-moment change are respectively 0.045351%, 0.024860% and 0.056944%. The corresponding largest tail change is 0.112113%.

The timestep sequence provides important qualification of these passes. For $\Delta t = 0.005 \to 0.0025$, entropy gain and active-moment change still differ by 1.30210% and 1.51359%. For the finest pair, $0.000625 \to 0.0003125$, the entropy-gain, final-production and active-moment discrepancies are 0.168910%, 0.131882% and 0.197278%. Successive absolute increments approximately halve, as expected for this first-order method. These are measured differences, not rigorous continuum error bounds; a first-order Richardson indicator additionally assumes the asymptotic expansion. The coarse-to-finest differences are still 2.45276%, 1.91519% and 2.85202%. A finest-pair pass therefore does not retrospectively certify the original coarse timestep. Spatial scans use the coarse timestep, whereas the time scan fixes spatial order five; joint refinement is not asserted.

At $\Delta t = 0.005$, the lagged baseline relative entropy decreases by 42.5839%, compared with 43.7427% in the historical same-grid discrete-gradient baseline at the same timestep. The finest lagged result, $\Delta t = 0.0003125$, decreases relative entropy by 43.6546%.

The smallest stored historical discrete-gradient timestep is $\Delta t = 0.0025$; its same-grid relative-entropy decrease is 43.7322%. Comparing this historical 0.0025 endpoint with the finest lagged 0.0003125 endpoint gives discrepancies of 0.177401% in entropy gain, 0.138914% in final production and 0.207378% in active-moment-change magnitude. Both comparisons hold all other grid, domain, initial-condition and coefficient inputs fixed. These cross-method differences do not identify either discrete endpoint with continuum truth. The separate numerical KL contribution to the entropy gain decreases from 3.51384% at the baseline timestep to 0.227342% at the finest; it is not included in the continuous nodal collision production.

Across the completed mirror histories, maximum number, energy and moment-bin errors normalized by the declared global scales are 1.648×10^{-12}, 4.312×10^{-12} and 4.862×10^{-13}. Every stored entropy identity defect is within its recorded residual bound. These bounds certify step budgets, not spatial or temporal truncation error. Sampled meridional constraints improve with spatial order: the mirror ψ^2 drift decreases from 1.363×10^{-10} at order three to 1.594×10^{-12} at order five and roundoff at order seven. This does not exhaust the disconnected Cartesian-cap population invariants or prove continuum mixing.

Accepted-step walls are observed costs of distinct shared Linux CI jobs, including each first-step compilation and excluding setup and separate initial/final diagnostics. The baseline four steps take 369.79 seconds; the finest timestep's 64 steps take 1456.37 seconds. These costs are not a matched-error speedup comparison with the historical method. Stored grid array bytes exclude other workspace and line factors; process RSS is a high-water mark including compiled runtime. No dipole or nonaxisymmetric campaign conclusion is inferred from these mirror results.

The public mirror audit retains the original per-case provenance and recomputes all eight comparisons, including every adjacent timestep difference and the coarse-to-finest bias.

A retained dipole solver failure. At the same frozen campaign revision `3d842bee`, the dipole $5^3 \times 25 \times 13$ case with $U = 4$, $M = 20$ and $\Delta t = 0.005$ accepts one step, then rejects a PCG correction. The native convergence status accompanies a recomputed relative residual 3.118×10^{-4}, above both the requested 10^{-6} and the unchanged 5×10^{-6} acceptance guard. The accepted first step uses seven Newton corrections and 7104 total PCG iterations, preserves every recorded moment-bin population, and has minimum nodal density 5.435×10^{-232}. Its entropy increase 6.9723×10^{-5} splits into frozen-mobility production 6.3797×10^{-5} and numerical KL 5.9261×10^{-6}. The full correction fractions and large negative log increments are retained in the public aggregate's partial-case record. The extreme evolved tails make finite-precision conditioning a concern. In particular, the next old-population norm divides each residual component by the square root of that evolved nodal population. A neighbouring weak-form flux can remain much larger than a tiny target population, so this scaling can strongly amplify small absolute equation errors. This is separate from the observed gap between native recursive and recomputed linear residuals; neither observation proves an unattainable precision floor or failure of the positive exact discrete root. A positive exact root does not guarantee convergence in binary64, and a different residual metric requires its own validation. This failure is not a physical decay measurement or a converged dipole refinement. No clipping, kernel change or relaxed tolerance is used to turn it into a passing result.

Fixed-reference pilot, kept separate from refinement. The old-population campaign was stopped after the scale-amplification reproduction, preserving sixteen completed cases and thirteen partial records. The replacement uses $n^{\mathrm{r}} = wf(0)$ without changing the equation or its physical inputs. Producer `3eb6a270` first runs one $n_\mu = 13$ case in each field before any full refinement campaign. Its completed mirror case has $5^3 \times 25 \times 13 = 40625$ nodes, $u \in [-4, 4]$, $\mu \in [0, 20]$, $D = 0.1$, $T = 0.02$ and $\Delta t = 0.005$, all normalized. Four accepted steps use twelve Newton corrections and 191 PCG iterations. Summed step wall time, including compilation and recorded diagnostics, is 74.1421 seconds on a four-logical-CPU AMD EPYC 9V45 Linux runner; cumulative process peak RSS is 874364928 bytes. Setup is separate, and these are not matched-run speedups.

Saved initial, previous and final densities are in `results/nonuniform_fixed_reference/`. An independent analytic-field NumPy pair assembly checks the final root with reference residual 7.09×10^{-16} and entropy-budget defect 1.62×10^{-17}. Independent population audits put cumulative number, energy and full moment-marginal errors below 2×10^{-15}. The entropy gain per particle is 0.00106530145521; selected bulk observables agree with the archived same-grid mirror case to rounding. Intermediate states have producer checks, not an independent full-pair replay. All three requested fields subsequently completed. A separate three-field spatial-block pilot at source `93bf752d` uses the same physical inputs and residual metric; its evidence is not merged into the line-preconditioned record. Its completed mirror case uses 156 PCG iterations and 110.54 seconds of summed accepted-step wall time on a four-logical-CPU AMD EPYC 7763 runner, with cumulative peak RSS 911200256 bytes. The single spatial factor occupies 40625000 bytes. These different-runner times do not establish a speedup. A separate final-step NumPy pair check gives residual 6.72×10^{-16}; selected bulk differences from the line pilot are at most 2.15×10^{-15} relatively. Weighted log endpoint difference is 1.54×10^{-15}, but the largest raw log difference is 7.69×10^{-8} in tiny tails: weighted agreement does not certify uniform pointwise relative accuracy. Both pilots now contain all three requested cases. Neither pilot is a continuum convergence result; the chosen numerical norm supplies no physical rate calibration.

The difficult-field histories contain four accepted steps each. Their measured costs are listed below; setup is excluded from accepted-step wall time and compilation is included. Different runners prevent a controlled speedup claim.

Auxiliary	Field	Newton	PCG	Accepted-step seconds
Line	dipole	60	22036	9705.441
Line	nonaxisymmetric	82	29582	17801.107
Spatial	dipole	60	16667	7912.900
Spatial	nonaxisymmetric	82	20009	9397.549

Each grid occupies 9431600 bytes; a spatial factor occupies 40625000 bytes, while process peak RSS across the six pilots is 874364928–946536448 bytes. Thus factor storage is not a measurement of total memory. Producer versions, hardware and provenance remain attached to every row.

Independent analytic-vacuum NumPy unordered-pair checks of the four new final steps give fixed-reference residuals 1.08–1.32 $\times 10^{-14}$. The maximum independently recomputed cumulative number, energy and globally normalized moment-marginal errors are 6.65×10^{-18}; separately, the maximum per-bin relative moment drift is 1.57×10^{-12}. The entropy-budget identity, including its residual contraction, closes within 7.95×10^{-21}. This independently audits the final saved steps, not all intermediate pair actions. The four pair evaluations take 3.18 seconds including their setup/checks on the shared Apple M4; their metadata distinguish postprocessing hardware from trajectory hardware.

The two dipole endpoints have weighted log difference 1.24×10^{-14}; for the perturbed field it is 1.50×10^{-12}. The largest raw log difference is respectively 1.25×10^{-4} and 0.0874, at reference population fractions 1.55×10^{-23} and 6.95×10^{-25}. This explicitly limits the agreement claim: bulk-weighted agreement does not certify uniform relative tail accuracy. The old-population residual metric can remain enormous; it is reported separately and is not used as an alternative passing criterion. Full saved inputs, states and independent receipts are in the two pilot result directories, with the cross-method comparison in `results/nonuniform_spatial_blocks/preconditioner_comparison.json`.

A separate bounded full-refinement attempt uses immutable producer `93bf752d` and workflow run `37803682825`. It retains its own artifacts and case ledger; pilot and full-campaign timings are not pooled. The current saved snapshot has three completed cases, three explicit failed partials and 42 missing final cases out of 45; a partial case also counts among the missing final cases. It does not establish full refinement convergence. The independent last-step pair checks and publication receipt are in `results/nonuniform_spatial_blocks_full/`. The 330-minute per-case execution bound is a feasibility limit, not a promise that every case will converge or finish. All partial/failure evidence must remain visible when interpreting final refinement coverage.

One difficult correction at matched error. Example 23 freezes a first-step dipole Newton iterate on the $5^3 \times 25 \times 13$ grid with $D = 0.1$ and $\Delta t = 0.005$. The two auxiliaries solve exactly the same original Hessian and right-hand side, without changing the old-population metric. If $s_i = \sqrt{w_i f_i^n}$, the measured correction error is

$$\epsilon_p = \frac{\|s\,(p - p_{\mathrm{ref}})\|_2}{\|s\,p_{\mathrm{ref}}\|_2}.$$

Tightened references at linear tolerances 10^{-9} and 10^{-11} require 429 and 540 iterations and differ by 2.53×10^{-12} in this norm. A separate analytic-dipole NumPy unordered-pair Hessian evaluation checks the fine reference residual at 2.08×10^{-11}. These supply empirical finite-dimensional reference accuracy, not a rigorous forward-error or pointwise tail bound.

Each route executes an excluded, checked runtime warm-up after explicit compilation, followed by three alternating paired repetitions. All measured corrections pass the same 10^{-7} error target and the unchanged native and fresh true-residual checks. On the recorded Apple M4:

Auxiliary	PCG	Median seconds	Maximum ϵ_p	Temporary MB
Line	1100	210.112	3.072×10^{-8}	163.821
Spatial	247	51.729	1.170×10^{-8}	241.821

The three line times are $171.94, 259.72, 210.11$ seconds and the paired spatial times are $43.60, 64.42, 51.73$ seconds. The ten-logical-CPU machine has one-minute load readings ranging from 12.32 to 32.40 at recorded timing boundaries. These are measured shared-machine costs, with no selected repeat or isolated/general speedup claim. The temporary-memory figures are XLA compiled buffer analyses in decimal MB; individual factors occupy 1.625 and 40.625 MB. Measured whole-process peak RSS is 900.60 MB, cumulative across both routes and references, so it is not attributed to either route alone. Warm kernel times include auxiliary factorization; preparation, compilation and excluded warm-up costs are recorded separately. The complete checks take 1187.05 seconds. The frozen producer is `58cc3e1`; fixture provenance, reference vectors, every timing and hardware/version record are in `results/difficult_correction/`. This is one linear correction, not an accepted trajectory or a nonuniform continuum-convergence benchmark.

A reproduced floating-point tail barrier. The full campaign's dipole $\Delta t = 0.0003125$ case accepts its first step with residual 2.54×10^{-14}, then rejects step two at Newton correction 18. The last inner true residual 8.19×10^{-7} passes; the nonlinear residual remains 2.886. Its saved first-step minimum density is 3.55×10^{-199}. The remote artifact does not contain every failed trial, so its particular guard cannot be reconstructed from that checkpoint alone.

An unchanged-equation, instrumented Apple M4 replay from the exact saved first-step state fails at correction 19 after 474.99 seconds. Its final 30 trial fractions produce four log-range rejections and 26 population rejections, with no objective test reached. At the last fraction 2^{-29}, the minimum trial log density is -708.396418985, just below $\log(\text{tiny}) = -708.396418532$, where "tiny" is the smallest normal binary64 number. NumPy gives a positive subnormal exponential $2.22507285 \times 10^{-308}$ and weighted population $7.550768 \times 10^{-314}$, while the actual JAX exponential is zero. The unchanged positivity guard therefore rejects the trial. An independent stable objective evaluation instead gives $\Delta\Phi = -5.63 \times 10^{-17}$, satisfying the Armijo right-hand side -5.63×10^{-21}. This replay localizes exhaustion to tail representability on the selected Newton ray, not PCG or non-descent.

The next previously untried fraction 2^{-30} has a minimum log value above the normal threshold. A subsequent direct check with the actual JAX population guard and objective accepts this trial, but its residual reduction is negligible. A separate continuation from the saved failed iterate permits 60 backtracks, retaining the same old state, reference population, mobility, PCG and nonlinear tolerances. Its declared 300-second bound expires during correction 40 after 21 accepted scalar updates. The reference residual changes from 2.887412924305 to 2.887412921250, a relative reduction of only 1.06×10^{-9}, while the required residual is 10^{-12}. Late fractions are 2^{-54}; from correction 30 through 40 only two represented log-density entries change, by at most 7.11×10^{-14}, with bitwise-flat recorded residual. These are recorded Armijo acceptances, not an independent proof of descent for the rounded represented update: at correction 39 the ideal ray's first-order decrement is about -1.68×10^{-24}, whereas direct contraction with the actual rounded update gives about -3.73×10^{-32}. The latter check does not assemble the quadratic term. No physical second step is accepted. Thus increasing the trial budget alone has not resolved this case; the bounded experiment does not prove that every possible continuation fails or that the desired root is unrepresentable. All inputs, final arrays, trials and the explicit continuation-only diff are retained in `results/nonuniform_spatial_blocks_full/dipole_dt16_replay/continuation60/`. The M4 replay is not bitwise identical to the Linux producer. The public solver now records counts of log-range, positive-population and Armijo rejections, together with

the last trial values, on exhaustion. Its 30 trial fractions and successful-step arithmetic are unchanged. A focused synthetic convex guard test verifies visible rejection without an accepted state; six successful routes retain bitwise-identical states and non-timing diagnostics. All 247 regression tests pass. No density floor, invariant projection, tolerance relaxation or SOLVAX change follows from this diagnosis. The independently failed $\Delta t = 0.000625$ case also rejects step two after an accepted first step; the fine timestep comparison remains unavailable. All partials and their producer/source hashes remain distinct from completed refinement cases.

An elementary obstruction to a relative maximum principle. Let the finite collision matrix K be symmetric with $K\mathbf{1} = 0$ and $KE = 0$. If all off-diagonal entries satisfy $K_{ij} \leq 0$, then

$$E^T K E = \sum_{i<j} (-K_{ij})(E_i - E_j)^2 = 0.$$

Every nonzero off-diagonal edge must therefore join equal discrete energies. Applying the same identity to each null moment-bin indicator forces every edge to stay within its moment-bin partition as well. Thus conservative coupling between unequal energies cannot have this graph-Laplacian sign pattern. Positive semidefiniteness of a weak Gram matrix does not impose it.

This sign condition would imply a relative maximum principle for the lagged entropy equation. Write $f = f_\star e^s$, where the positive stationary reference's log is a linear combination of represented invariants, so $K \ln f_\star = 0$ for every frozen mobility, and $n_{\star i} = w_i f_{\star i}$. The equation is $n_{\star i}(e^{s_{\mathrm{new,i}}} - e^{s_{\mathrm{old,i}}}) + \Delta t (K s_{\mathrm{new}})_i = 0$. At a maximum of s_{new} the graph sign gives $(K s_{\mathrm{new}})_i \geq 0$, and at a minimum it gives the opposite inequality, bounding the new ratios by their initial envelope. Conversely, if the matrix frozen at the stationary population has $K_{ij} > 0$, its linearized resolvent satisfies $[(M_\star + \Delta t K)^{-1} M_\star]_{ij} = -\Delta t K_{ij}/n_{\star i} + O(\Delta t^2) < 0$ for sufficiently small positive Δt, with $M_\star = \mathrm{diag}(n_\star)$. It is not order preserving. Conservation, entropy increase and positive nodal populations consequently do not by themselves imply initial bounds on $f/f_\star$. For a rare node, a large negative log change δ_i contributes only $n_{\mathrm{old},i}(e^{\delta_i} - 1 - \delta_i) \sim n_{\mathrm{old},i}|\delta_i|$ to the numerical KL budget; weighted control supplies no useful mass-independent pointwise bound. This is a finite-dimensional structural statement, not an attribution to spatial discretization, a physical positivity failure, or a diagnosis of the particular dipole tail magnitude above.

Solver-tolerance audit on one fixed mirror grid. Example 19 at source revision `e1bc1ab` retains every state in five four-step runs on $3^3 \times 25 \times 21$ nodes, with $D = 0.1$, $T = 0.02$ and $\Delta t = 0.005$. Outer Newton tolerances $10^{-10}, 10^{-11}, 10^{-12}$ at inner tolerance 10^{-6} give byte-identical endpoints; this is not an outer-error-order estimate. At outer tolerance 10^{-12}, the separate inner sequence is $10^{-4}, 10^{-5}, 10^{-6}$. An independent NumPy saved-state audit reconstructs all 20 nonlinear residuals and entropy/invariant budgets, with explicit unordered-pair production checks. The maximum independently recomputed nonlinear relative residual is 2.162×10^{-13}; maximum cumulative number, energy and full moment-bin errors are respectively 1.019×10^{-14}, 1.579×10^{-14} and 4.135×10^{-15}. The last inner pair has entropy-scaled log endpoint difference 1.393×10^{-14} and maximum independently recomputed relative change in the three principal observables 5.27×10^{-13}, well below the unchanged 10^{-8} endpoint and 10^{-6} observable targets. The fixed finite-dimensional reference is not continuum truth. Krylov correction vectors are not archived, so their true residual scalars are checked as producer records rather than independently replayed vectors. The five recorded run costs sum to 89.698 seconds on the Apple M4; the separate 3.670-second setup/initial-diagnostic cost is excluded from that sum. These observed costs retain first-use compilation, evidence writes and concurrent load, and supply no speedup claim. Exact normalized inputs, versions and hardware are retained in the producing summary.

Bounded Maxwellian encounter flux and thermal center motion

Let $\theta = k_B T/m$ denote a single-particle velocity variance. For two equal Maxwellian particles, $\boldsymbol{w} = \boldsymbol{v}_1 - \boldsymbol{v}_2$ and $\boldsymbol{V} = (\boldsymbol{v}_1 + \boldsymbol{v}_2)/2$ are independent, with covariances $2\theta I$ and $\theta I/2$. The transformation has absolute Jacobian one. The relative distribution and equations for a screened pair potential are

$$F_{\mathrm{rel}}(\boldsymbol{w}) = (4\pi\theta)^{-3/2} e^{-|\boldsymbol{w}|^2/(4\theta)}, \tag{126}$$

$$\dot{\boldsymbol{r}} = \boldsymbol{w}, \qquad \dot{\boldsymbol{w}} = \Omega \boldsymbol{w} \times \boldsymbol{e}_z + \frac{2\kappa}{m} \frac{e^{-r/L_s}(1 + r/L_s)}{r^3} \boldsymbol{r}, \qquad \Omega = qB/m. \tag{127}$$

The conserved relative energy is $m|\boldsymbol{w}|^2/4 + \kappa e^{-r/L_s}/r$; center motion follows an independent free helix. The new six-dimensional relative integrator is checked against the independent twelve-dimensional pair integrator, including nonunit masses and both charge signs.

For guiding-center impact b, the translation to the incoming perpendicular position has unit Jacobian. Axisymmetry integrates its orientation to $2\pi b\,\mathrm{d}b$. On a plane normal to the field, the two-sided incoming crossing measure per partner density is

$$\frac{\mathrm{d}\nu}{n} = 2(2\pi b\,\mathrm{d}b)\, w_\parallel F_{\mathrm{rel}}(\boldsymbol{w}) w_\perp\, \mathrm{d}w_\perp\, \mathrm{d}w_\parallel\, \mathrm{d}\varphi, \qquad w_\parallel > 0. \tag{128}$$

The factor two includes the opposite incoming direction. There is no unordered-pair half factor for this tagged-particle flux, and the plane flux uses $|w_\parallel|$, not the cross-section flux $|\boldsymbol{w}|$. For impact area $A = \pi(b_{\max}^2 - b_{\min}^2)$, the bounded integral is $2A(2\pi)J_\parallel J_\perp/(4\pi\theta)^{3/2}$, where $J(a,b) = 2\theta[e^{-a^2/(4\theta)} - e^{-b^2/(4\theta)}]$. This supplies an independent analytic normalization.

Undo outgoing free gyromotion when defining $\Delta\boldsymbol{w}_\perp = \overline{\boldsymbol{w}}_{\mathrm{out},\perp} - \boldsymbol{w}_{\mathrm{in},\perp}$. With $c = m\Delta|\boldsymbol{w}_\perp|^2/(8B)$, the individual changes are $\Delta\mu_{1,2} = c \pm [m/(2B)]\boldsymbol{V}_\perp \cdot \Delta\boldsymbol{w}_\perp$. Exact Gaussian center averaging therefore gives

$$\langle \Delta\mu_1 \rangle_V = c, \tag{129}$$

$$\langle (\Delta\mu_1)^2 \rangle_V = c^2 + \sigma_c^2, \qquad \langle \Delta\mu_1 \Delta\mu_2 \rangle_V = c^2 - \sigma_c^2, \tag{130}$$

$$\sigma_c^2 = \frac{m^2\theta}{8B^2} |\Delta\boldsymbol{w}_\perp|^2. \tag{131}$$

A separate Gaussian quadrature of the actual individual initial/final velocities verifies these raw moments. A zero-center trajectory omits σ_c^2. The drift and diffusion moment integrals are $A_\mu/n = \int \langle \Delta\mu \rangle_V \mathrm{d}\nu/n$ and $D_{\mu\mu}/n = \frac{1}{2}\int \langle (\Delta\mu)^2 \rangle_V \mathrm{d}\nu/n$; the latter half is the second-moment convention, not pair counting.

The executed normalized case uses $m = q = 1$, $B = 2$, $\kappa = 0.03$, $L_s = 3$, $\theta = 0.5$, $b \in [1.2, 2.4]$, $w_\parallel \in [0.7, 1.5]$ and $w_\perp \in [0.2, 0.8]$. The $4^3 \times 16$ base quadrature is independently refined in each impact/speed coordinate through orders $3, 4, 5$, in gyrophase through $8, 16, 32$, and by a half-grid phase shift. Moving the planes through $L = 18, 24, 32$ preserves the same incoming free helix. The first outward exit is recorded; reflected trajectories are retained, and failure to exit within $6L/w_\parallel$ raises. Eight held-out states additionally check active maximum steps $0.2, 0.1, 0.05$. Direct quadrature changes are below 0.021%; analytic flux normalization error is 1.62×10^{-9}. The bounded results are $\nu/n = 1.2601061$, $A_\mu/n = 3.11546 \times 10^{-6}$ and $D_{\mu\mu}/n = 8.00765 \times 10^{-7}$ in the stated normalized units.

A supplementary multilinear scattering table is tested on 64 uniformly sampled states in the inner quadrature-node hull (seed 701). Its held-out second-moment relative RMS error is 20.5%, exceeding the predeclared 10% target; this surrogate remains unresolved despite converged direct integrals. These speed bands contain only 11.6% of the full thermal incoming flux in the same annulus. Omitted impact/speed regions and successive encounter correlations remain untested. Truncation can exclude reverse encounters, so the positive conditional drift is not a full Maxwellian-plasma drift. Neither these bounded crossing coefficients nor the interpolation checks calibrate the constrained scalar D or establish a dipole lifetime.

Physical applicability: flux, screening and individual moments

The crossing coefficients above have units $[\nu/n] = L^3/T$, $[A_\mu/n] = \mu L^3/T$, and $[D_{\mu\mu}/n] = \mu^2 L^3/T$. They are conditional on the stated potential and incident domain. Integrating the same plane measure over all relative velocities gives

$$\frac{\nu_{\text{plane}}}{n} = \pi(b_{\max}^2 - b_{\min}^2)\,\mathbb{E}|w_\parallel| = 2\pi(b_{\max}^2 - b_{\min}^2)\sqrt{\theta/\pi}. \tag{132}$$

For the original annulus this is 10.82863415, whereas the narrow speed bands give 1.260106087, or 11.6368% of the crossing flux. This normalization is a tagged-particle crossing frequency. An unordered pair count per volume has a separate half factor; a second-moment diffusion coefficient also has its own half factor. Neither changes (132). A finite guiding-center annulus is not a complete collision volume in a general finite-field kinetic theory: side-surface contributions can matter. The surface-flux formulation and its cylindrical extreme-field limit are treated explicitly in JB20, Eqs. (5), (16)–(17) in the inspected arXiv version [20].

For an equilibrium Maxwellian plasma, the complete incident measure and outgoing map must preserve the equilibrium pair distribution and include inverse states. At a finite plane, the canonical equilibrium pair weight contains $\exp[-V(r)/(m\theta)]$ in addition to the velocity Maxwellians; using asymptotically free data instead requires receding-plane convergence. Hamiltonian phase-volume preservation and pair energy conservation then imply a zero net equilibrium mean individual magnetic-moment increment. This is a stationarity statement and does not require fixed-B microscopic time reversal. Nonzero second moments can coexist with this zero drift. A truncated rectangle of impact parameters and speeds need not be invariant under the outgoing map, so its signed conditional drift cannot test the full equilibrium balance. Nor does zero equilibrium drift imply preservation of an arbitrary nonequilibrium $G(\mu)$.

If the supplied Yukawa length L_s is interpreted as the classical one-component Debye length, $\kappa = q^2/(4\pi\epsilon_0)$ and $\theta = k_B T/m$ require

$$L_s^2 = \frac{m\theta}{4\pi n\kappa}, \qquad a = \left(\frac{3}{4\pi n}\right)^{1/3}, \qquad \Gamma = \frac{\kappa}{am\theta}, \qquad N_D = \frac{4\pi n L_s^3}{3}, \qquad \omega_p^2 = \frac{4\pi n\kappa}{m}. \tag{133}$$

These equations constrain density; it cannot be chosen independently to assign a physical lifetime while retaining the same Debye potential. For $m = 1$, $\theta = 0.5$, $L_s = 3$, $\kappa = 0.03$, they give $n = 0.147365688$, $\Gamma = 0.0510873$, $N_D = 16.6667$ and $|\Omega|/\omega_p = 8.48528$ at $B = 2$. These are consistency diagnostics under a hypothetical Debye interpretation, not a measured density or a validated many-body closure. A multicomponent plasma requires the screening contributions of all species. Static screening, dynamic screening and a potential of mean force are distinct approximations [22, 23, 20]. A weakly coupled Debye sphere contains many particles; isolated two-body trajectories alone cannot test correlations or repeated encounters.

Use the single-particle rms perpendicular radius $\rho_{\text{rms}} = \sqrt{2\theta}/|\Omega|$, the relative radius $\rho_{\text{rel}} = w_\perp/|\Omega|$, the Coulomb reference $b_{90} = 2\kappa/(m|\boldsymbol{w}|^2)$, and an actual sampled closest separation. The literature also uses $r_c = \sqrt{\theta}/|\Omega|$; these conventions must not be mixed. A field uniform on the system scale makes $\rho/L_B = 0$ here, but supplies no close-collision adiabatic estimate. At the original parameters $\rho_{\text{rms}} = 0.5$ and $b_{90} = 0.02$ at rms relative speed $\sqrt{6\theta}$, hence $\rho_{\text{rms}}/b_{90} = 25$. In the bounded band, $b - \rho_{\text{rel}} \geq 0.8$, but the indicative lower bound $|\Omega|(b - \rho_{\text{rel}})/w_\parallel \geq 1.0667$ is not an asymptotically large encounter-time ratio. Small force increments on this safe band and adiabatic protection of close collisions are different tests. The hierarchy $r_c < L_s$ alone places an encounter in a magnetized screening regime; it does not enforce $r_c \ll b_{90}$ [20].

A screened equal-frequency exchange diagnostic. Individual moments require care even if the relative perpendicular action is approximately preserved. This distinction is already present in

NZ09: its Eq. (14) gives individual energy exchange through $\Delta(\boldsymbol{V}\cdot\boldsymbol{w})$ while the total relative and center energies are separately conserved for identical particles [22]. Its Eq. (23) develops the first-order free-helix response; the second-order energy and squared-transfer quantities in Eqs. (27)–(30), (42)–(46) are established antecedents. The following calculation specializes that mechanism to the present singular repulsive Yukawa potential away from close encounters. It is an audit of the approximation, not a novelty claim.

Let $J\boldsymbol{v}=\boldsymbol{v}\times\boldsymbol{e}_z$ on the perpendicular plane, so $J^2=-I$, and write $\boldsymbol{f}=-(2/m)\nabla V$ for the relative acceleration. Then

$$\boldsymbol{R}_{\mathrm{rel},\perp}=\boldsymbol{r}_\perp+J\boldsymbol{w}_\perp/\Omega,\qquad \dot{\boldsymbol{R}}_{\mathrm{rel},\perp}=J\boldsymbol{f}_\perp/\Omega. \tag{134}$$

For a weak interaction, nonzero fixed guiding-center impact b, and large $|\Omega|$ with $w_\perp/(|\Omega|b)\ll 1$, expand the force about $\boldsymbol{r}_\perp=\boldsymbol{b}-J\boldsymbol{w}_\perp/\Omega$, $z=w_\parallel t$. The rotating-frame velocity receives a gyroaveraged phase increment

$$\Delta\overline{\boldsymbol{w}}_\perp\simeq\alpha J\boldsymbol{w}_{\mathrm{in},\perp},\qquad \alpha=\frac{1}{m\Omega w_\parallel}\int_{-\infty}^{\infty}\Delta_\perp V(b,z)\,\mathrm{d}z. \tag{135}$$

Indeed the conjugated perpendicular Hessian has gyroaverage $(\operatorname{tr}H_\perp/2)I$; the factor two in the relative force cancels this half. For Yukawa screening and $b>0$,

$$\int_{-\infty}^{\infty}V(b,z)\,\mathrm{d}z=2\kappa K_0(b/L_s), \tag{136}$$

$$\alpha=\frac{2\kappa}{m\Omega w_\parallel L_s^2}K_0(b/L_s), \tag{137}$$

$$|\Delta\boldsymbol{R}_{\mathrm{rel},\perp}|\simeq\frac{4\kappa}{m|\Omega|w_\parallel L_s}K_1(b/L_s). \tag{138}$$

Here K_j are modified Bessel functions. The leading phase term is specifically screened: in the unscreened Coulomb limit its transverse Hessian integral is zero for $b>0$, while the guiding-center kick tends to $4\kappa/(m|\Omega|w_\parallel b)$. Finite-$B$ oscillatory response, finite endpoints, and nonlinear parallel deceleration remain separate corrections; a screening scan and a force-strength scan are needed before assigning their relative sizes. Direct scalar quadrature of the Yukawa Hessian and radial force at four impacts and $L_s=3,6,\infty$ agrees with these integral identities with maximum absolute discrepancy 2.34×10^{-18} in the declared normalized test; it is not an ODE or ensemble-convergence test.

With the thermal scale $\mu_{\mathrm{ref}}=m\theta/B$, the exact center-average second moment has leading contribution $\sigma_c^2/\mu_{\mathrm{ref}}^2=\alpha^2w_\perp^2/(8\theta)$. The individual spatial kick is one half the relative kick. Define

$$r_\mu=\frac{2D_{\mu\mu}}{\mu_{\mathrm{ref}}^2},\qquad r_R=\frac{\int|\Delta\boldsymbol{R}_{\mathrm{rel},\perp}|^2\,\mathrm{d}\nu}{4L_s^2}. \tag{139}$$

The spatial quantity is a fixed-distance kick proxy; it is not a certified mixing eigenrate. At each incident state the leading integrands obey

$$\frac{r_\mu\text{ integrand}}{r_R\text{ integrand}}\simeq\frac{w_\perp^2}{8\theta}\left[\frac{K_0(b/L_s)}{K_1(b/L_s)}\right]^2. \tag{140}$$

Both normalized rates scale as Ω^{-2} on a fixed, screened weak-force band. Increasing B alone therefore does not make this particular ratio vanish. Its size depends on the speed band, impact band and stated spatial scale. This is not an impossibility theorem for time separation in a physical plasma. In particular the Coulomb cancellation above and the omitted parts of the incident ensemble prevent that extrapolation.

Nonzero thermal-center second moments do not alone prove that the Maxwellian moment marginal changes. A rotation of relative perpendicular velocity in a rotationally invariant Gaussian ensemble leaves

equilibrium stationary. A nonequilibrium diagnostic is a perpendicular ring with $|\boldsymbol{v}_{1\perp}| = |\boldsymbol{v}_{2\perp}| = v_0$, independent initial angles, and $\mu_1 = \mu_2 = \mu_0$. Here $\boldsymbol{V}_\perp \cdot \boldsymbol{w}_\perp = 0$, but a small relative phase rotation gives

$$\Delta\mu_1 = -\Delta\mu_2 \simeq \frac{m\alpha}{2B}\boldsymbol{V}_\perp \cdot J\boldsymbol{w}_\perp, \qquad \Delta(\mu_1^2 + \mu_2^2) = 2(\Delta\mu_1)^2, \tag{141}$$

where $|\boldsymbol{V}_\perp \cdot J\boldsymbol{w}_\perp| = v_0^2|\sin(\varphi_1 - \varphi_2)|$. A smooth narrow ring approximates this limiting distribution. Equation (141) describes the leading pure-phase approximation. For direct finite-field trajectories, write $\delta_i = \mu_{i,\mathrm{out}} - \mu_0$ and retain the exact identity

$$\Delta(\mu_1^2 + \mu_2^2) = 2\mu_0\Delta(\mu_1 + \mu_2) + \delta_1^2 + \delta_2^2. \tag{142}$$

The signed first term is a change of pair-total moment; the positive second term includes individual exchange. The centered outgoing spread is $(\delta_1 - \delta_2)^2/2$ and need not equal either full second-moment change. Thus conservation of pair-total moment and relative perpendicular energy can coexist with broadening of the individual moment marginal. The constrained closure conserves every represented $G(\mu)$ exactly. Its physical applicability to such a nonequilibrium population therefore needs a distribution-level error estimate over its actual mixing time, including the signed drift and correlations, rather than a zero equilibrium mean or a small per-encounter increment.

Acceptance and remaining physical scope. An independently validated free-helix force response can support a weak, bounded screened-scattering claim. It must predict held-out trajectories or weighted observables, and the exact moment integrals must pass impact, speed, phase, endpoint and timestep comparisons. This does not calibrate the interaction tensor in source Eq. (119), its scalar/local reduction, or the prescribed D in Eq. (181). A physical constrained interval further needs a reverse-complete population model, an interaction potential tied to stated plasma parameters, coefficient/correlation evidence for that population, and independently measured $\tau_{\mathrm{mix}} \ll \tau_\mu$. None of those requirements is replaced by a chosen numerical density.

Direct primary comparisons are available beyond binary perturbation theory. JWTT25, Eqs. (6), (10)–(11), computes anisotropic energy-transfer coefficients by integrating full two-particle trajectories with a potential of mean force and perturbed velocity distributions [25]. JLB25 compares a generalized collision theory for one-component anisotropy relaxation with experiments and molecular dynamics [26]. WJTB26, inspected arXiv version, Sec. IV and Eqs. (15)–(22), explicitly introduces a memory kernel and a matrix of temperature correlations before extracting long-time relaxation coefficients [27]. These references provide benchmarks for a future plasma-rate calculation. Their demonstrated suppression of temperature anisotropy relaxation does not establish conservation of the full individual magnetic-moment marginal for the present local scalar surrogate. The present outcome can be a supported failure of a selected applicability test while leaving stronger or different physical regimes open.

Completed screened validation and its failed gates. Example 34, producer `57a8d4e`, keeps the original band, $\theta = 0.5$, $\kappa = 0.03$, $L_s = 3$ and $m = q = 1$. It completes 21,524 distinct relative encounters, retaining their original weights, and scans $B = 2, 4, 8, 16, 32$ with $4^3 \times 16$ nodes. Each field has a $3^3 \times 16$ coordinate comparison, an eight-phase comparison and a half-grid phase shift. At $B = 2$, a separate $5^3 \times 32$ calculation, planes $L = 18, 24, 32$, sixteen held-out active maximum-step controls $0.2, 0.1, 0.05$, and three Cartesian pair controls test different errors. The criterion is 1% for moment and spatial-kick quadrature comparisons; the signed mean is scaled by M_2/μ_{ref}, and the cross moment by M_2, rather than by a nearly zero signed result. The active step target is 0.1%, and the Cartesian position and velocity targets are each 10^{-8}.

Every planned trajectory exits, the runtime cap is not exhausted, and the base energy errors are below 1.93×10^{-11}. Eleven of the thirteen required gates pass, including the incoming-flux normalization,

receding planes, $B = 2$ final refinement, active timestep, weak-force response, ring controls, Cartesian trajectories and full-moment consistency. The experiment nevertheless exits nonzero and remains unresolved. At $B = 4$, the coordinate comparison changes the second and cross moments by 1.93161% and 1.41436%, exceeding the unchanged 1% target. Phase comparisons pass at every field; they cannot replace this failed coordinate comparison. The $B = 2$ final comparison changes the second moment by 0.014625%, and moving $L = 24 \to 32$ changes it by 0.000358%. Active step halving changes the scaled moment vector by at most 4.35×10^{-9}, with all actual maximum-step ratios equal to 1/2. Cartesian position and velocity discrepancies are below 1.11×10^{-13} and 2.78×10^{-14} respectively.

The conditional $5^3 \times 32$ interpolation table is evaluated on 128 held-out states, seed 3401, within its impact/speed node hull and over the full periodic phase. Its second-moment relative RMS error is 0.35466%, below the 2% target, but its 95th-percentile pointwise relative error is 7.89330%, above the 5% target. This is the second failed gate. A passing bulk RMS does not certify this tail target. The direct $B = 2$ integral is separately resolved at the declared comparisons, with $A_\mu/n = 3.115997 \times 10^{-6}$ and $D_{\mu\mu}/n = 8.008824 \times 10^{-7}$ on the final grid. This signed conditional drift still does not test equilibrium balance or a complete velocity ensemble.

An independently integrated first-order free-helix force response tests the same 128 states at $B = 2$ and weaker strengths $\kappa = 0.003, 0.0015$. At the first strength, second/cross-moment RMS discrepancies are 0.61877%/0.91470%; at half strength they are 0.30830%/0.45726%. All lie below 3%, and the independent $256 \to 512$ force-quadrature changes are below 0.065%, against a 0.1% target. The approximately halved discrepancies support the force expansion on these held-out states. First-order phase-averaged drift vanishes by symmetry, so this check does not predict the exact ensemble drift. Nor does weak-force accuracy establish the original $\kappa = 0.03$ amplitude at high B.

B	r_μ/n	r_R/n	r_μ/r_R
2	2.56245×10^{-5}	2.55222×10^{-5}	1.00401
4	6.19780×10^{-7}	6.36900×10^{-6}	0.097312
8	4.53882×10^{-8}	1.59308×10^{-6}	0.028491
16	1.12629×10^{-8}	3.98322×10^{-7}	0.028276
32	2.81494×10^{-9}	9.95836×10^{-8}	0.028267

These are the fixed $4^3 \times 16$ sampled rates of Eq. (139); the $B = 4$ row retains its failed coordinate refinement. The $B = 16 \to 32$ exponents are 2.00040 and 1.99995 for the moment and spatial rates. The ratio consequently approaches a nonzero value on this selected screened band. At $B = 32$, the leading weak-force formulas instead give 2.61834×10^{-9} and 9.58363×10^{-8} per density: the direct values exceed them by 7.50889% and 3.91014%. Similar offsets already occur at $B = 16$. Convergence of the power of B is therefore distinct from convergence to the first-order force amplitude. Nonlinear parallel deceleration and finite endpoints are candidate corrections; this fixed-strength scan does not assign the remaining error between them. A matched high-B strength scan is the next direct test of the force-order interpretation. The separately recorded $L_s = 6, 12, \infty$ controls are analytic leading references only, with the spatial normalization held at length three; there are no direct changed-screening trajectories in this campaign.

At $B = 32$, sampled relative radii satisfy $\rho_{\mathrm{rel}}/L_s \le 0.00790$ and $\rho_{\mathrm{rel}}/(b - \rho_{\mathrm{rel}}) \le 0.01882$; the sampled near-pass time ratios $2|\Omega|(b - \rho_{\mathrm{rel}})/w_\parallel$ range from 55.81 to 195.60. These support a slowly varying, distant encounter interpretation on the selected band. They do not give uniform extreme magnetization of close collisions: the thermal radius to b_{90} ratio still ranges from 0.3277 to 1.3862 on the same nodes, and close impacts are excluded. The crossing measure gives $\nu/(n|\Omega|) = 0.0393783$ at $B = 32$; the actual ordering $\nu/|\Omega| \ll 1$ additionally depends on density. The quantity $r_\mu/(n|\Omega|) = 8.79670 \times 10^{-11}$

describes weighted moment kicks, not the crossing frequency.

The six separate $B = 2$ cutoff probes are coarse conditional integrals, without cutoff refinement or a bound on omitted corners. For example, the inner impact interval $[0.6, 1.2]$ gives a moment rate about 10.47 times the base rate, and the perpendicular-speed interval $[0.8, 1.6]$ about 4.10 times it. These probes show why the original narrow band cannot define a plasma coefficient; they do not establish convergence after adding those intervals.

Direct nonequilibrium ring evidence. The ring uses $v_0 = 1$, $b = 1.8$, $w_\parallel = 1.1$, angle differences $\pi/3, \pi/2, 2\pi/3$ and four common phases $0, 0.8, 1.6, 2.4$. At $B = 16, 32$ all 24 original/refined pairs pass the unchanged 10^{-9} normalized increment target, with an active refined maximum step 0.0025. The largest increment difference divided by μ_{ref} is 1.67×10^{-10}. The full normalized $\Delta(\mu_1^2 + \mu_2^2)$ is positive at every refined state and exceeds ten times its refinement difference. Its ranges are $[1.27951, 1.82407] \times 10^{-7}$ at $B = 16$ and $[3.47073, 4.88297] \times 10^{-8}$ at $B = 32$.

The direct pair-total correction is retained in Eq. (142). At $B = 16$, its normalized contribution $2\mu_0\Delta(\mu_1 + \mu_2)/\mu_{\mathrm{ref}}^2$ ranges from -7.63373×10^{-9} to 2.02416×10^{-9}; at $B = 32$ it ranges from 8.27255×10^{-10} to 4.47729×10^{-9}. The normalized sum of squared individual increments ranges from 1.35545×10^{-7} to 1.80756×10^{-7} and from 3.38789×10^{-8} to 4.51748×10^{-8} respectively. Thus the positive full second-moment change survives the signed total correction; it is not inferred solely from the spread or from an assumed exact pair-total invariant. The sampled outgoing moments differ from the equal incoming ring moments. This supplies a finite-encounter counterexample to exact preservation of every individual $G(\mu)$ by the selected screened model, consistent with the established center/relative exchange mechanism in NZ09. The 12 selected angle pairs are not a converged ring collision integral or a many-body relaxation rate. An approximate constrained interval still requires an error bound for the actual population over an independently measured mixing time.

All primitive encounters, ring states, failed gates, normalized inputs, and software/hardware records are retained under `results/physical_final_campaign/`. The recorded campaign cost is 872.94 seconds on the shared Apple M4. Parent and largest worker process-lifetime memory peaks are 1.0860 and 0.1139 GB separately; they are not a simultaneous aggregate peak. At this stage the failed numerical gates, incident-domain errors, dynamic screening and correlations prevent a calibrated source coefficient or plasma lifetime. The restricted large-B and ring results support a failure of the selected exact moment-preservation claim, while leaving a different population, potential or asymptotic regime open.

Focused refinement of the failed numerical checks. Example 37 is separately declared at producer `04d0bf4`, preserving the original example-34 failure. It completes 18,736 distinct encounters on the same screened band. At $B = 4$ it compares $5^3 \times 16$ and $6^3 \times 16$ grids. The scaled mean, second moment, cross moment and spatial-kick changes are respectively 0.004954%, 0.007741%, 0.005685% and 0.000069%, below the unchanged 1% criterion. The finer moment and spatial rates per density are 6.20658×10^{-7} and 6.36909×10^{-6}. This resolves the selected final coordinate comparison; it does not alter the failed $3^3 \to 4^3$ comparison in the original record.

At $B = 2$, a $7^3 \times 32$ table is tested both on the original 128 held-out states and on 128 fresh states, seed 3701. Each sample lies within the table's impact/speed node hull and covers periodic phase. Second-moment RMS errors are 0.085818% and 0.113739%; the corresponding 95th-percentile pointwise errors are 1.88279% and 2.44985%. Both pass the unchanged 2% RMS and 5% tail targets. These are held-out validation statistics on the declared hull, without a supremum error bound or omitted-domain coverage. The largest individual relative errors remain 96.2132% and 67.2155%. Their direct second moments are respectively 0.0312318% and 0.0213705% of the largest value in each sample. The full

sorted error distributions are retained in the focused figure; passing the RMS and percentile tests does not certify every table query.

The follow-up also repeats the matched $B = 32$, $4^3 \times 16$ band at $\kappa = 0.003$ and 0.0015. The direct moment-rate discrepancies from the screened leading reference fall to 0.70178% and 0.34806%; the spatial-rate discrepancies fall to 0.37402% and 0.18722%. They pass the separately declared 2% amplitude target. Together with the original $\kappa = 0.03$ offsets of 7.50889% and 3.91014%, this reduction supports a finite-force correction to the leading large-field amplitudes. A field scan at fixed force alone had left that question unresolved. These are force-strength controls for moment and spatial second moments; they do not resolve the tiny signed high-field drift or identify a unique correction mechanism. They also do not test the changed-screening or close-impact regimes.

All six follow-up gates pass, including energy, flux and source integrity. The largest recorded energy error is 1.95×10^{-11} and the largest flux normalization error 1.63×10^{-9}, against targets 10^{-9} and 10^{-5} respectively. Outputs are retained under `results/physical_refinement/`. This distinct campaign takes 439.49 seconds on the shared Apple M4, with parent and largest worker process-lifetime memory peaks 1.0248 GB and 0.03044 GB separately. It supplies passing final microscopic checks on this finite screened band; the original campaign remains unresolved in its own record. Successful numerical refinement and weak-force agreement do not supply a reverse-complete plasma population, a collision closure, successive encounter correlations or an independently measured mixing time. Thus the exact ring exchange counterexample and the physical applicability limits remain after the two numerical failures are resolved.

Independent retained-state reconstruction. Example 35, frozen at `e77cff3`, imports no producing scientific kernels. Its separate reports under `results/encounter_audit/` reconstruct the positive crossing quadrature, Gaussian center moments, normalized coefficients, ring budgets, force-response references, periodic interpolation tables and every declared acceptance decision. The original and follow-up maximum row-moment absolute discrepancies are 5.56×10^{-17} and 1.39×10^{-17} respectively; their second/cross-moment discrepancies scaled by the positive second moment are below 7.46×10^{-12} and 5.07×10^{-11}. The original integrated moment-vector discrepancy is below 4.04×10^{-18}. Cancellation-prone signed quantities have separate dimensionless arithmetic checks, including a 10^{-6} positive-moment scale target. These audit tolerances do not change the scientific acceptance targets or replace a failed gate.

For the follow-up, saved positions and velocities additionally reconstruct the relative guiding-center kick, with maximum component discrepancy 4.45×10^{-16}. The original campaign's general rows omit positions, so their stored guiding-center kicks remain primitive inputs: squaring those kicks and weighting them is independently checked, rather than claiming an independent trajectory reconstruction for every row. General-row energy and actual-step diagnostics are retained producer records. Three separately solved Cartesian relative paths, compared with the saved individual-particle paths, have maximum position and velocity differences 1.64×10^{-13} and 1.09×10^{-14}. These controls use DOP853 as the producer does. The reconstructed refined ring second-moment signal is at least 517 times its refinement difference. The audit preserves the original two failed gates and independently confirms all six passing follow-up gates.

The precision report independently checks decimal path serialization, exact binary64 initial inputs, outgoing-event brackets, energy and canonical angular momentum, and recomputes the 50/70-digit comparisons. Across all six saved paths, the maximum relative energy drift is 2.80×10^{-37} and the maximum absolute drift of relative canonical angular momentum is 6.54×10^{-37}. This checks retained high-precision paths; it does not perform a new arbitrary-precision solve or certify a continuous phase integral. Source-byte and input-artifact hashes in the three execution receipts distinguish the actual

producers and the audit. All three independent reports pass. The single audit takes 4.40 seconds with a process-lifetime peak of 0.2691 GB; the three receipt timings belong to this one execution and are not added together or pooled with producer costs.

Broader encounter coverage and microscopic reversal

Example 13 retains the same screened pair model and thermal center average, while widening the tested speed band to $w_\parallel \in [0.35, 3]$, $w_\perp \in [0, 2.5]$ and splitting the impact interval into $[0.15, 0.4]$, $[0.4, 0.8]$, $[0.8, 1.2]$, $[1.2, 2.4]$, $[2.4, 4.8]$ and $[4.8, 8]$. The impact variable b^2 is sampled uniformly; each squared speed is drawn from its truncated exponential incoming-flux law. These inverse cumulative distributions implement the positive crossing measure above. Scrambled Sobol orders $32, 64, 128, 256$ share one scramble; a separate 128-point scramble checks replication. Gyrophase uses 16 equally spaced points and an independent $8 \to 16$ refinement. The two-scramble spread is a finite numerical comparison, not a statistical confidence interval. The speed band retains 88.8641% of the complete thermal crossing flux in the same impact annuli; this flux fraction does not bound the omitted moment contribution.

Microscopic reversal is checked on 16 separate transmitted encounters. Starting at the actual outgoing position, apply parity and time reversal, $(\boldsymbol{r}_R, \boldsymbol{w}_R) = (-\boldsymbol{r}_{\mathrm{out}}, \boldsymbol{w}_{\mathrm{out}})$, and reverse the charge, hence $\Omega_R = -\Omega$. A common axial rotation restores the incoming guiding-center orientation, and the free-helix phase is recovered from that state. The relative equations then retrace the encounter. The thermal-center mean changes sign, while its second and cross moments are unchanged. Maximum absolute discrepancies are respectively 2.87×10^{-11}, 1.09×10^{-13} and 7.52×10^{-14}. This tests the two-body inverse map; it does not establish detailed balance for a truncated incident ensemble that omits reverse states.

The broad campaign evaluates 38,220 distinct trajectories. All base trajectories exit, but one independent-replica state fails to reach an outgoing plane within $6L/w_\parallel$:

$$(b, w_\parallel, w_\perp, \varphi) = (0.9938225266, 1.9056642462, 1.9668376625, \pi/2), \qquad L = 24. \tag{143}$$

Missing samples are not renormalized away. The $[0.8, 1.2]$ annulus and total conditional drift/diffusion are therefore reported as unresolved, with no aggregate coefficient. The final empirical diffusion spreads in the six annuli are 10.4%, 11.6%, 65.9%, 8.62%, 3.95% and 4.85%. Only the outer two meet the stated 5% finite-comparison target; the inner four do not. Positive contributions from completed nodes are lower bounds only on their specified finite positive quadrature sums, not certified lower bounds on a continuous integral.

Even in the previously tested $[1.2, 2.4]$ impact annulus, the wider speed band gives $D_{\mu\mu}/n = 2.78117 \times 10^{-5}$, about 34.7 times the earlier narrow-band value, and reverses the conditional drift sign to $A_\mu/n = -1.01609 \times 10^{-5}$. This row is itself unresolved by the 8.62% spread and 5.26% phase change; the comparison demonstrates cutoff sensitivity rather than a converged full rate. Twelve separate states give relative L^2 changes of the twelve-state second-moment vector of 4.57×10^{-5} on moving the planes from 24 to 32 and 1.35×10^{-10} on decreasing the active maximum timestep from 0.1 to 0.05. These state checks do not settle the remaining ensemble quadrature error or its failed trajectory.

For a repulsive uniform-field pair,

$$\ddot{z} = k(t) z, \qquad k(t) = \frac{2\kappa}{m} \frac{e^{-r/L_s}(1 + r/L_s)}{r^3} \geq 0. \tag{144}$$

On $z < 0$, the incoming positive parallel velocity decreases monotonically. If it turns, the particle thereafter recedes toward the incoming exit and cannot turn again. If it crosses $z = 0$ with positive velocity, that velocity increases on $z > 0$. Reaching $(z, \dot{z}) = (0, 0)$ at a finite time is excluded by

uniqueness of the regular equations unless the trajectory stays in that plane. Thus an exiting trajectory has at most one parallel turn. A finite flight-budget failure can involve slow outward travel or long near-turning motion; it does not establish a bound pair or a failure of molecular chaos. The omitted impact and speed tails, reverse-complete incident statistics and successive-encounter correlations remain unchecked. No physical constrained coefficient, plasma relaxation time or dipole lifetime follows.

Resolving the failed encounter's flight budget

Example 17 preserves the original broad-campaign metadata and replays its exact failed node. The declared flight factors $6, 12, 24, 48$ vary only the finite integration budget. Factor 6 ends at $t = 75.5642$, after reflection, with $z = -19.268$ and $\dot{z} = -0.320016$; the particle is already travelling toward the incoming plane. Factor 12 reaches the reflected outgoing event $z = -24$ at $t = 90.3503$, with one parallel turn. Factors 24 and 48 reach the same event. This diagnoses an insufficient original numerical flight budget, not a trapped trajectory.

Active maximum steps $0.2, 0.1, 0.05$ give a final conditional second-moment change 2.13×10^{-10}; independently moving the planes through $L = 24, 32, 40$ gives a final change 2.24×10^{-5}. The original comparison of the 0.2 relative solve to the Cartesian pair path has velocity error 1.69×10^{-8}, above the unchanged 10^{-8} absolute target, and is retained as an unresolved coarse comparison. The finest declared relative solve instead uses maximum step 0.05 and its own event time $t = 90.350330908$. Independent twelve-dimensional pair paths with maximum steps 0.05 and 0.025 agree with it in relative velocity to 5.15×10^{-12} and 4.34×10^{-11}, and in relative position to 2.54×10^{-10} and 4.02×10^{-9}. Their velocity plateau is 4.63×10^{-11}. Both satisfy the unchanged 10^{-8} position and velocity targets. No discrepancy is removed by an invariant correction.

Restoring this one node changes only the chosen replica's finite quadrature. Its conditional thermal-center mean, second and cross moments are $(0.2205711, 0.1127298, -0.0154266)$. The positive increment to that replica's finite $D_{\mu\mu}/n$ sum, using the half-second-moment convention, is 4.90439×10^{-5}, equal to about 20.12% of the *base* annulus's reported finite diffusion sum, even though the node represents only $1/2048 = 0.04883\%$ of the replica count. The two populations are distinct; this ratio is a diagnostic of numerical leverage, not a percentage correction to a converged continuous coefficient. For this node the unperturbed relative gyro-radius is $\rho_{\mathrm{rel}} = w_\perp/|\Omega| = 0.9834188$. At the declared zero-plane phase $\varphi = \pi/2$ and $\Omega = 2$, $\boldsymbol{r}_\perp = (b - w_\perp \sin\varphi/\Omega, w_\perp \cos\varphi/\Omega)$; the free relative orbit's closest perpendicular gap is only $b - \rho_{\mathrm{rel}} = 0.0104037$. The interacting independent Cartesian path has sampled minimum separation 0.0225124. These are relative-orbit geometrical quantities, not individual thermal gyro-radii. The Coulomb reference length is $b_{90} = \kappa/[(m/2)|\boldsymbol{w}_{\mathrm{in}}|^2] = 0.00799999$, so the free gap is $1.30046 b_{90}$, despite the much larger guiding-center impact. This is the unscreened Coulomb 90-degree reference length, not an exact deflection parameter for the screened magnetized trajectory. At the sampled closest point, the actual relative perpendicular velocity gives $\rho_{\mathrm{kin}} = |\boldsymbol{w}_\perp(t)|/|\Omega| = 0.683913$ and $\rho_{\mathrm{kin}}/r = 30.3794$; these differ from the incoming ratio $\rho_{\mathrm{rel}}/r = 43.6835$. The measured full-speed ratio is $|\Omega| r/|\boldsymbol{w}(t)| = 0.0302907$, while $|\Omega| b_{90}/|\boldsymbol{w}_{\mathrm{in}}| = 0.00584237$. These local scales place this concrete close encounter outside a verified grazing or close-collision adiabatic ordering. They help interpret its large finite moment contribution, but are not a counterexample to an asymptotic SM working regime or an ensemble relaxation law. Resolving the trajectory does not settle the independent Sobol/phase spread, omitted bands or kinetic closure. Example 13's original aggregate remains unresolved and its metadata and failed comparisons remain available.

The same final audit gives a deterministic inverse-cutoff counterexample, with the field explicitly fixed at $B = 2$ in both trajectory and thermal-center moment calculations. The incoming node $(b, w_\parallel, w_\perp, \varphi) = (1.2, 0.7, 0.8, \pi/2)$ maps under the tested microscopic inverse to

$(1.20082758, 0.69429998, 0.80495188, 3.94743171)$. Both inverse speeds lie outside the original narrow bands. Its forward mean and second moment are 4.96721×10^{-4} and 7.44799×10^{-6}. The reverse mean negation, second-moment equality and cross-moment equality have absolute errors respectively 5.07×10^{-16}, 3.19×10^{-18} and 2.18×10^{-18}; energy errors are below 1.52×10^{-14}. This is a fixed boundary counterexample, not an independent statistical validation or a fitted state. The outgoing event is transverse, so continuity also gives nearby interior incoming states with inverse speeds outside the bands. The incident rectangle therefore is genuinely not closed under this inverse. Conditional drift on that rectangle is not a full Maxwellian drift or evidence against microscopic reversibility.

Positive periodic scattering table: bulk accuracy and a failed tail target

The refined table interpolates the signed conditional mean c and $\ln \sigma_c^2$ with cubic interpolation in impact and speeds and a truly periodic cubic spline in gyrophase. Reconstruction as $c^2 + \sigma_c^2$ and $c^2 - \sigma_c^2$ preserves the positive variance and thermal-center identities. A $4^3 \times 16$ version, tested at 256 fresh uniform states with seed 1301, gives a second-moment relative L^2 error 1.038%, but a 95th percentile pointwise relative error 9.869%; the prescribed 5% tail target fails.

Example 16 refines to $6^3 \times 32 = 6912$ training states on the original narrow bands, and evaluates 256 independent uniform states (seed 1311) inside the training-node hull, with no extrapolation. Its relative L^2 error is defined as

$$e_{\text{bulk}} = \frac{\sqrt{\sum_i (\widehat{M}_i - M_i)^2}}{\sqrt{\sum_i M_i^2}}, \qquad e_i = |\widehat{M}_i - M_i|/M_i, \tag{145}$$

where $M_i = \langle (\Delta\mu)^2 \rangle_V > 0$ is the direct moment. This first sample gives $e_{\text{bulk}} = 0.1328\%$ and an observed 95th percentile 4.480%, meeting the original observed-error criteria. However, the two-sided 95% binomial order-statistic interval for the population 95th percentile is $[2.273\%, 30.303\%]$; 256 states do not establish a 5% population tail bound. Eight independent impact-direction derivative checks give relative L^2 error 0.3632%, with direct centered-difference plateau change 6.73×10^{-7}. This is one moment derivative on eight states, not a general kinetic sensitivity result.

The same table is then tested without retraining on 4096 new independent uniform states (seed 1321). All direct trajectories exit. The original 256-state measurements and training archive remain separate; cached nodes, producer metadata, dependency hashes and interpolant values are verified. After a duration-helper change invalidated that cache's controls hash, the complete training table, original 256-state stage and fresh 4096-state stage were regenerated at source revision `cdb577b`. All eleven saved training, moment, derivative and validation arrays are identical in shape, dtype and element bytes to the archived `90abe1d` measurements; the negative tail result is reproduced without retuning. The fresh bulk error is 0.2282%, with an empirical 2000-resample bootstrap interval $[0.1912\%, 0.2700\%]$. The observed 95th percentile is 6.2610% and its two-sided 95% order-statistic interval is $[5.3207\%, 7.2237\%]$. Thus the predeclared 5% tail target fails in the larger sample, including its confidence interval. Of these states, 5.884% have relative error above 5% and 0.2930% above 100%; small direct moments permit large relative outliers despite small bulk error.

The order-statistic interval uses the binomial distribution of the count below a fixed population quantile, valid for these independent uniform samples and a continuous error distribution. It describes that declared sampling distribution, not every state, a flux-weighted ensemble or omitted domain regions. The bootstrap bulk interval is an empirical uncertainty estimate. A failed tail target is retained as a result, with no parameter retuning on those validation states. Table resolution and conditional moment accuracy do not supply kinetic closure, screening validation, inter-encounter independence or a constrained collision coefficient.

Predeclared eight-node refinement on the unchanged validation domain

A separate, predeclared $8^3 \times 32$ table uses 16384 direct training states, with the same signed-mean/log-variance interpolation, physical inputs, integration tolerances and accuracy targets. The validation domain remains exactly the original six-node inner hull: the impact and two speed intervals are approximately $[1.2405183, 2.3594817]$, $[0.7270122, 1.4729878]$ and $[0.2202591, 0.7797409]$, respectively, with uniform gyrophase on $[0, 2\pi)$. The larger eight-node training hull is not substituted for this fixed test domain. Gauss-six and Gauss-eight training nodes have no bit-identical full states, so no earlier trajectory is reused. The failed six-node record above remains available and is not revised by the new outcome.

Source revision `e1bc1ab` declares 4096 fresh primary states with seed 1331 and bootstrap seed 1332, and 256 fresh secondary states with seed 1333 and bootstrap seed 1334, before training or inspecting their outcomes. The normalized inputs remain $m = q = 1$, $B = 2$, $\kappa = 0.03$, $\lambda_s = 3$, $\theta = 0.5$, incoming/outgoing planes $L = 24$, maximum DOP853 step 0.2 and relative tolerance 10^{-10}. The exact hull, source, versions and hardware are recorded in the public normalized-input metadata.

All 4096 primary direct encounters exit. Their second-moment bulk relative error is 0.11506%, with 2000-resample bootstrap interval $[0.08909\%, 0.14119\%]$. The observed 95th percentile is 2.53760%; the two-sided 95% order-statistic interval is $[2.05643\%, 3.36706\%]$. Thus the unchanged criteria—bulk error at most 2%, observed 95th percentile at most 5%, and confidence upper limit at most 5%—all pass for this declared sampling distribution. The secondary sample has bulk error 0.05963% and observed 95th percentile 3.90539%, but its much wider interval $[1.25255\%, 10.00161\%]$ illustrates why the larger primary sample is the confidence evidence. The eight impact-direction checks give derivative relative error 0.47585%, below the original 5% target; their direct finite-difference plateau change is 6.7149×10^{-7}. Maximum saved training, secondary and primary energy errors are respectively 5.76×10^{-10}, 3.59×10^{-10} and 4.39×10^{-10}, below the unchanged 10^{-8} guard applied jointly to all three stages.

The pass is not a pointwise accuracy bound: 140 of 4096 states (3.4180%) exceed 5% relative error, and eight (0.1953%) exceed 100%; the maximum relative error is 293.95%. Small direct moments still permit large relative outliers. An independent saved-array audit reconstructs the seeds and fixed domain, verifies no overlap with earlier states, rebuilds the interpolant and reproduces the reported statistics and derivative comparisons. It verifies source dependencies against the producer revision. Training energy maxima are replayed from the saved checkpoint; secondary and primary energy checks use saved producer maxima rather than rerunning individual trajectories.

The calculation comprises $16384 + 4096 + 256 + 48 = 20784$ direct solves, including 48 finite-difference solves for the eight derivative states, and takes 354.638 seconds on the recorded Apple M4. This is one observed whole-run cost including concurrent machine load, not a matched-cost speedup over the older table. The successful bounded conditional interpolation test does not validate a physical kinetic closure, a calibrated collision coefficient, inter-encounter independence or accuracy outside the declared hull.

Reproducible evolution visualization

The README animation is a solution of the uniform linear surrogate at finite velocity quadrature, not a nonlinear encounter or physical-rate simulation. The nine positive velocity masses M_j are proportional to $\exp(-v_j^2/2)$, with v_j equally spaced on $[-2, 2]$ and $\sum_j M_j = 1$. Define

$$p_j = \frac{v_j^2 - \sum_i M_i v_i^2}{\max_i |v_i^2 - \sum_l M_l v_l^2|}, \qquad h_j(x, 0) = (0.25 + 0.20 p_j) \cos x. \tag{146}$$

The density and velocity-neutral amplitudes are evolved using matrix exponentials of an independently assembled ordered-pair Fourier matrix ($N_x = 64$, Gaussian range $\ell = 0.6$) and the analytic local matrix. For $D = 0.8$, $B = 1.7$, $q = -1.3$, and $\tau = \lambda t$, the exact comparisons are

$$h_{j,\mathrm{local}}(x,t) = [0.25 + 0.20e^{-\tau}p_j]\cos x, \tag{147}$$

$$h_{j,\mathrm{range}}(x,t) = [0.25e^{-(1-e^{-0.6^2/2})\tau} + 0.20e^{-\tau}p_j]\cos x. \tag{148}$$

All 65 saved frames on $0 \le \tau \le 8$ are checked against these formulas; $1 + h_j$ stays positive. Fixed color limits prevent apparent relaxation caused by automatic rescaling. The displayed velocity coordinate is a discrete node index, not a physical three-dimensional velocity grid. GIF and MP4 use the same frames, and the arrays are saved in a compressed archive. The accompanying validation and cost panels read the existing experiment JSON files. Their metadata retain each source file's hash and source commit, rather than attributing old measurements to the visualization revision.

Continuous local geometric sensitivity

Example 14 differentiates an initial local entropy-production density, not a kinetic trajectory or an implicitly defined steady state. Its prescribed distribution is

$$\ln f = -E - 0.2\mu + p, \qquad p = 0.1\sin(x)u + 0.04y\mu. \tag{149}$$

At $\mathbf{X} = (1, 0.12, 0.3)$, $m = 1.3$, $q = -0.8$, $D = 0.1$, define

$$\mathcal{P}(\mathbf{X}) = \frac{DB^2}{2}\iint ff'\,|I_X P\Delta A \ln f|^2\,\mathrm{d}u\mathrm{d}\mu\mathrm{d}u'\mathrm{d}\mu'. \tag{150}$$

Energy and moment degeneracy give $P\Delta A \ln f = P\Delta A p$. The independent NumPy reference reconstructs Ap from the analytic gradient $\nabla_X p = (0.1\cos(x)u, 0.04\mu, 0)$, $p_u = 0.1\sin(x)$ and the vacuum flow (58); it shares only the field derivative data. Both evaluations retain the two local B Jacobians. Automatic differentiation translates the sample in x while holding the analytic distribution definition, velocity bounds and field parameters fixed; hence it differentiates B^2, f, f', both observable actions and the energy-flow projector. No part of the geometry is stopped in this derivative.

On the base 25×21 velocity quadrature, $|u| \le 4$, $0 \le \mu \le 20$, the mirror, dipole and perturbed dipole respectively give

$$(\mathcal{P}, \partial_x\mathcal{P})_m = (0.00142987874, -0.00316286298), \tag{151}$$

$$(\mathcal{P}, \partial_x\mathcal{P})_d = (0.00528569785, 0.0324274535), \tag{152}$$

$$(\mathcal{P}, \partial_x\mathcal{P})_{na} = (0.00507509906, 0.0310717576). \tag{153}$$

The independent pair values differ by at most 1.03×10^{-15} relative. Sixteen independent centered-difference steps from 0.04 to 10^{-8} show the derivative plateau; the last-three-step maximum relative discrepancy is below 1.54×10^{-8}. Separate velocity-order checks, expanded $|u| \le 5$ and $\mu \le 24$ bounds, and quadrature refinement on each expanded domain give final derivative changes below 0.661%. The local maps are injective on these domains, as established above. This is a checked derivative of the specified normalized weak observable, with no general steady-state sensitivity or physical-rate inference.

Positive dipole distributions with incompatible flux populations

Example 15 constructs a finite-domain counterexample without a time solve. Use poloidal flux ψ, magnetic latitude λ, and periodic azimuth θ. With $r = C\cos^2\lambda/\psi$,

$$\left|\frac{\partial\mathbf{X}}{\partial(\psi,\lambda,\theta)}\right| = \frac{C^3\cos^7\lambda}{\psi^4}, \qquad B\mathrm{d}\mathbf{X} = \frac{C\cos\lambda\sqrt{1+3\sin^2\lambda}}{\psi}\mathrm{d}\psi\mathrm{d}\lambda\mathrm{d}\theta. \tag{154}$$

An independent Cartesian automatic-differentiation determinant checks both identities at two nontrivial coordinates and verifies the inverse flux map. The finite domain is $0.4 \le \psi \le 1$, $|\lambda| \le \pi/6$, $|u| \le 4$, $0 \le \mu \le 12$; $C = 1$ and $f_* = \exp(-E - 0.2\mu)$. Angular integration is exact over 2π.

Let conditional means under $f_* \mathrm{d}\Gamma$ at fixed μ be denoted by an overbar, and set

$$r_E = E - \overline{E}_\mu, \quad r_\psi = \psi - \overline{\psi}_\mu, \quad r_2 = \psi^2 - \overline{\psi^2}_\mu. \tag{155}$$

Solve the positive two-dimensional covariance system

$$\int f_* \begin{pmatrix} r_E^2 & r_E r_\psi \\ r_E r_\psi & r_\psi^2 \end{pmatrix} \mathrm{d}\Gamma \begin{pmatrix} a \\ b \end{pmatrix} = \int f_* \begin{pmatrix} r_E r_2 \\ r_\psi r_2 \end{pmatrix} \mathrm{d}\Gamma, \qquad h = r_2 - a r_E - b r_\psi. \tag{156}$$

The Gram matrix is nonsingular on this positive product domain: an affine relation between E and ψ would require a vanishing u^2 coefficient, then a constant flux. The residual h is nonzero because ψ^2 cannot be an affine function of ψ on a nonzero-width flux interval after the u^2 coefficient has vanished. Define $f_\pm = f_*(1 \pm \epsilon h)$. The conservative bound

$$|h| \le (\psi^2_{\max} - \psi^2_{\min}) + |a| E_{\max} + |b|(\psi_{\max} - \psi_{\min}) =: H \tag{157}$$

holds over the entire truncated domain, since conditional means lie in the same global intervals as their observables. Taking $\epsilon = 0.4/H$ guarantees $f_\pm \ge 0.6 f_* > 0$. Conditional centering preserves every $G(\mu)$; the covariance projection also preserves total energy and mean flux. Number follows from the moment marginal. Nevertheless, writing $N = \int f_* \mathrm{d}\Gamma$,

$$\frac{1}{N} \int \psi^2 (f_+ - f_-) \mathrm{d}\Gamma = \frac{2\epsilon}{N} \int f_* h^2 \mathrm{d}\Gamma > 0. \tag{158}$$

Thus the flux populations differ even after matching the mean canonical toroidal momentum. This is stronger than a mismatch of mean momentum alone.

For mass-normalized distributions let ρ_ψ be the flux pushforward density. The conditional Jensen inequality gives the information bound

$$\frac{1}{N} \int f \ln(f/f_*) \mathrm{d}\Gamma \ge \int \rho_\psi \ln(\rho_\psi / \rho_{\psi,*}) \mathrm{d}\psi. \tag{159}$$

Conservation of the full flux population fixes its strictly positive right-hand side. Therefore any closed evolution satisfying (80) cannot reach f_* from f_+. Ten independent quadratures refine each integration coordinate and all coordinates jointly. The $32 \times 24 \times 32 \times 32$ base and $40 \times 30 \times 40 \times 40$ joint refinement give $\Delta\langle\psi^2\rangle = 2.5868566 \times 10^{-4}$ and a flux information floor of 1.5140605×10^{-5} per particle. Discrete number, energy, each moment-bin population and mean flux match to rounding; the independent squared-residual identity has about 4×10^{-12} relative discrepancy. These are matched *discrete* constraints; replacing sums by conditional integrals gives the continuous construction. The quadrature convergence is on the stated finite domain, not a tail limit.

The finite-domain distributions are not asserted to satisfy a confined ideal evolution. In particular, $u = \pm 4$ walls are not tangent to the mirror force. Collision-only natural boundaries suffice for their local invariant algebra; the combined theorem separately requires closed flux-compatible ideal boundaries. Specular reversal of u at latitude endcaps preserves energy, moment and flux, but does not by itself close finite-u walls. No kinetic trajectory, approach rate or physical dipole confinement time is inferred from this constructive obstruction.

Separating trajectory accuracy from incoming-phase resolution

Example 24 fixes the original close-encounter triple

$$(b, w_\parallel, w_\perp) = (0.9938225266, 1.9056642462, 1.9668376625)$$

and averages over its incoming phase. For each phase let $M_2(\varphi) = \langle(\Delta\mu_1)^2\rangle_V$ be the exact Gaussian center average already derived above. The conditional target and its nested periodic quadrature are

$$I_2 = \frac{1}{2\pi}\int_0^{2\pi} M_2(\varphi)\,\mathrm{d}\varphi, \qquad I_{2,N} = \frac{1}{N}\sum_{j=0}^{N-1} M_2(2\pi j/N). \tag{160}$$

This is a conditional second moment, not an impact/speed integral or a collision rate. The declared $N = 32, 64, 128, 256, 512$ study requires two successive relative changes below 1%, without renormalizing missing phase weights. All 512 phases reach outgoing events within the fixed flight budget, but successive changes are 56.34%, 15.62%, 1.741% and 3.943%. The original run therefore fails its phase criterion and exits nonzero. The last finite sum is $I_{2,512} = 0.002194413424$; it is not a converged value of I_2.

Repeating the entire 512-phase sum with maximum step reduced from 0.05 to 0.025 changes it by 3.13×10^{-11} relatively. Three declared phases also pass independent individual-particle Cartesian trajectory checks at both steps; the largest position discrepancy is 4.27×10^{-9}, velocity discrepancy 4.63×10^{-11} and magnetic-moment discrepancy 1.09×10^{-11}. These use distinct equations and coordinates but the same DOP853 integrator. An independent saved-array audit recomputes the thermal-center moments, all phase sums and individual path energies; its moment and sum discrepancies are below 5.56×10^{-17} and 5.21×10^{-18} respectively. Accurate trajectories do not resolve an undersampled phase integral.

The largest 6 of the 512 phase-node contributions account for 54.71% of this positive finite sum; 26 nodes account for 88.36%. These are finite quadrature concentration diagnostics, not certified fractions of the continuous integral. The 1024 relative solves and six Cartesian controls take 59.19 seconds on the recorded shared Apple M4, including intermediate checkpoint/path writes but excluding imports and final plotting/output. The frozen producer is `58cc3e1`; inputs, every phase result, failed criterion and independent paths remain under `results/encounter_phase/`. No coefficient calibration or lifetime follows from this conditional study.

A separately declared second-moment extension. Producer `3adea8c` extends the nested hierarchy through a declared 8192-node cap, reusing the original 1024 relative-solver rows after exact phase mapping and physical-source hash checks. Its unchanged second-moment criterion stops at $N = 2048$: the $512 \to 1024$ and $1024 \to 2048$ changes are 0.243192% and 0.254258%, respectively. All 2048 sampled phases exit within the fixed budget. Repeating the whole final sum at maximum step 0.025 gives

$$I^{\mathrm{fine}}_{2,2048} = 0.00218353791028, \qquad \frac{|I^{\mathrm{fine}}_{2,2048} - I_{2,2048}|}{I^{\mathrm{fine}}_{2,2048}} = 4.10 \times 10^{-11},$$

passing the declared 0.1% timestep check. The three original anchors and three largest new second-moment contributors also pass Cartesian checks at both steps, with maximum position and velocity differences 6.66×10^{-9} and 1.33×10^{-10}. This run performs 3072 new relative solves in 142.44 seconds on the recorded shared Apple M4; imports and final plots/output are excluded. Its outputs are in `results/encounter_phase/refined8192/`; 4096 and 8192 phases were not run after the predeclared early stop. Independent saved-array receipt `32ac054c` checks the reused rows, moment algebra, phase sums and Cartesian controls without importing production ODE kernels.

Only the second-moment angular criterion passes. The same last angular comparison changes the mean increment by 1.683% and the cross moment by 4.801%; their phase convergence remains unresolved. The original failed 512-node study is preserved. Neither the passing sampled second-moment checks nor all sampled exits certify a continuous outgoing integral, almost-everywhere exits, an impact/speed-integrated rate or a source coefficient.

All three sampled moments remain unresolved. A subsequent source-frozen extension (producer `b7c68b9`) requires both successive angular changes below 1% for each of the mean increment, second moment and cross moment. Its final whole-grid maximum-step halving must change each by less than 0.1%. Both exact-zero successive values have zero relative change; a zero current denominator with nonzero previous value fails. The 8192-node and 1200-second caps are fixed before execution. The audited 4096 relative-solver rows from the earlier 2048-phase study are reused by exact phase mapping. The independent-control set is the three original anchors plus the union of the three largest new absolute contributors to each component, excluding the actual old 2048-node lattice. No failed control is replaced.

All 8192 sampled phases exit at both maximum steps, with 19 reflected outcomes. At step 0.025, the sampled moment vector is

$$(\langle\Delta\mu_1\rangle, \langle(\Delta\mu_1)^2\rangle, \langle\Delta\mu_1\Delta\mu_2\rangle) = (0.000822975985467,\ 0.002182972585931,\ 0.000118812188764).$$

The last angular relative changes are 2.5563%, 0.09989% and 7.1372%. The whole-grid half-step relative changes are 5.90×10^{-10}, 2.61×10^{-10} and 4.82×10^{-9}. The mean and cross moment therefore fail the angular criterion even though the sampled timestep comparisons pass.

Of 24 Cartesian checks at 12 selected phases, 22 pass. At master index 2054, the step-0.05 position, velocity and moment discrepancies are 8.99×10^{-8}, 7.48×10^{-8} and 1.89×10^{-9}; the maximum thermal-center-moment discrepancy is 5.10×10^{-9}. At step 0.025, position and velocity pass but the center-moment discrepancy 1.05709×10^{-10} exceeds the unchanged 10^{-10} target. Both failures are retained. Thus the whole experiment is unresolved and exits nonzero, independently of the angular failure.

There are 12288 new and 4096 reused relative solves. The measured total is 477.437 seconds: 448.766 for new relative solves, 6.891 for Cartesian controls and 21.779 for checkpointing and other overhead. Imports and final plots/output are excluded. Peak process memory is 475955200 bytes on the recorded shared Apple M4. The numerical budget is not exhausted. The independent saved-array audit recomputes all 16384 row moments, cached payloads, sums, control selection and stopping rules; the maximum independent moment discrepancy is 8.33×10^{-17}. Inputs, versions, hardware, arrays, failed controls and audit are in `results/encounter_phase/all_moments8192/`. A separate plotting receipt records the nonnegative error-axis correction at `fc20e88`; it leaves the numerical evidence unchanged. These are finite sampled sums, with neither continuous-coverage certification nor an impact/speed-integrated physical rate.

A continuity obstruction to finite-budget phase coverage. For the exact repulsive screened relative equations,

$$\ddot{z} = a(r)z, \qquad a(r) = \frac{2\kappa}{mr^3}e^{-r/L_s}(1 + r/L_s) > 0. \tag{161}$$

Initial speed is fixed and initial radius satisfies $r \geq L$ uniformly in phase. Hence relative energy obeys

$$H \leq H_{\max} = \frac{m}{4}(w_\parallel^2 + w_\perp^2) + \frac{\kappa e^{-L/L_s}}{L}, \qquad |w|^2 \leq W^2 = 4H_{\max}/m. \tag{162}$$

Since the potential diverges at $r = 0$, energy also excludes a common positive radius around the origin. The full IVP, continued through event planes, has a smooth global flow and continuous phase dependence at each finite time. An outgoing plane crossing is transversal: a zero-velocity first contact would have outward acceleration and would already lie outside just before contact. Once $z = L, \dot{z} > 0$ or $z = -L, \dot{z} < 0$, the acceleration has the same outward sign and reentry is impossible.

Suppose two phases have opposite outgoing exits before a common time T. Their continued positions satisfy $z(T) > L$ and $z(T) < -L$. Continuity supplies an intermediate phase with $z(T) = 0$, and an open neighborhood with $|z(T)| < L$. Irreversibility implies that none of these trajectories has exited before T. Thus opposite observed channels imply a positive-measure finite-budget nonexit interval, even if every node of a finite phase grid has exited. The 512-node evidence brackets such an interval between $\varphi = 1.5585244805$ (transmitted at $t = 90.45180$) and $\pi/2$ (reflected at $t = 90.35033$), for $T = 151.12841$. This theorem is conditional on the exact endpoints having the displayed classifications; the numerical trajectories have independently checked, substantial transverse margins but are not interval-certified solutions.

More generally the finite-time transmitted and reflected phase sets are disjoint open sets. If both meet a connected bracket, they cannot cover it; at least one phase has no finite outgoing-plane exit. This does not prove bounded three-dimensional motion, a positive infinite-time nonexit measure, or a trapped physical pair. In particular, the outgoing scattering integral on such phases needs an almost-everywhere finite-exit result or an explicit extension convention. A finite timeout alone decides none of these issues.

For any finite terminal time, or an eventual finite outgoing event, the center-averaged second moment obeys the uniform bound

$$M_2 \le \left[\frac{m}{8B}\max(W^2 - w_\perp^2, w_\perp^2)\right]^2 + \frac{m^2\theta}{8B^2}(W + w_\perp)^2 =: M_{2,\max}. \tag{163}$$

Here $M_{2,\max} = 0.40441415$. The whole displayed bracket can therefore contribute at most 0.0007898714 under uniform phase weighting, if its outgoing moments exist. This is about 36% of the unresolved finite sum, not a small omission estimate. It neither controls phase quadrature outside the bracket nor defines moments of infinite-time nonexit trajectories. The mixed channels have the antecedents described in [22, 23]; the continuity argument is an elementary coverage limitation, with no priority claim for a new scattering mechanism.

Searching the unresolved phase interval. Example 25 retains three successive, separately committed searches with the same physical inputs. The original producer `687be127` performs 14 bisections of outgoing-event labels and stops at its declared width 7.49×10^{-7} radians without finding a finite-budget nonexit. Producer `661e3c7` instead bisects the continued full-IVP observable $F(\varphi) = z_\varphi(T)$, with $T = 151.1284061$, maximum step 0.025 and relative integration tolerance 10^{-12}. It seeks $|F| < 12$ rather than assuming any monotonicity of exit times. After 31 bisections the phase bracket reaches one binary64 spacing, 2.22×10^{-16} radians, without verifying an interior trajectory. Both searches exit nonzero; neither final bracket is a measured nonexit-interval width.

Producer `17ec229` repeats this full-IVP search and then checks both final endpoints with maximum step 0.0125, relative tolerances 10^{-12} and 10^{-13}, and independent individual-particle Cartesian equations at the tighter setting. All six controls retain the same physical inputs and exactly the same saved binary64 initial arrays. All use DOP853; this is an equation/coordinate and refinement check, not an independent integration-method comparison. The unchanged absolute endpoint targets are 10^{-8} in position and velocity.

Computed continued $z(T)$	Left phase	Right phase
Primary relative trajectory	40.045	−134.563
Half step, tolerance 10^{-12}	226.447	221.736
Half step, tolerance 10^{-13}	97.894	172.542
Individual Cartesian, tolerance 10^{-13}	185.530	67.227

All six endpoint checks fail. Their relative energy errors are only approximately 10^{-12} (or 3×10^{-13} at the tighter settings), yet refined endpoints disagree by tens to hundreds of length units. The right endpoint changes from a computed reflected outcome to transmission. Thus small energy drift does not certify trajectory accuracy; the final machine-width bracket does not verify opposite exact exits. Adding the free-flight angle also loses low phase bits in binary64 input preparation, but controls at identical initial arrays show that this alone does not explain the disagreement. No chaos, trapping, phase-coverage bound or physical collision coefficient follows. The coarse independent endpoint checks of example 24 remain separate evidence for the conditional continuity argument.

The precision-checked search takes 27.31 seconds on the recorded shared M4, including numerical checks and checkpoint writes, excluding imports, metadata initialization and final plots/output. All attempted paths, hexadecimal initial values, six failed controls and source hashes are retained under `results/encounter_censoring/precision_checked/`; earlier failed searches remain in the parent and `refined/` directories.

Arbitrary-precision checks of the saved initial states. Example 33 converts each saved binary64 state component and physical parameter exactly to arbitrary precision, without preparing new trigonometric initial data. It uses Cartesian six-dimensional Taylor integration at 50 and 70 decimal digits, with orders 28 and 36, tolerances 10^{-30} and 10^{-44}, and maximum step 0.5. The first outgoing event is located within the step polynomial; the two saved search endpoints separately continue to the identical binary64 terminal time $T = 151.1284060524644$. The unchanged absolute position, velocity and first-exit-time comparison targets are each 10^{-8}. These are two precision/order/tolerance refinements of one method, not interval-certified exact solutions.

All three fixed-state comparisons pass. Master phase index 2054 remains reflected at $t = 27.8214996438276$; the 50/70-digit position and velocity differences are below 1.24×10^{-32}. For the two final saved machine-width search endpoints, the largest position, velocity and first-exit-time differences are 1.39×10^{-21}, 1.10×10^{-23} and 1.45×10^{-22} respectively. Both transmit: their first exits occur at 41.3437052727771 and 43.0504086092726, and their continued $z(T)$ values are 188.230142729897 and 170.171953545402. Thus the original opposite exit classification is not reproduced for those exact saved binary64 states. Their common transmitted outcome corrects any description of this last bracket as a validated transmission/reflection boundary. The original low energy errors and machine-width phase spacing were insufficient to validate that interpretation.

The two saved states differ initially by at most 6.98×10^{-15} in a Cartesian component and finally by 18.0582; the recorded finite-difference amplification is 2.59×10^{15}. This is evidence of large finite perturbation amplification at these states, without a Lyapunov exponent, an interval error bound, or a new opposite-exit bracket. It does not establish a nonexit interval or continuous phase convergence. The original example-25 failures and example-24 conditional continuity argument remain as recorded; the arbitrary-precision calculation resolves only these three saved IVPs. Its six runs take 164.21 seconds on the recorded shared Apple M4. Full decimal paths, exact binary64 input values and the precision comparisons are retained under `results/encounter_precision/saved_binary64/`. No phase integral, physical coefficient, trapping or lifetime follows from this correction. The producing metadata records revision `9fa28f3` with uncommitted precision source; the retained byte hash matches its later committed

version `5189c0d`. This source match does not relabel the original producer revision. A producer memory peak was not recorded for this run.

16 Reader-facing analytical figures

`examples/22_first_principles.py` generates the introductory README figures. Its helical trajectory exactly solves $m\dot{v} = qv \times B$ for $m = q = B = 1$, $u = 0.4$, $v_\perp = 0.8$ and $0 \le t \le 4\pi$; it checks the Lorentz equation, constant energy and constant magnetic moment. This uniform orbit illustrates the coordinates, not adiabaticity in a varying field. The two schematic moment-bin populations $(0.25, 0.50, 0.25)$ and $(0.40, 0.20, 0.40)$ at $\mu = (0.5, 1.5, 2.5)$ both have number one and mean 1.5, but different full marginals. They are not a time evolution. The mode-decomposition panels read the saved, independently checked uniform local and finite-range operator evolution from example 09 and verify both neutral branches and the two density branches against their exact formulas. The guide states the collision-only, uniform-field, homogeneous-reference and closed-boundary assumptions before interpreting these equations. All input arrays, source hashes, normalized units, versions, hardware and figure digests are recorded in `results/first_principles/summary.json`.

Examples 26 and 27 render existing checked arrays without solving new trajectories. Example 26 reconstructs each of the 30 final Newton-ray trials from saved g, direction and fractions, checks them against the rejection ledger, and plots the smallest trial log density relative to the binary64 normal/subnormal bounds. Example 27 animates the independently checked individual-particle paths at phases 0 and $\pi/2$ from example 24. With $m = 1$, $B = 2$ and stationary center of mass, it recomputes $\mu_1 = m(v_{1x}^2 + v_{1y}^2)/(2B)$ along each path, checking final increments and path energies against the earlier independent evidence. The final ratios $\mu_1/\mu_1(0)$ are 1.0199971 and 1.9122873, with exit times 25.34561 and 90.35033. Both panels share actual normalized time; each curve stops at its own outgoing event. No extrapolated segment, thermal center average, phase integral or calibrated physical timescale is implied. Movie frame count, playback speed, source hashes, rendering versions and hardware are saved in `results/encounter_movie/summary.json`.

Example 36 renders the screened-encounter evidence from immutable example-34 and example-37 arrays. It checks their exact input hashes before writing figures and records the renderer revision, source bytes, versions, hardware, cost and output hashes. The ring panels use twelve prescribed pairs per field; their empirical individual marginal has equal weight $1/24$ on each outgoing particle. The field-scan panels instead use the thermal crossing quadrature. The focused held-out plot retains all 128 error ranks in each sample, including the isolated large relative errors. Original and follow-up acceptance states appear in the figure captions. Equation text uses SVG paths for consistent README rendering. Separate fresh output directories preserve the original and refined evidence.

Spatial views of the recorded calculations

Example 27 also renders both individual-particle trajectories in three spatial dimensions, with identical coordinate bounds, camera and physical clock in the transmitted and reflected columns. The long z axis is visually compressed; the stored box aspect and limits give its relative scale. The lower x–z projections use equal physical length scales and show $|x|, |z| \le 1.5$. Each frame reveals only original saved points with time at or before the displayed clock. The moving marker uses the last such point and disappears at exit; the outgoing endpoint remains marked. The starred close approach is the minimum sampled separation, not a continuous-time minimization. These values are approximately 1.0734 and 0.02251. The animation does not introduce new trajectories or a physical collision rate. The GIF and MP4 encodings use the same 120 frames at 10 frames per second. The static poster shows the final

frame.

Example 28 shows the dipole field and the archived dipole pilot states. For unit strength, a field line at fixed azimuth has

$$r = L\sin^2\theta, \qquad R = L\sin^3\theta, \qquad z = L\sin^2\theta\cos\theta, \qquad \psi = R^2/r^3 = 1/L.$$

Differentiation gives $R' = 3L\sin^2\theta\cos\theta$ and $z' = L(2\sin\theta\cos^2\theta - \sin^3\theta)$. Thus its tangent is parallel to the prescribed dipole vector, in the increasing-θ direction. The renderer checks this identity against the implemented field and verifies constant flux. It clips these drawing curves at $r = 0.25$, outside the source; this drawing cutoff does not change the collision box or simulation data. The 3D field axes have equal physical length scales. The meridional panel uses a logarithmic $|B|$ color scale.

The recorded pilot has $5^3 \times 25 \times 13$ nodes, with spatial box $[0.8, 1.2] \times [-0.2, 0.2] \times [0.1, 0.5]$, $u \in [-4, 4]$, $\mu \in [0, 20]$, $D = 0.1$ and $\Delta t = 0.005$. Only initial, previous and final spatial states are archived: $t = 0, 0.015, 0.02$. No intervening spatial states are reconstructed. Using the same positive quadrature, the plotted quantities are

$$\begin{aligned} n(\mathbf{X}) &= B(\mathbf{X}) \sum_{a,b} w_a^u w_b^\mu f_{\mathbf{X}ab}, \\ \overline{u}(\mathbf{X}) &= \frac{B(\mathbf{X})}{n(\mathbf{X})} \sum_{a,b} w_a^u w_b^\mu u_a f_{\mathbf{X}ab}, \\ p(u_a \mid \mathbf{X}) &= \frac{B(\mathbf{X})}{n(\mathbf{X})} \sum_{b} w_b^\mu f_{\mathbf{X}ab}, \\ N_b &= w_b^\mu \sum_{\mathbf{X},a} w_{\mathbf{X}}^X B(\mathbf{X}) w_a^u f_{\mathbf{X}ab}. \end{aligned}$$

The conditional velocity density obeys $\sum_a w_a^u p(u_a \mid \mathbf{X}) = 1$. Its plotted reference $p_M \propto e^{-u^2/2}$ is normalized on the same finite u quadrature. The maps use the central $z = 0.3$ node; local velocity curves use $x = 1, z = 0.3$ and the two opposite interior y nodes. Display tiles are associated with the quadrature nodes and are bounded by node midpoints and the box edges. They do not represent finite-volume cell averages. These views reveal the spatially varying reduction of the parallel shear, while the global moment-bin populations remain fixed. They retain the pilot's unresolved grid/tail/timestep scope. Inputs, plotted arrays, tests of number, energy and all moment bins, field-line checks, source commits, versions and hardware are in `results/spatial_relaxation/`.

17 Collision operators on a common homogeneous initial state

Example 29 uses one physical three-velocity density throughout:

$$F_0(\mathbf{v}) = \frac{\exp[-(v_x^2 + v_y^2)/(2a) - v_z^2/(2c)]}{(2\pi)^{3/2} a\sqrt{c}}, \qquad a = 1.15, \quad c = 0.7.$$

Here $n = m = B = 1$, $\mathbf{U} = 0$, and $\theta = (2a + c)/3 = 1$. The energy is $3/2$. For a gyrotropic density, $u = v_z$ and $\mu = v_\perp^2/2$, so $d^3v = 2\pi\, du\, d\mu$ and the native guiding-center density in the repository's measure is $f = 2\pi F$. The magnetic-moment marginal is

$$G(\mu, t) = 2\pi \int_{-\infty}^{\infty} F(u, \sqrt{2\mu}, t)\, du, \qquad G(\mu, 0) = a^{-1} e^{-\mu/a}.$$

Thus all operators start with the same distribution, number, momentum, energy and pressure anisotropy, rather than merely matching these moments between different test cases. Homogeneous states are

stationary for the uniform local constrained kernel: its projected spatial derivative vanishes. The example checks this with the actual nonlinear weak implementation at a finite-amplitude state. It also agrees with the homogeneous null branch of source Eq. (181). This statement concerns the implemented reduced kernel, not a claim that ordinary Coulomb collisions leave anisotropic homogeneous plasmas unchanged.

17.1 Constant-frequency Lorentz and conserving Dougherty controls

The gyroangle-independent angular operator is

$$C_L[F] = \frac{\nu_L}{2}\partial_\xi[(1-\xi^2)\partial_\xi F], \qquad \xi = v_z/|\mathbf{v}|.$$

Its Legendre degree ℓ decays at $\nu_L\ell(\ell+1)/2$ and each speed-shell population is conserved. The constant frequency used here is a model choice; the Coulomb deflection frequency in Ref. [28], Eqs. (3) and (10), depends on speed. The long-time state is

$$F_{L,\infty}(r) = \frac{1}{2}\int_{-1}^{1} F_0(r,\xi)\, d\xi.$$

It is isotropic but generally non-Maxwellian. Angular scattering can transfer the species' momentum to its scattering reservoir; the common zero-drift initial state does not test this distinction, which is covered separately by the degree-one control.

The single-species conserving Dougherty operator is

$$C_D[F] = \nu_D \nabla_v \cdot [(\mathbf{v}-\mathbf{U})F + \theta\nabla_v F].$$

Its drift and temperature are the distribution's own conserved moments, not externally fixed bath parameters; see Ref. [29]. The Gaussian covariance evolves exactly as $A_D(t) = I + (A_0 - I)e^{-2\nu_D t}$, and the full density follows the Gaussian-mixture solution already tested against the strong equation. The final state is $F_M = (2\pi)^{-3/2}e^{-|\mathbf{v}|^2/2}$.

17.2 A rate convention that does not equate the operators

With $\Gamma = 1$, the independent Laplace and spherical physical-3V Landau integrals give

$$\dot{A}_{\mathrm{La}}(0) = \mathrm{diag}(-0.0740392537562, -0.0740392537562, 0.1480785075124).$$

For $\delta = T_\perp - T_\parallel = a - c = 0.45$, define

$$\alpha = -\frac{\dot{\delta}_{\mathrm{La}}(0)}{\delta(0)} = 0.493595025041\ldots, \qquad \nu_L = \alpha/3, \quad \nu_D = \alpha/2, \quad \tau = \alpha t.$$

Both conventional controls then have $\delta(\tau) = 0.45e^{-\tau}$ and $\langle\mu\rangle = 1 + 0.15e^{-\tau}$ at every time. Matching this one initial Landau slope neither equates subsequent kinetic evolution nor calibrates a constrained collision coefficient. The time units remain normalized; no density, Coulomb logarithm or magnetized encounter rate has been inferred from this matching.

Higher moments resolve the difference. Wick's rule gives

$$\langle r^4\rangle_0 = (\mathrm{tr}\, A_0)^2 + 2\,\mathrm{tr}(A_0^2) = 15.27.$$

Lorentz preserves this value, whereas Dougherty gives $\langle r^4\rangle_D = 15 + 0.27e^{-2\tau}$. Using $(1-\xi^2)^2 = 8/15 - 16P_2(\xi)/21 + 8P_4(\xi)/35$ and the initial moments $\langle r^4 P_2\rangle_0 = -3.015$, $\langle r^4 P_4\rangle_0 = 0.6075$, one obtains

$$\langle \mu^2\rangle_L = 2.036 + 0.574285714285714\, e^{-\tau} + 0.034714285714286\, e^{-10\tau/3},$$
$$\langle \mu^2\rangle_D = 2(1 + 0.15e^{-\tau})^2.$$

The equilibria therefore have second magnetic-moment moments 2.036 and 2, despite identical pressure relaxation. This is a benchmark consequence of the known operators and their invariants, not a novelty claim. The constrained state retains $\langle \mu^2\rangle = 2a^2 = 2.645$.

The plotted relative entropy is

$$H(F \mid F_M) = \int [F \log(F/F_M) - F + F_M]\, d^3v.$$

For Dougherty it equals $-\frac{1}{2}\log[(1 + 0.15e^{-\tau})^2(1 - 0.3e^{-\tau})]$. Independent scalar integration gives $H(F_{L,\infty} \mid F_M) = 0.000294144974219\ldots$; the initial value is $0.0385755295942\ldots$. The nonzero Lorentz limit follows from its conserved non-Maxwellian speed distribution.

17.3 Numerical representation and interpretation

The conventional time factors are analytic. Example 29 reconstructs Lorentz densities with a finite Legendre expansion and positive velocity quadrature; it does not claim that truncation preserves positivity at every velocity. The retained production expansion uses degree 56 and 96 angular projection nodes. Spherical observables use 64 radial and 64 pitch nodes with radius 10; $G(\mu)$ uses independent parallel quadrature on $[-10, 10]$. The displayed marginal covers $0 \le \mu \le 6$; its second moment is calculated from the full spherical quadrature, not from this plotting window.

Refinement checks include $\tau = 0, 0.05, 1, 3$, including the first positive movie time when high angular degrees have decayed least. They vary quadrature order, tail radius, retained angular degree and projection order separately, and check both the plotted density and the marginal. Errors in densities are scaled by the initial peak density; they are not uniform relative errors in exponentially small tails. No negative reconstruction is clipped. The failed coarse degree-24 reconstruction is retained in the ledger. The final two levels in every accepted direction must pass the declared absolute 2×10^{-8} tolerance in normalized units. The movie evaluates analytic modal time factors at each of its 61 times, with fixed color limits and explicitly saturated colors. All plotted arrays, failed coarse checks and source/runtime metadata are saved under `results/operator_comparison/`.

18 Full three-velocity Landau discretization and consistency

18.1 Unprojected finite weak operator

The physical homogeneous Landau equation, with a specified coefficient Γ, is

$$\partial_t F = \Gamma \nabla_v \cdot \int_{\mathbb{R}^3} U(v - v') F F' \big(\nabla_v \log F - \nabla_{v'} \log F'\big)\, d^3v', \tag{164}$$

$$U(r) = \frac{I - \widehat{r}\widehat{r}^{\,T}}{|r|}. \tag{165}$$

Here F is density per physical velocity volume. It has no gyroangle or magnetic-moment reduction. Positive tensor Gauss weights w_i on a finite cube give populations $n_i = w_i F_i$. Let D be a nodal gradient,

exact on constants, linear functions and quadratics, and set $\Delta_{ij}Dh = (Dh)_i - (Dh)_j$. The matrix-free finite weak form is

$$h^T K(F)k = \Gamma \sum_{i<j} n_i n_j (\Delta_{ij}Dh)^T U(v_i - v_j)(\Delta_{ij}Dk), \qquad w_i \dot{F}_i = -[K(F)\log F]_i. \tag{166}$$

All pairs and all cross terms are retained. The coincident-node contribution is zero. For smooth tests the weak pair integrand tends to zero at coincidence, but this fact supplies no error bound for the near-diagonal quadrature. Optional softening replaces only $|r|$ in the denominator by $\sqrt{|r|^2 + \epsilon^2}$; it defines a different kernel until convergence to $\epsilon = 0$ is demonstrated.

The five functions $1, v_x, v_y, v_z, |v|^2/2$ lie in the raw nullspace: the energy gradient difference is $v_i - v_j$, which U annihilates. They are never imposed by a posteriori projection. Independently assembled small matrices have exactly these five numerical null modes; this is a finite-grid statement. Symmetry and positivity give

$$\frac{d}{dt}\left[-\sum_i w_i F_i \log F_i\right] = (\log F)^T K(F) \log F \geq 0. \tag{167}$$

The default derivative is global polynomial interpolation. The explicit `local_quadratic` option uses three-point quadratic interpolation and is retained as a consistency diagnostic. Both have the five invariants, but they define different finite weak equations. In particular, the local derivative does not differentiate fourth-degree test functions exactly. Natural weak no-flux conditions close the cube; these require tail checks against the infinite-domain reference.

18.2 Full-density implicit step

For $g = \log F$, freeze K at the previous density and solve

$$w \odot (e^{g^{n+1}} - F^n) + \Delta t K(F^n) g^{n+1} = 0. \tag{168}$$

This is the stationary equation of the convex objective $\sum_i w_i[e^{g_i} - F_i^n g_i] + \Delta t\, g^T K(F^n) g/2$. Its Newton Hessian is $\mathrm{diag}(w \odot e^g) + \Delta t K(F^n)$. The existing SOLVAX conjugate-gradient and line-search host evaluates this Hessian without storing the full pair matrix. The diagonal preconditioner is an auxiliary approximation; it changes no pair interaction in the residual. Every nodal log density is an unknown, so the scheme imposes no Gaussian closure.

The discrete entropy budget contains the frozen-mobility production, a nonnegative time-step relative-entropy term, and the measured residual defect. Frozen production is a numerical budget, rather than the instantaneous production of the unfrozen physical equation. Independent dense convex solves and finite-grid ODE integrations check accepted steps, conservation, positivity, entropy and first-order time accuracy. A converged nonlinear solve of this finite equation is insufficient to establish physical velocity-space accuracy.

18.3 An independent continuum Gaussian collision term

Let F be a normalized Gaussian with mean b and positive diagonal covariance $A = \mathrm{diag}(A_1, A_2, A_3)$. Write $x = v - b$, $s = -A^{-1}x$, and $\mathcal{D}(v) = \int U(v - v')F(v')\, d^3v'$. Integration by parts gives $\int U \nabla' F'\, d^3v' = \nabla \cdot \mathcal{D}$. The derivatives of the two flux terms cancel their first-derivative cross terms, leaving

$$\frac{C[F](v)}{F(v)} = \Gamma\left[\mathcal{D}(v) : (ss^T - A^{-1}) + 8\pi F(v)\right]. \tag{169}$$

The sign and factor follow from $\partial_a \partial_b U_{ab} = -8\pi\delta(r)$ in three dimensions. This formula is instantaneous. A nonisotropic Gaussian does not remain Gaussian under Landau evolution, as its nonzero fourth-cumulant production already demonstrates.

Using $1/r = (2/\sqrt{\pi}) \int_0^\infty e^{-z^2 r^2}\, dz$, define

$$d_i = 1 + 2A_i z^2, \qquad V_i = A_i/d_i, \qquad m_i = x_i/d_i, \tag{170}$$

$$Z = \prod_i d_i^{-1/2} \exp\left[-z^2 \sum_i x_i^2/d_i\right]. \tag{171}$$

Completing the Gaussian square yields

$$\mathcal{D}(v) = \frac{2}{\sqrt{\pi}} \int_0^\infty Z \left[I - 2z^2\{\mathrm{diag}(V) + mm^T\}\right] dz. \tag{172}$$

Example 30 uses $z = q/(1-q)$ and separately refines the scalar Gauss integral. The independent audit instead uses adaptive vector quadrature at selected velocities, checks a separate fourth-order finite difference of the original flux divergence, and integrates the strong expression with Gaussian Hermite quadrature. Isotropic Gaussians at three temperatures with nonzero drift give zero to 1.84×10^{-15}. The strong covariance rates agree with the independent weak Gaussian oracle to 1.32×10^{-13} relatively; fourth-cumulant rates agree within 1.9×10^{-14} absolutely. The independent flux-divergence check differs by at most 2.21×10^{-12} at the sampled velocities.

18.4 What the consistency test measures

For $A = (1.15, 1.15, 0.7)$ and $\Gamma = 1$, example 30 compares the complete initial collision term at the same velocity nodes, without rescaling or repairing the sampled Gaussian. Its primary error is

$$E_1 = \frac{\sum_i w_i |C_i^{\mathrm{discrete}} - C[F](v_i)|}{\sum_i w_i |C[F](v_i)|}. \tag{173}$$

It also reports the population-weighted RMS error in C/F, the core $|v| \le 3$ error, separate core/tail contributions to E_1, pressure and fourth-cumulant rates, and the initial quadrature mass defect. The RMS is especially sensitive to small tail populations; neither it nor a pointwise maximum replaces the weighted distribution error. Grid, cube extent and softening scans are recorded separately. Varying the extent at fixed point count changes both tail truncation and interior resolution.

The independent 16^3 calculation already shows the distinction: global polynomial differentiation has 1.98% pressure-rate error but 37.72% full collision-term error E_1. Local quadratic differentiation has the same pressure rate but 49.31% full-term error. Its second magnetic-moment rate is -0.31479, versus the continuum -0.26075. A private five-point quartic diagnostic improves fourth moments but still has 34.09% full-term error at this grid. Conservation alone therefore cannot qualify any of these trajectories as a physical Landau reference. The public refinement arrays and figures are under `results/landau_consistency/`; independent structural and continuum checks are under `results/landau_evolution/`.

The completed public scan at 28^3 nodes and extent 5 gives polynomial $E_1 = 0.0072544$, pressure-rate error 0.0023141 and fourth-cumulant-rate error 0.0053675. The local quadratic values are respectively 0.2459123, 0.0023141 and 0.4930163. The polynomial result passes this single initial weighted L^1 target; its tail-sensitive RMS error in C/F is still 0.2100, and no trajectory accuracy follows. The entire 18-case study takes 183.07 s, including compilation, three warm repeats per case, references and rendering, with peak process RSS 781156352 bytes. The polynomial 28^3 warm median is 8.384 s. These are observed CPU costs on the recorded Apple M4 environment; the two derivatives have not reached matched accuracy. Example 30 retains all measurements and exits with an accuracy failure because the local scheme has not met the declared 1% full-term target.

18.5 Separating pair quadrature from the weak adjoint

The exact continuum Gaussian flux is $J = F(\mathcal{D}s - \nabla \cdot \mathcal{D})$, with

$$\nabla \cdot \mathcal{D} = -\frac{8}{\sqrt{\pi}} \int_0^\infty z^2 Z m \, dz. \tag{174}$$

An independent diagnostic samples this flux on the same cube. Write $\mathcal{A}_h J = -W^{-1} D^T W J$ for the finite weak divergence, with $W = \mathrm{diag}(w)$, and J_h for the sampled pair convolution. The signed collision-term error decomposes exactly as

$$\mathcal{A}_h J_h - \nabla \cdot J = \underbrace{\mathcal{A}_h J - \nabla \cdot J}_{e_D} + \underbrace{\mathcal{A}_h (J_h - J)}_{e_Q}. \tag{175}$$

The first term includes finite derivative reconstruction and the weak cube boundary; the second propagates pair quadrature error through that same derivative. Their norm percentages are not additive. The audit retains the signed weighted L^2 cross term as well as separate norms.

At 28^3 the pair flux itself has vector L^1 error 0.309%. The local quadratic derivative applied to exact flux gives 24.371% collision-term error; the propagated pair-error norm is only 0.291%, compared with total 24.591%. Thus its failure is dominated by the finite weak derivative and boundary reconstruction. For the global polynomial derivative, the corresponding exact-flux and propagated-pair errors are 0.484% and 0.400%, with total 0.725%. The independent 28^3 action agrees with the public saved action within 1.43×10^{-13} in weighted relative L^1. The diagnostic takes 38.33 s on the recorded M4 and changes no production operator. Its inputs, formulas, signed error decomposition and source hashes are in `results/landau_consistency/independent_flux_audit.json`.

19 Hermite quadrature and full-density solver conditioning

19.1 Physical-volume measure and weighted integration by parts

Let z_a, q_a be the n-point Gauss–Hermite nodes and positive weights for $e^{-z^2/2}\,dz$, so $\sum_a q_a = \sqrt{2\pi}$. For a fixed scale $\theta > 0$, define

$$v_a = \sqrt{\theta} z_a, \qquad w_a = \sqrt{\theta} q_a e^{z_a^2/2}, \qquad \int_{\mathbb{R}} \phi(v)\, dv = \sqrt{\theta} \int_{\mathbb{R}} e^{-z^2/2} [e^{z^2/2} \phi(\sqrt{\theta} z)]\, dz.$$

The three-dimensional weights are tensor products of these physical-volume weights. This is a quadrature transformation, not a Gaussian density closure: every nodal log density remains an independent unknown. Nor does nodal positivity establish global integrability of the exponential of a polynomial interpolant. Quadrature and decay accuracy require order and scale tests.

For the polynomial nodal derivative D_z and $Q = \mathrm{diag}(q_a)$, Gaussian integration by parts and exact quadrature through degree $2n - 1$ give

$$Q D_z + D_z^T Q = Q\, \mathrm{diag}(z_a).$$

Indeed, the product of two degree-$n-1$ basis functions times z has degree at most $2n-1$, and their differentiated product has degree at most $2n-3$; the Gaussian-weighted polynomial boundary term vanishes. This identity explains the boundary-free weak adjoint for Gaussian-reference-times-polynomial fluxes, but is not an error estimate for the nonlinear collision flux. With $D_v = D_z/\sqrt{\theta}$, constants, physical velocities and energy still have exact polynomial gradients in real arithmetic. The same

Coulomb pair Gram therefore retains its five raw invariant directions. Independent small-matrix checks verify finite numerical nullity; no continuum nullspace theorem is inherited from a different basis or quadrature.

For extreme Hermite nodes the polynomial derivative can have large entries. The implementation tests each raw reproduction error with the componentwise allowance $10^{-11} + 8n\epsilon_{\mathrm{mach}}(|D_v||h| + |h'|)$. No correction to D_v is applied. Weighted raw invariant actions and trajectory drifts remain separate measured checks.

19.2 Relation to existing structure-preserving formulations

The positive exponential representation, quadratic reproduction, collision invariants and entropy mechanism parallel Hirvijoki–Burby–Kraus [16, Eqs. (7)–(17)]. Their compactly supported second-order Galerkin basis and exact mass/collision bilinear forms differ from the present Hermite physical-volume quadrature, nodal gradients and diagonal population mass. Their basis-specific equilibrium and nullspace argument is consequently not an inherited theorem for this finite pair operator. The present raw checks are independent finite-dimensional statements. No novelty of Gauss–Hermite quadrature or the positive exponential representation is asserted.

19.3 Completed independent checks and the strong oracle

The shared public helper `landau_gaussian_relative_rate` evaluates Eq. (169) at arbitrary physical velocities. It is an instantaneous nonisotropic Gaussian oracle, with scalar-quadrature refinement separate from velocity discretization; evolving densities are not replaced by Gaussians. An independent implementation uses SciPy probabilists' Hermite roots, direct barycentric gradients and explicit NumPy Coulomb pairs. For $A = (1.15, 1.15, 0.7)$, $\Gamma = 1$, the completed measurements are

n	θ	initial E_1	RMS error in C/F	$\max\lvert C_h/F\rvert$
16	0.7	0.0071857	0.0012145	3.5397
20	0.7	0.0047427	0.0007923	19.9347
24	0.7	0.0033262	0.0005563	137.8144
20	0.65	0.0041851	0.0006928	1.3452
20	0.6	0.0047481	0.0008047	0.7664
24	0.6	0.0031602	0.0005283	0.8486
20	0.5	0.0596696	0.0115032	0.8852

The RMS is $[\sum_i w_i F_i (C_{h,i}/F_i - C[F]_i/F_i)^2]^{1/2}$; these are errors at the same nodes, without mass renormalization. Order changes both the largest sampled velocity and core resolution. The 20^3, $\theta = 0.5$ example shows why a small maximum relative rate alone cannot establish collision accuracy. At $\theta = 1$, the 20^3 weighted E_1 is 0.0094182 but its maximum relative rate is 1.2062×10^8; a weighted density norm can conceal severe tail rates. The independently checked weighted integration-by-parts defect is at most 1.34×10^{-15} for the reported $\theta = 0.7$ orders.

A separate 4^3 dense Gram using a positive non-Gaussian density has five numerical zero eigenvalues at threshold $10^{-11}\|K\|_2$, first positive eigenvalue 0.0224943 and largest eigenvalue 56.0472; its raw invariant-action relative norms are at most 1.22×10^{-16}. This small-grid nullity check is not a proof for every order. Endpoint Gauss–Lobatto SBP was also tested: at 28^3 and extent 5 it gives $E_1 = 0.0136215$ and maximum relative rate 1.2028×10^5. Its exact SBP relation does not cure collision-flux interpolation error, so this candidate was not adopted.

One accepted full-density 16^3, $\theta = 0.7$, $\Delta t = 0.05$ step was independently checked against the original frozen weak root. Its weighted root norm is 3.50×10^{-13}, maximum absolute drift among the five invariants 4.78×10^{-15} and minimum new nodal density 4.63×10^{-23}. Its entropy gain 0.00195281655 splits into frozen Gram production 0.00191820352 and time-step KL $3.46130287 \times 10^{-5}$; the entropy identity including the measured residual defect differs by 6.82×10^{-17}. A separate accepted 20^3, $\theta = 0.65$, $\Delta t = 0.05$ endpoint using an auxiliary tensor preconditioner also passes the original unregularized Coulomb equation independently: weighted root norm 1.28×10^{-13}, largest absolute invariant drift 2.98×10^{-14} and minimum density 6.35×10^{-28}. Its entropy gain is 0.00195647958, with Gram production 0.00192170246 and KL $3.47771210 \times 10^{-5}$; the corrected identity defect is 3.01×10^{-16}. The tensor approximation changes only the PCG preconditioner; all original pair interactions and residuals are retained. The next 20^3 step reaches $t = 0.1$. Independent evaluation gives fixed-initial-population root norm 2.11×10^{-11}, maximum cumulative invariant drift 1.28×10^{-11} and positive minimum density 3.02×10^{-28}. Its entropy gain 0.00183222774 contains frozen production 0.00180039917 and KL $3.18285460 \times 10^{-5}$. The residual entropy defect is -2.50×10^{-11}; retaining it gives corrected identity defect 1.72×10^{-16} in magnitude. These checks verify three finite equations, including every stage of the two-step fine-grid pilot, with no finite-time continuum trajectory claim. Inputs, source hashes, signed root and entropy budgets, and negative candidate checks are retained in `results/landau_evolution/hermite_independent_method_audit.json`.

19.4 Auxiliary tensor inverse in population coordinates

Let the positive correction metric be m_i (the old population or a fixed reference with the same total population), with $M = \sum_i m_i$. Write its one-axis population marginals as $m^{(a)}$ and define

$$s_{ijk} = m_i^{(1)} m_j^{(2)} m_k^{(3)} / M^2, \qquad L_a = \operatorname{diag}(m^{(a)})^{-1/2} D^T \operatorname{diag}(m^{(a)}) D \operatorname{diag}(m^{(a)})^{-1/2}.$$

The auxiliary population s has total M; it is used only for the inverse, not as a replacement for the evolved density. Each L_a is positive semidefinite. The frozen local diffusion diagonal is $q_{i,a} = \Gamma n_i \sum_j n_j (U_{ij})_{aa}$, and

$$\kappa = \frac{\sum_{i,a} q_{i,a}}{3 \sum_i n_i}$$

is its population-weighted mean trace divided by three. It has units velocity squared per time, whereas L_a has units inverse velocity squared. If $L_a = Q_a \Lambda_a Q_a^T$, the tensor inverse in coordinates $x = \sqrt{m} \odot d$ is

$$P_m^{-1} = A \mathcal{Q} \operatorname{diag}\left[\frac{1}{1 + \Delta t \kappa (\lambda_{1,i} + \lambda_{2,j} + \lambda_{3,k})} \right] \mathcal{Q}^T A, \tag{176}$$

$$\mathcal{Q} = Q_1 \otimes Q_2 \otimes Q_3, \qquad A = \operatorname{diag}(\sqrt{m/s}). \tag{177}$$

This is the inverse of the congruence-scaled auxiliary logarithmic Hessian $\operatorname{diag}(s) + \Delta t \kappa \sum_a D_a^T \operatorname{diag}(s) D_a$; therefore the scaling also applies to nonseparable m. All computed spectral denominators and scaling factors must be finite and positive. Invalid factors reject the correction; no eigenvalue is clipped and no invariant direction is removed. The default inverse remains diagonal. The residual, full pair Hessian, objective and true PCG residual use the original weak operator for both choices.

19.5 Damped Newton acceptance

The optional Newton globalization chooses the initial common line-search fraction $\alpha_0 = \min\{1, c/\|d\|_\infty\}$, then performs ordinary Armijo backtracking. It changes neither the convex objective nor its root, and

no nodal value is clipped. For strictly positive finite old populations the finite convex objective is coercive and its Hessian is positive definite; its minimizer is unique. Residual and entropy tests still decide acceptance.

19.6 Archived finite stages and the representation limit

The interrupted $\theta = 0.65$ campaign retains ten accepted advances: four for 16^3, four for 12^3 and two for 20^3. The first two cases reach $t = 0.2$; the fine case reaches only $t = 0.1$. Independent raw-pair replay verifies positivity, five unprojected invariants, the original root and the entropy–Gram–KL identity at every saved advance. The 12^3 initial strong error is 0.0115355, so passing its discrete equations does not satisfy the separate one-percent instantaneous accuracy target. Measured campaign wall time is 661.7 seconds (baseline 45.46, 12^3 10.27, fine 605.71 including its interrupted third attempt), not an isolated performance benchmark. The portable independent source is `examples/32_landau_audit.py`; its inputs and hashes are retained in `results/landau_trajectory/independent_portable_audit.json`.

At shared times 0.05 and 0.1, independent interpolation of $\log(F/F_0)$ and Legendre cubature on $[-5, 5]^3$ give 16^3 versus 20^3 full-density differences of respectively 0.2176–0.2247 percent and 0.2020–0.2024 percent of the actual baseline relaxation increment, across orders 32, 48 and 64. Both node sets enclose this box; no extrapolation or time interpolation is used. Outside-5 sampled populations are about 6.3×10^{-6} and 7.5×10^{-6}; outside 4 they are about 2.7×10^{-4} and 1.7×10^{-4}. These raw quadrature sums are not rigorous continuum tail bounds. The missing fine endpoint and unexecuted time, scale and kernel scans leave the campaign unresolved.

A bounded continuation from the immutable fine $t = 0.1$ state removes the optional initial direction cap ($\alpha_0 = 1$) and tightens inner tolerance to 10^{-12}, retaining the original tensor inverse, Hessian, objective, population reference and acceptance. For $m = wF_0$ its original equation and measured norm are

$$R(g) = w(e^g - F^{\mathrm{old}}) + \Delta t K(F^{\mathrm{old}})g, \qquad \rho = \|R(g)/\sqrt{m}\|_2 / \sqrt{\sum_i m_i}.$$

It accepts $t = 0.15$ after three Newton corrections and 317 PCG iterations in 485.5 seconds, all with fraction one despite negative log directions of magnitudes 3.86, 95.13 and 183.61. Independent replay gives $\rho = 1.27 \times 10^{-13}$, minimum positive density 4.86×10^{-151} and cumulative invariant drift at most 4.39×10^{-11}. Entropy gain 0.00171836379 contains frozen production 0.00168914213 and KL $2.92216659 \times 10^{-5}$; the identity including $(g+1)^T R$ has defect 4.82×10^{-17}. The shared $t = 0.15$ density difference on the 5-box remains 0.1886–0.1906 percent of the actual baseline increment.

No $t = 0.2$ root is accepted. True inner residuals stay below 10^{-12}, but fractions fall from one half to 1/32 as outer values approach the actual population representation limit; the last completed correction starts at $\rho = 0.001935$. Trials with minimum logs -727.8, -715.1 and -708.7 pass the finite log-range check but yield zero populations in the backend and are rejected by the separate positivity guard; the selected minimum is -705.5. The probe is deliberately interrupted at this representation-driven slowdown, without clipping or accepting a failed iterate. This is a numerical outer-node limit, not nonexistence of the positive minimizer in real arithmetic. Launch source, accepted arrays, trial history and independent replay are archived in `results/landau_evolution/uncapped_theta065/`.

19.7 Finite-time refinement protocol and timestep preflight

The physical initial density is the same sampled anisotropic Gaussian in all cases: covariance $(1.15, 1.15, 0.7)$, density and particle mass one, $\Gamma = 1$, and magnetic field one for $\mu = (v_x^2 + v_y^2)/2$. Quadrature variance θ changes the sampling measure and outer node extent, not this physical covariance.

The initial strong-RHS oracle is an instantaneous continuum comparison; subsequent accepted states have independent positive nodal densities. An initial oracle pass does not verify a finite-time trajectory.

For the baseline b, a comparison case c and a shared positive saved time, the bounded-box density error is

$$E_{c,L}(t) = \frac{\int_{[-L,L]^3} |F_c(t) - F_b(t)|\, dv}{\int_{[-L,L]^3} |F_b(t) - F_0|\, dv}, \qquad L \in \{4, 5\}.$$

The integrals use Legendre orders 32, 48 and 64 after tensor interpolation of $\log(F/F_0)$. Both node axes must enclose the entire box; 12^3 therefore retains its 4-box diagnostic and skips the 5-box. No extrapolation or time interpolation is allowed. Every shared positive saved time, including the earliest, is tested. The Legendre 48-to-64 change in the relative error must be below 10^{-4} and the relative change in its baseline increment denominator below one percent. Neither the boxes nor the separately reported raw outside-box sampled populations constitute a rigorous continuum tail bound.

For a whole-quadrature observable M, write $\delta M_c(t) = M_c(t) - M_c(0)$ and test

$$E_c^M(t) = \frac{\|\delta M_c(t) - \delta M_b(t)\|_\infty}{\|\delta M_b(t)\|_\infty}.$$

The observables are pressure anisotropy, $\langle \mu^2 \rangle$, $\langle |v|^4 \rangle$, and the vector of centered fourth cumulants $\langle (v_a - \langle v_a \rangle)^4 \rangle - 3\Sigma_{aa}^2$. Each case uses its own raw sampled initial moments. Expectations divide by its actual quadrature number, with no normalization of the evolved population. Denominators at or below 10^{-14} reject an unresolved comparison rather than being clipped. The refinement target is strictly one percent. Velocity-grid, temporal, quadrature-scale, softening and separate linear/nonlinear tolerance controls are retained individually.

The twelve declared cases use $T = 0.2$, the same original tensor-PCG residual and objective, and the following parameters. A dash means the baseline value in that column; the table specifies inputs, not a claim that every trajectory has passed.

Case	n	θ	Δt	root tol.	PCG tol.	log cap
Baseline	16	0.6	0.025	10^{-10}	10^{-8}	2
Coarse grid	12	0.6	0.025	–	–	–
Fine grid	20	0.6	0.025	–	10^{-12}	none
Coarse timestep	16	0.6	0.05	–	–	–
Fine timestep	16	0.6	0.0125	–	–	–
Narrow quadrature	16	0.585	0.025	–	–	–
Wide quadrature	16	0.625	0.025	–	–	–
Softening 0.2	16	0.6	0.025	–	–	–
Softening 0.1	16	0.6	0.025	–	–	–
Both tolerances	16	0.6	0.025	10^{-12}	10^{-10}	–
Nonlinear tolerance	16	0.6	0.025	10^{-12}	–	–
Linear tolerance	16	0.6	0.025	–	10^{-10}	–

The coarse grid, coarse timestep and softening 0.2 cases remain explicit diagnostics; the finer grid/timestep, both quadrature scales, softening 0.1 and each tolerance control determine the declared refinement gate. Every accepted stage additionally requires its original fixed-population root to satisfy the row's nonlinear tolerance, positive density, raw five-invariant absolute drift below 10^{-9}, and the entropy–Gram–KL identity including its residual term.

The earlier complete $\Delta t = 0.05$ baseline study saved 42 accepted advances but remained unresolved: comparison to $\Delta t = 0.025$ at the earliest shared time gave a full-density increment error above one

percent. Independent replay confirms all 42 discrete equations and the producer's density and whole-moment comparisons; passing these equations does not repair the failed accuracy gate. At $T = 0.2$, the empirical native-density ratio for timesteps $0.1, 0.05, 0.025$ is 1.903880, corresponding to $\log_2$ ratio 0.928943. Generally, for three factor-two levels one measures

$$p_{\mathrm{obs}} = \log_2\left(\frac{\|F_{2h}(T) - F_h(T)\|}{\|F_h(T) - F_{h/2}(T)\|}\right),$$

using native weighted L^1, bounded-box L^1, or observable maximum norms separately. This is a finite-interval difference diagnostic, not a convergence theorem or an absolute continuum-error oracle.

A predeclared 16^3, $\theta = 0.6$, $\Delta t = 0.0125$ preflight then saves 16 advances to $T = 0.2$. Its eight comparisons to the immutable $\Delta t = 0.025$ reference, starting at $t = 0.025$, have maximum density error 0.62909 percent across both boxes and all Legendre orders; the largest of the four moment errors is the fourth cumulant at 0.93138 percent. Every quadrature-refinement gate also passes. Independent replay checks 24 original roots (eight reference and sixteen preflight), with maximum root norm 3.82×10^{-11}, raw invariant drift 3.12×10^{-10} and corrected entropy identity defect 1.15×10^{-16}. The preflight costs 50.36 seconds, 32 Newton corrections and 369 PCG iterations; measured process peak RSS is 351,469,568 bytes on Apple M4 under shared load. Exact source, launch commit, versions, inputs, accepted arrays and independent receipts are retained in `results/landau_evolution/timestep_preflight/`. This is a timestep preflight, not a substitute for the other declared controls.

19.8 Short-time Landau results

The twelve-case study completes all 100 advances to $T = 0.2$. Every row has $\epsilon = 0$ except the two explicitly named softening controls. Independent NumPy raw-pair replay verifies each original fixed-population equation at its declared nonlinear tolerance, positive nodal density, and all five unprojected invariants. Across every stage, the largest root norm is 3.82×10^{-11}, the largest absolute invariant drift 3.12×10^{-10}, and the largest corrected entropy–Gram–KL identity defect 1.46×10^{-16}. The minimum saved density remains positive at 7.00×10^{-26}.

All eight required controls pass the separate density, moment and quadrature-refinement gates at every shared saved time. Across both boxes and all three Legendre orders, the fine-grid density maximum is 0.34682 percent and its largest moment error, the fourth cumulant, 0.29647 percent. The fine-timestep maxima are 0.62909 and 0.93138 percent. The quadrature-scale density maxima are 0.10817 and 0.13278 percent; softening 0.1 gives 0.14075 percent, with largest moment error 0.19591 percent. Separate and combined tolerance-control moment differences are below 8×10^{-13} in the defined normalized norm. The coarser $\Delta t = 0.05$ diagnostic remains outside the accuracy target, with 1.17751 percent density and 1.77881 percent fourth-cumulant differences; it is retained explicitly rather than averaged into the passing controls. The baseline and fine-grid initial strong-RHS L^1 errors are 0.77758 and 0.47481 percent respectively, distinct from these trajectory tests.

For timesteps $0.05, 0.025, 0.0125$, the final native-density ratio is 1.948942 and $p_{\mathrm{obs}} = 0.962691$. The 4- and 5-box GL64 values are 0.963120 and 0.963098. Observable orders range from 0.946871 to 0.979140. These are empirical refinement trends for this fixed interval. At $T = 0.2$, raw outside-5 sampled populations are 1.56×10^{-5} for the baseline and 1.81×10^{-5} for the fine grid. Neither those sums nor the integrated error tests bound continuum tails or large relative rates at vanishing-density outer nodes. The supported result is short-time numerical agreement under the declared refinements, without a rigorous infinite-domain or finite-time continuum error bound.

The producer costs 2011.94 seconds under shared load, with 284 Newton corrections, 5751 PCG iterations and measured process peak RSS 602,128,384 bytes on Apple M4. Independent replay costs 139.68

seconds. The actual producer launch commit is `4304054`; scientific source and core hashes, Python and library versions, units, hardware, all input/output hashes and costs are retained in `results/landau_trajectory_dt_refined/execution_receipt.json` and `results/landau_trajectory_dt_refined/independent_portable_audit.json`. The portable audit source is `examples/32_landau_audit.py`.

For the same initial covariance and $\Gamma = 1$, matching the initial pressure-anisotropy slope gives $\alpha = 0.4935950250$ and $\tau = \alpha t$. This is a comparison time scale for the chosen input. The exact Lorentz and Dougherty anisotropies are $0.45e^{-\tau}$; their second magnetic moments are respectively

$$2.036 + 0.5742857143e^{-\tau} + 0.0347142857e^{-10\tau/3}, \qquad 2(1 + 0.15e^{-\tau})^2.$$

At $t = 0.2$ ($\tau = 0.0987190050$), independently checked values are:

Operator	anisotropy/0.45	$\langle\mu^2\rangle$	fourth cumulant in v_z
Constrained local	1	2.645000	0
Lorentz	0.905997	2.581282	0.002188
Dougherty	0.905997	2.580536	0
Landau saved state	0.908435	2.595895	−0.020160

Independent control-formula and source-plot replay is retained in `results/landau_trajectory_dt_refined/independent_comparison_media_receipt.json`. Matching one initial slope does not equate the subsequent operators. The Landau values come from the evolved full nodal density, not a Gaussian moment closure. Its nonzero fourth cumulants are well separated from the sampled initial offsets (at most 1.54×10^{-5}), which are preserved in the raw evidence and subtracted in increment comparisons. The standard operator distinctions shown here provide an implementation comparison for this input, not a new physical collision-rate or lifetime claim.

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